



A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference

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Chapter 1

Introduction

1.1 Background of the Study

A functional legal system requires strict consistency so that citizens can clearly understand their rights and obligations [62]. However, the sheer volume of modern legislation has grown so rapidly that human practitioners can no longer track every active rule [161]. As governing bodies continuously pass new regulations to address emerging issues, they frequently enact laws that inadvertently contradict older ones [95]. Because the total number of documents is overwhelming, identifying these hidden conflicts through manual review is computationally impossible for legal staff. This issue is particularly severe in local government units, where historical ordinances are often scattered across fragmented files rather than maintained in a structured digital database [50].

Within the Philippine legal framework, local governments like Davao City operate under delegated authority and can only enact laws permitted by the national government [71]. Due to this structural limitation, the *Magtajas v. Pryce* Doctrine strictly prohibits any local municipal ordinance from contradicting established national statutes [188]. Consequently, in the local context, city councils must rigorously audit every newly proposed draft ordinance to ensure it logically aligns with existing laws before final approval. If reviewers overlook a normative conflict, such as a local ordinance prohibiting an activity that a national law explicitly permits, the resulting local legislation is legally void. Passing invalid laws creates severe jurisdictional disputes, wastes administrative resources, and ultimately compromises the constitutional rights of the public.

To address these legal risks, the legislative process must utilize an *ex-ante* evaluation: a term derived from Latin meaning 'before the event.' Unlike an *'ex-post'* review, which happens after a law is already in effect and often after legal damage has occurred, an *ex-ante* audit acts as a first line of defense. By checking a proposed ordinance for conflicts while it is still in the drafting phase, the Sangguniang Panlungsod can proactively align local rules with national statutes. This preventative approach is consistent with modern Philippine standards like Republic Act No. 11032 or the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which encourages government agencies to perform regulatory assessments to ensure that new rules do not create unnecessary or

conflicting burdens on the public.

The decentralized drafting process of local ordinances further complicates manual review because the resulting texts are often highly unstructured [166]. City councils frequently splice and amend different drafts, producing verbose documents that blend formal legal English with regional dialects like Cebuano and administrative "Taglish" [54]. Furthermore, because municipal archives heavily rely on digitized PDF scans, the text suffers from severe Optical Character Recognition (OCR) errors. While OCR is typically 99 percent accurate for clean English text, studies show accuracy can drop to 78 percent for low-resource languages, with Cebuano specifically seeing word error rates around 6.9 percent [138]. These errors cause an "error cascade" that shatters standard vocabulary, making it difficult for traditional search engines to semantically differentiate between a statutory obligation and a strict prohibition.

In light of these challenges, the proponents seek to develop a specialized application capable of retrieving relevant legal documents and comparing them against proposed ordinances to detect potential conflicts.

1.2 Problem Statement

This study is conducted to investigate the use of a multi-stage architecture to automate the manual auditing of proposed local legislation in Davao City.

Specifically, the study aims to answer the following research questions:

- What are the specific linguistic parameters and structural markers that define a "semantic conflict" between a proposed local ordinance and existing local or national laws?
- How can a Coarse-to-Fine system be designed to efficiently retrieve relevant national and local statutes before performing deep logical inference on constrained LGU hardware?
- Which combination of a hybrid Information Retrieval (IR) algorithms and Natural Language Inference (NLI) models provides the highest accuracy when processing the specific hierarchy of Philippine legal texts?
- What is the empirical performance of the proposed pipeline when evaluated against a manually curated Ground Truth dataset of local and national conflict pairs?
- How usable and viable is the final system interface for key stakeholders within the Davao City Sangguniang Panlungsod?

1.3 Objectives of the Study

The primary goal of this research is to develop a computationally efficient, two-stage semantic conflict detection system for proposed Davao City ordinances.

Specifically, this study aims to:

- Identify the specific linguistic characteristics and structural constraints of Davao City’s legislative archive that contribute to semantic ambiguity and retrieval failure;
- Design a Coarse-to-Fine computational pipeline that decouples document retrieval from logical inference to optimize resource usage for local deployment;
- Implement a hybrid Information Retrieval (IR) algorithm and Natural Language Inference (NLI) system specifically calibrated for the Philippine legal hierarchy;
- Evaluate the system’s reliability by comparing its automated detections against a verified set of known legal conflicts to measure how accurately it identifies contradictions; and
- Develop a user-friendly web interface for the Davao City Sangguniang Panlungsod designed to accept uploaded draft documents and provide real-time conflict alerts.

1.4 Significance of the Study

The Davao City Sangguniang Panlungsod stands as the primary beneficiary of this automated consistency checking system. By transitioning from manual review to computational auditing, the tool saves legislative staff hundreds of hours of reading time. This technological intervention actively prevents the city from enacting logically flawed laws that invite future litigation or administrative deadlocks. It empowers the local government to draft regulatory measures with a significantly higher degree of legal certainty and operational speed. Ultimately, the system modernizes the legislative lifecycle by embedding rigorous logical checks directly into the initial drafting phase.

This study also provides substantial value to legal researchers and the local Natural Language Processing community. It establishes an empirical baseline for processing messy, localized Philippine legal corpora that frequently contain “Taglish” and archaic Spanish loanwords. The research demonstrates that efficient, edge-deployable encoder models can successfully parse OCR-degraded documents without relying on expensive, cloud-based supercomputers. Beyond academic and governmental applications, the general public benefits directly through the consistent implementation of coherent public policy. The tool acts as a procedural safety net, protecting citizens from the legal traps and frustrations of contradictory municipal obligations.

1.5 Scope and Limitations

The scope of this study focuses on developing an automated semantic conflict detection system to evaluate draft ordinances from Davao City against a dual statutory search space. This collection encompasses national statutory enactments compiled to the extent accessible from public legal repositories, paired alongside the historical archive of enacted Davao City municipal ordinances collected to the fullest extent possible from local legislative records. Operationally, the system audits draft measures across two complementary dimensions: vertical statutory preemption, ensuring local provisions do not contradict superior national statutes under the *Magtajas* doctrine and Republic Act No. 7160; and

horizontal consistency, verifying that proposed clauses do not contradict, duplicate, or inadvertently cause implied repeals of active municipal ordinances.

To establish a feasible engineering boundary, several explicit delimitations govern this research, serving as baseline operational constraints while defining clear avenues for future work:

Statutory Text Boundary. The knowledge base is strictly bounded to codified national statutory instruments and enacted municipal ordinances. To keep retrieval indexes manageable and avoid ambiguous litigation records, judicial case law from Supreme Court decisions and departmental administrative circulars are excluded from the primary retrieval index. Synthesizing sprawling judicial case law alongside statutory enactments remains an open research frontier recommended for future large-scale legal informatics projects.

Municipal Archive Gaps. The historical municipal archive reflects the practical realities of local record-keeping. While recent legislative terms benefit from digitized municipal database records, earlier decades rely primarily on curated landmark ordinances due to physical archive limitations. The system evaluates drafts strictly against the collection assembled through public portals and institutional coordination with the Sangguniang Panlungsod, leaving the OCR recovery of older paper records as an ongoing digitization task.

Linguistic Scope. The system operates exclusively on English-language texts, following Section 20, Chapter 5, Book I of Executive Order No. 292, which establishes that the English text controls in legal interpretation. While local drafts occasionally include regional terms or administrative Taglish, filtering the analysis to English avoids semantic drift and translation distortion. Extending conflict detection to multi-lingual and regional dialect reasoning is designated as a recommendation for future research.

Single-Premise Asymmetric Pairing. The inference module evaluates proposed ordinance sections against individual candidate provisions using asymmetric span-level pairing (1-vs- N). The pipeline deliberately excludes multi-hop reasoning across multiple disparate enactments simultaneously. Identifying intricate legal contradictions that emerge only when synthesizing two or more separate national statutes exceeds the computational capacity of standard workstation hardware and is recommended for future exploration.

Bounded Open-Weight Model Envelope. The evaluation is intentionally restricted to open-weight transformer encoder architectures capable of executing within the computational envelope of standard Local Government Unit (LGU) desktop computers (sub-600M parameters, sub-2GB inference VRAM). Massive autoregressive large language models (LLMs) and proprietary cloud APIs are excluded to ensure privacy, zero recurring cloud costs, and local offline deployment. Extending the pipeline to multi-billion parameter foundation models is reserved for future enterprise-grade studies.

Analytical Diagnostic Role. The system functions strictly as a diagnostic decision-

support tool providing white-box explainability, such as confidence metrics and attention heatmaps. The framework does not rewrite, generate, or alter legislative text, preserving human legal judgment in the hands of legislative researchers.

Direct Normative Scope vs. Longitudinal Statutory Construction. The operational scope of the natural language inference engine is strictly delimited to direct normative and modal contradictions arising from explicit textual collisions (such as local prohibitions of federally authorized activities, unauthorized procedural preconditions, or explicit statutory penalty ceiling infractions). The system deliberately does not model longitudinal legislative histories, historical tax rate compounding, or multi-year financial calculations across unprovided statutory baselines, as observed in complex judicial tax disputes [192]. Modeling such multi-hop temporal statutory construction and arithmetic accounting exceeds the ex-ante text-matching role of cross-encoder architectures and is preserved as an avenue for specialized fiscal analytics.

1.6 Definition of Terms

Coarse-to-Fine Architecture. A two-step computer process. First, the system quickly searches through thousands of documents to find "candidates" (coarse), then it carefully analyzes those specific candidates for logical conflicts (fine).

Contradiction. A relationship in NLI where the second sentence is false if the first sentence is true, indicating a logical conflict.

Davao City Ordinances. These are written local laws created by the Davao City government that are intended to be permanent and enforceable.

Entailment. A relationship in NLI where the first sentence (premise) proves that the second sentence (hypothesis) is true.

Ex-Ante. Derived from Latin meaning "before the event". Operationally, this refers to the evaluation or checking of a proposed city ordinance before it is officially signed into law.

Information Retrieval (IR). The process of searching for and finding relevant documents (like specific laws) within a massive collection of text.

Local Government Code of 1991 (RA 7160). The national law that gives local governments the power to create their own ordinances for the "General Welfare" of their citizens.

Magtajas v. Pryce Doctrine. A legal rule from the Supreme Court stating that local ordinances are inferior to national laws and cannot go against any Act of Congress.

Natural Language Inference (NLI). A task where the computer compares two sentences—a "premise" and a "hypothesis"—to see if they agree or disagree with each other.

Neutral. A relationship in NLI where the two sentences are unrelated and do not affect each other's truth.

OCR Noise. Text errors (like misspellings or broken words) that happen when physical paper documents are scanned into digital files.

Sangguniang Panlungsod (SP). The local legislative body or "City Council" of Davao City, responsible for creating and approving local laws.

Semantic Conflict. A logical clash where two different laws give opposite instructions. For example, a conflict occurs if one law permits an activity while another strictly prohibits it.

Chapter 2

Literature Review

2.1 Legal Informatics Conflict Detection

For a legal system to work properly, its laws must be consistent. This means that citizens and the government should be able to understand their rights and duties without running into laws that give opposite instructions [62]. Semantic Conflict Detection—also called the detection of statutory or normative conflicts in recent studies—is the computer’s task of finding these logical clashes between different legal rules [111, 3]. Specifically, it identifies cases where two or more valid laws give opposite orders to the same person under the same situation, such as when one law says you “must” do an act while another law “forbids” it [3, 89]. It is also important to distinguish between a new law that is meant to change an old one (a valid amendment) and a law that accidentally goes against an old one without saying so (a legal error) [84]. In the world of computer science and law, the move from manual reading to automated conflict detection is necessary because there are simply too many laws for humans to track. Ruhl and Katz [161] found that the number of new regulations has grown so fast that human experts can no longer keep up with all of them. As Shao et al. [175] points out, while lawyers are good at resolving known conflicts, it is physically impossible for them to cross-reference thousands of old laws at the same time to find hidden contradictions. Therefore, Semantic Conflict Detection acts like a “health check” for laws, using math to map out where different rules overlap so that problems can be fixed before they lead to court cases or government confusion. To understand how a computer can detect these conflicts, we must look at the factors that make it difficult. This section will examine the challenges of “legal inflation” (the growing number of laws), the limits of manual auditing, and the confusion caused by legal language. This section will also look at conflict detection research within the Philippines, specifically focusing on the hurdles found in local government archives.

2.1.1 The Unstructured Text Problem in Legal Corpora

Managing unstructured legal text—text not stored in structured databases—represents a fundamental maintenance challenge for statutory systems. Practical automated auditing requires addressing rapid textual accumulation alongside the linguistic complexity of

legislative drafting. These structural realities drive the research transition toward neural retrieval and inference tools for statutory verification.

2.1.1.1 Legal Inflation

We refer to "Legal inflation" as to how the number of laws and their connections keep growing. Katz et al. argue that the sheer volume of laws has made it too difficult for humans to check for consistency manually [95]. As new rules are made to keep up with technology, they create "regulatory debt," where new laws might conflict with older ones without anyone noticing. This is a big problem for local governments, because unlike national laws, local ordinances are often not organized or scanned properly. Dias et al. [50] notes that turning these laws into machine-readable data is hard because they are often scattered across different files, from formal resolutions to quick executive orders.

The "search space" for finding conflicts is therefore a messy collection of different texts rather than a neat database. Zhong et al. [229] describe this as a "dynamic consistency problem" where facts and rules are always changing. Because this data is not organized, government agencies cannot be proactive. They only realize there is a conflict after something goes wrong. This is why the field has moved from old AI systems that use simple rules to new "neural models" that can learn from data. These systems are needed to process huge collections of laws that are too big for the human brain to handle [27].

2.1.1.2 The Limits of Manual Review

Scanning documents into a computer is no longer the main challenge; the primary operational hurdle is now checking if those documents agree with each other. As legislative bodies pass laws faster, they create a massive pile of unorganized data. The main difficulty is telling the difference between a "good" overlap (where a law is meant to cover multiple areas) and a "bad" conflict (where a law is written poorly and contradicts another). Recent surveys show that standard search engines usually fail to understand these subtle distinctions [9]. As a result, many conflicts stay hidden until someone gets sued. This moves the burden of finding conflicts from the people writing the laws to the judges in court [202].

2.1.1.3 Semantic Ambiguity

Old computer search systems look for exact word matches. However, this fails to catch the real meaning of human language. Legal language is like a specialized code. Words like "shall," "must," or "may" have very strict meanings that keyword searches often mix up. Chalkidis et al. [27] show that this "ambiguity" makes exact-match techniques fail at complex legal tasks. For example, a law banning "motor vehicles" might conflict with a law allowing "electric scooters." A keyword search for "vehicle" will not find the problem if the words do not match exactly.

Legal consistency often depends on "implied" rules—things the law suggests but doesn't say directly. Rigano et al. [157] argue that finding these contradictions requires a system that evaluates logical relationships rather than surface string comparisons. "Transformer" AI models are an example of a solution that can solve this gap by creating

mathematical representations (vectors) that capture deep meaning [228], which shall be discussed in later sections. However, there is a trade-off. Sovrano et al. [181] argue that while these models are very accurate, their "black box" nature is a problem because they don't explain *why* they found a conflict. In a legal setting, the reasoning is just as important as the detection [172].

2.1.2 Semantic Conflict Detection in the Local Context

While researchers around the world are building conflict detection tools using benchmarks like LexGLUE, these tools are still in the early stages in less prominent countries like the Philippines [27]. Since the Local Government Code of 1991, cities like Davao have created thousands of ordinances that are not well-organized [71]. Although the math for detecting conflicts exists in general AI research, it has rarely been applied to Philippine laws.

2.1.2.1 Legal-Related Semantic Conflict Detection in the Philippines

When Philippine AI research does focus on law, it is usually about organizing data rather than deep reasoning. Early work by Virtucio et al. [210] used basic classifiers to predict Supreme Court decisions, while Sagum et al. [163] used AI just to summarize national court rulings.

More recently, studies have started using "Transformer" models, a method that shall be discussed in later sections, but they are still used mostly for search. Roque et al. showed in 2024 [159] that "dense embedding" models are much better than keyword searches at finding specific tax laws. However, these embedding models are basically just advanced search engines. They can find two laws about "business taxes" because they are topically similar, but they lack the logic layers needed to tell if one law actually contradicts the other.

Looking at these trends, there is a clear gap in the research. The Philippine AI community has successfully used computers to categorize, summarize, and search national laws. They have also proven that NLI models can understand logical contradictions in local dialects for fact-checking [210, 163, 159, 136]. However, actually using these advanced NLI models to find conflicting rules in the messy, unorganized ordinances of a local city like Davao remains almost entirely unexplored.

2.1.2.2 The Landscape of Logic and Conflict Detection in the Philippines

In the Philippine AI community, the tools needed for conflict detection, specifically Natural Language Inference (NLI), are already being used, but mostly for other tasks. Recent local studies have used NLI to find logical contradictions in things like online news and misinformation. For example, Nuñez et al. proved that these AI models can handle the logic needed for Tagalog and "Taglish" (a mix of Tagalog and English) [136].

However, when researchers try to analyze local government records, they usually go back to traditional methods. Most document analysis in the Philippines still relies on "Boolean" search (keyword matching) and simple rule-based AI. As recent Philippine studies show, these old systems are "lexically fragile" [129]. They fail if two laws use

different words for the same thing (like “fireworks” vs. “pyrotechnics”) and they break down when there are typos or scanning errors (OCR noise) in the digital files. Trying to use manual rules to find clashes in thousands of messy city ordinances is simply too hard for a computer or a person to handle alone.

2.1.2.3 Gaps in Local Legislative Conflict Detection

While the Philippine AI community has successfully automated the search and summarization of national laws, a significant gap remains in using deep logic models (NLI) to audit local municipal ordinances. Research suggests this gap exists because local legislation is far more difficult for a computer to process than national statutes. This section outlines the specific reasons why this area of research remains largely unexplored.

Dataset Gap. The primary barrier to research is the lack of “ground truth” datasets for Philippine ordinances. Unlike national laws, which are well-documented, local ordinances are considered “low-resource” data [149]. Standard AI models are trained on massive Western datasets like LexGLUE, but there are no equivalent datasets that provide annotated pairs of conflicting Philippine ordinances to train an NLI model [63, 132]. Without a standardized benchmark to measure progress, local researchers have found it difficult to build systems that move beyond simple keyword matching to actual logical reasoning [63].

Resource Gap. Deductive logic models like Cross-Encoders require significant computing power that many Local Government Units (LGUs) simply do not have. Research on Philippine public administration shows that AI adoption is still in its early stages due to limited funding, a lack of skilled IT personnel, and outdated computer hardware [63]. Studies have shown that in some municipal councils, only 20 percent of legislative staff have received formal ICT training, and most still rely heavily on printed documents rather than structured databases [63]. This lack of digital infrastructure makes it difficult for researchers to justify developing complex AI models that would be too “heavy” to run on actual LGU workstations [86, 63].

Linguistic Gap. Local ordinances are linguistically more complex than national laws because they frequently blend English with regional dialects like Cebuano and administrative “Taglish” [54]. Most Natural Language Processing (NLP) tools are designed for standard English and often break down when they encounter local terms or Spanish legal loanwords common in the Philippines [123]. Research on Philippine social media AI has proven that NLI can handle local dialects for simple fact-checking, but applying this same logic to the dense, multi-generational language of local ordinances is a much harder task that requires specialized domain knowledge that has not yet been fully researched [31, 74].

Trust Gap. There is also a trust gap within the legal profession. As noted by Supreme Court Senior Associate Justice Marvic Leonen, the legal community has not yet fully embraced AI because current models are often “black boxes” that do not explain their reasoning [109]. Lawyers are cautious about using automated tools because inaccuracies could lead to legal errors or the passage of void laws [36]. Consequently, research has focused more on simple organization and retrieval tasks—where the

computer acts as a simple search engine—rather than deep reasoning tasks where the computer makes a judgment about a legal conflict [11].

2.2 The Philippine Legislative Domain

The Philippine legal system is a unitary state. This means that Local Government Units (LGUs) only have powers that are given or delegated to them by the national government [71]. This setup creates a strict top-down hierarchy that limits how much freedom cities like Davao City have when making laws. Any computer system built to check these laws must understand this specific legal environment. This section explains the barriers in Philippine municipal law. First, it looks at the strict rules that make local ordinances follow national laws. Second, it examines the difficulties and language issues in the local drafting process. Finally, it reviews the current methods used to check for consistency, which shows why we need to move toward automated auditing.

2.2.1 Hierarchical Constraints

The legal system in the Philippines depends on the rule that national laws always override local ones. To understand the limits of writing local laws, we must look at the specific rules that put ordinances below national statutes. This part describes the centralized framework of the government, the powers given by the Local Government Code, and the tests used by the Supreme Court to cancel local laws that conflict with national ones.

Unitary Principle in the Philippines. Research on Philippine political law shows a rigid pyramid of power: the 1987 Constitution is at the very top, followed by national laws (Republic Acts) passed by Congress, international treaties, executive orders, and finally, local ordinances [71]. Under this setup, no LGU can ignore or contradict an Act of Congress. The rule of "national preemption" is a major constraint; when a national law already covers an entire subject—such as firearms or immigration—local governments are not allowed to make different rules [71].

The Local Government Code of 1991. The main law for LGUs is the Local Government Code (LGC) of 1991. Section 16, known as the General Welfare Clause, gives LGUs the power to create ordinances for the safety and success of their citizens, but only if these rules stay "within the framework of the law" [71]. Since Davao City is a "Highly Urbanized City" (HUC), it is independent of any province and its ordinances are not checked by a provincial board. This makes the city more responsible for checking its own laws for conflicts before they are passed [71].

Magtajas vs. Pryce Doctrine. Perhaps the most important test for whether an ordinance is valid comes from the Supreme Court case *Magtajas v. Pryce Properties Corporation, Inc.* The Court listed six requirements that a local law must meet to be valid [188]:

- It must not contravene the constitution or any statute.
- It must not be unfair or oppressive.

- It must not be partial or discriminatory.
- It must not prohibit but may regulate trade.
- It must be general and match public policy.
- It must not be unreasonable.

Local ordinances are considered lower in rank than state laws. Therefore, LGUs cannot try to regulate activities that the national government has already permitted or regulated [71].

Vertical Preemption and Historical Judicial Clashes. When a local ordinance impermissibly enters an area governed by national legislation, Philippine courts consistently apply the *Magtajas* doctrine to declare the local measure invalid. In *Magtajas v. Pryce Properties Corporation, Inc.* [188], the City Council of Cagayan de Oro enacted ordinances prohibiting casino operations to protect public morals. The Supreme Court struck down the local enactments as *ultra vires*, ruling that a municipal council cannot prohibit commercial gambling activities that a national statute expressly authorized through the Philippine Amusement and Gaming Corporation (PAGCOR) charter.

Similar preemption disputes have directly affected Davao City legislation. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [191], the Supreme Court invalidated Davao City Ordinance No. 0309-07, which imposed a municipal ban on aerial pesticide spraying across agricultural plantations. While recognizing the environmental protection objectives of the city government, the Court ruled that the ordinance exceeded municipal police powers and collided with the regulatory authority granted to the national Fertilizer and Pesticide Authority under Presidential Decree No. 1144. In *City of Manila v. Laguio, Jr.* [190], the Court likewise struck down an ordinance prohibiting motels and massage parlors, holding that local police powers cannot arbitrarily prohibit lawful commercial activities under the guise of zoning regulation. These historical rulings demonstrate that when local drafts exceed statutory boundaries, the resulting enactments face invalidation in judicial review.

Horizontal Coherence and Legislative Saving Devices. In contrast to vertical statutory preemption, municipal lawmaking also involves horizontal consistency, where newly proposed ordinances must align with existing, fellow local enactments. When drafting local legislation, the Sangguniang Panlungsod routinely uses the Three-Reading legislative process, particularly the committee debates and line-by-line amendments during the Second Reading, to harmonize new proposals with older municipal codes [166]. Unlike vertical conflicts that represent fatal legal defects, inconsistencies between a new draft and an older municipal ordinance frequently reflect intentional legislative updates, such as revising local tax rates or adjusting traffic routes.

To manage these horizontal modifications systematically, legislative drafters rely on standardized legislative saving devices recommended by the Department of the Interior and Local Government (DILG) [45]. Every formal ordinance incorporates

an express or general repealing clause, declaring that all prior local enactments inconsistent with the newly approved measure are amended or repealed accordingly. Furthermore, ordinances feature a separability clause, acting as a structural legal safeguard ensuring that if any single section is declared invalid by judicial authorities, the unoffending sections remain in full legal force [45]. However, because municipal legal researchers must review hundreds of active local enactments under strict legislative session schedules, subtle inconsistencies often go unnoticed during the Second Reading, allowing problematic clauses to proceed to final enactment.

Normative Authority of Executive Orders vs. Judicial Adjudication. A fundamental design decision in legal informatics is delineating which positive legal instruments establish binding preemption standards. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III, Title I, Chapter 2, Section 2 of the Administrative Code of 1987 (Executive Order No. 292) [153], Executive Orders (EOs) are acts of the President providing rules of a general or permanent character in implementation or execution of constitutional or statutory mandates. Unlike Supreme Court decisions—which function as *ex-post* judicial dispute resolutions settling adversarial controversies under the doctrine of *stare decisis* (Article 8, Civil Code)—Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force and effect of law. Furthermore, during both the Martial Law period (1972–1986) and the Revolutionary Government under the Freedom Constitution (Proclamation No. 3, 1986–1987), the President exercised direct legislative powers through Executive Orders and Presidential Decrees, creating substantive national enactments that remain operative law today. Modern Executive Orders frequently establish binding nationwide regulatory standards—such as Executive Order No. 26 (s. 2017) establishing nationwide public smoking prohibitions, or Executive Order No. 168 (s. 2014) creating the Inter-Agency Task Force for the Management of Emerging Infectious Diseases (IATF)—that legally preempt conflicting municipal legislation. Excluding Executive Orders would therefore omit critical nationwide regulatory mandates that local ordinances are legally bound to respect.

2.2.2 Historical Evolution and Dual Stylistic Heritage

The structural complexity of Philippine legislative text stems directly from the nation’s historical experience as a mixed legal jurisdiction [4, 40]. Philippine jurisprudence combines continental civil law traditions with Anglo-American statutory common law conventions. Understanding why Philippine statutes and local ordinances are drafted in their characteristic syntax requires examining how these colonial and post-colonial traditions fused over more than a century of codification.

The Continental Civil Law Heritage. During more than three centuries of Spanish rule, the Philippine legal system was rooted firmly in the Roman-Germanic civil law tradition [4]. This continental tradition emphasized systematic codification, deductive legal reasoning, and broad normative principles organized into structured books, titles, and chapters. Foundational enactments, such as the Spanish Civil Code of 1889 and the Penal Code of 1870, relied on generalized legal mandates rather than

granular factual definitions. Furthermore, this civilian heritage embedded a rich vocabulary of Latin doctrinal maxims into Philippine jurisprudence, including *in pari materia* (construing related laws together), *generalia specialibus non derogant* (a general law does not override a specific one), and *ultra vires* (acts beyond legal authority) [4]. Modern Philippine codes retain this civilian structure, expecting adjudicators to derive specific legal duties from comprehensive normative commands.

The Anglo-American Statutory Drafting Overlay. Following the Treaty of Paris and the establishment of the American Insular Government (1900–1935), American legal administrators introduced common law statutory drafting habits without fully displacing the civilian codification base [40]. Enacted by the Philippine Commission and later the Philippine Legislature, early statutes introduced formal drafting conventions borrowed directly from United States federal and state legislation. These conventions included explicit statutory title preambles, exhaustive definitions sections, severability clauses, and general repealing provisions [4]. Significantly, American administrators introduced the practice of drafting extensive conditional provisos (introduced by the traditional formula “*Provided, that...*” and “*Provided, further, that...*”), designed to restrict, qualify, or create exceptions to preceding operative clauses within the same section. The Revised Administrative Code of 1917 (Act No. 2711) and the contemporary Administrative Code of 1987 (Executive Order No. 292) solidified this mixed tradition, confirming English as the controlling statutory language while maintaining civil law institutional categories [153].

Structural and Linguistic Implications. This dual heritage produces a legislative drafting style marked by high syntactic density and multi-clause complexity. Unlike standard conversational English, a single statutory provision frequently spans one hundred to two hundred words contained within a solitary grammatical sentence [15]. These sentences embed multiple dependent clauses, qualifying prepositional phrases, and recursive proviso exceptions. At the municipal level, local legislative staff adopt this formal national drafting template while adding procedural boilerplate, resulting in verbose ordinances where the operative regulatory command is separated from its governing subject by dozens of intervening words [45, 166]. From an analytical and document processing perspective, this hybrid syntax presents substantial structural hurdles: archaic legal terms resist straightforward vocabulary normalization, long-distance dependencies obscure the core legal directive, and qualifying provisos invert the practical meaning of preceding clauses unless the text is examined in its full structural context.

2.2.3 Procedural and Linguistic Friction

Besides the hierarchy of laws, the actual way local laws are written and passed creates a lot of inconsistencies in language and format. National laws are usually written by specialized committees using standard legal terms. However, local ordinances go through a localized process that often breaks up formal language. To detect conflicts accurately, the computer system must be adjusted to handle the “noise” and confusion caused by using two languages in the Davao City Council (Sangguniang Panlungsod).

Ordinance Creation Process. In Davao City, the life of a proposed law is guided by the Internal Rules of Procedure, found in Resolution No. 001-22 [166]. The drafting process is decentralized, meaning different council members or their staff write their own laws rather than having one main office do all the writing [71, 166]. Because of this, the style and words used across different ordinances can be very inconsistent. For example, a transportation law might use completely different terms than an environmental law, making it hard for computer models to see how they relate.

Textual Fragmentation and the Three-Reading Rule. According to the law, no ordinance can be passed unless it goes through three separate readings [71, 166]. This process is democratic, but it often breaks the text apart. During the First Reading, the draft is sent to a committee. The Second Reading is where the most intense checking happens: the text is put into a period of interpolation, debates, and amendments [166]. This constant changing of the text can lead to disjointed rules. For instance, a new exception added in one section might accidentally contradict a general rule in another section, which may contain a hidden conflict that stays buried in the text after final approval on the Third Reading.

Bilingual Mandates and the Risks of Translation. A unique challenge within the Philippines is the legal requirement to make laws accessible in two languages. Under Section 469 of the Local Government Code, the Secretary must translate all approved ordinances into the local dialect (e.g. Cebuano for Davao City) [71, 166, 54]. However, this creates a legal risk. The Administrative Code of 1987 (Executive Order No. 292) says that if the English and local versions disagree, the English version is the one that counts [152]. Consequently, if a translation error happens, a citizen following the local language could be punished for breaking the English version of the law. This relates to the Supreme Court ruling in *Tañada v. Tuvera*, which says laws are void without proper public notice [186]. A mistranslated law is essentially "false notice." Despite new efforts like the *Batas sa Sariling Wika Act* to mandate more regional translations [150], there is still no technical system to check if these translations are actually correct. As researchers have noted, the small details in local languages are often ignored, leading to "semantic drifts" where the original legal meaning in English is lost in regional translation [123].

2.2.4 Ex-Ante in Legislation

The term "ex-ante" comes from Latin and means "before the event." In the context of lawmaking, it refers to the process of evaluating a proposed law or regulation *before* it is officially enacted [60]. This subsection examines how statutory frameworks utilize ex-ante reviews to prevent conflicting or invalid enactments prior to promulgation.

Ex-Ante in the Philippine Context. In the Philippines, ex-ante review has become more important since the passage of Republic Act No. 11032, or the "Ease of Doing Business and Efficient Government Service Delivery Act of 2018" [72]. Under this law, the Anti-Red Tape Authority (ARTA) and the Department of the Interior and Local Government (DILG) mandates that government agencies and local government units (LGUs) conduct a Regulatory Impact Assessment (RIA) for any

proposed regulation that might significantly affect the public [72]. The goal is to ensure that new rules have a solid legal basis and to avoid creating overlapping or conflicting regulations that could make it harder for citizens and businesses to operate.

Ex-Ante vs Ex-Post. The main difference between the two is timing and purpose. Ex-ante evaluation is predictive; it looks forward to stop problems before they happen. It acts as a “first line of defense” against poor lawmaking [60]. In contrast, ex-post evaluation happens *after* the law is already active, focusing on how the law has actually performed or whether it has caused damage [204]. While ex-post reviews are useful for fixing existing mistakes, they are often difficult because it is much harder to revoke a law once it is already being enforced [204]. Ex-ante detection is considered more efficient because it saves the government from the cost and trouble of litigation later on.

Conflict Detection in Local Ex-Ante Legislation. Using automated tools for the ex-ante Conflict Detection can help the local legislative branch identify “normative collisions” during the drafting phase. Because legislative bodies now produce a massive amount of text, manual review often fails to catch hidden clashes between a new ordinance and older statutes [95]. By treating Conflict Detection as an automated diagnostic triage tool, computational systems can flag potential contradictions while the ordinance remains in draft form. This facilitates alignment with the national legal hierarchy, mitigating the risk of promulgating municipal enactments that are legally void *ab initio* [188].

2.2.5 Consistency Checking in Legislative Processes.

Because local governments are lower in rank and their drafting process is fragmented, checking for consistency is a critical duty. However, checking ordinances for conflicts is not as simple as matching titles, as it requires understanding complex legal meaning and identifying subtle oversteps in authority. This part examines current methods of consistency checking, highlighting how confusing language and the reliance on simple tracking systems slow down the legislative review process.

Legalese Linguistic Ambiguity. Discourse analysis of Philippine laws reveals many ambiguities in how words and sentences are used [187]. Specifically, the word “shall” is often confusing because it can be interpreted as either a mandatory command or a simple suggestion depending on the sentence. Studies show that these language issues make it hard to interpret laws correctly, even when they follow standard formats [15].

Legislative Gaps and Implied Preemption. The Philippine legal framework recognizes that conflicts between laws are rarely obvious [40]. A local law almost never says, “This law contradicts a national law.” Instead, conflicts often appear as “implied preemption.” This happens when a national law already covers a subject, but a local government tries to add extra hurdles—like requiring a local fee to do something already allowed by the state [187]. In these cases, the Supreme Court rules that the local ordinance is unlawfully trying to “amend” a national law under the

guise of local regulation [187]. For a human reviewer or a basic keyword search, this is very hard to find because the laws might use different words. The conflict is in the actual effect of the rules, which requires deeper semantic checking to identify.

Local Context. The Davao City Sangguniang Panlungsod passed nearly 400 ordinances and over 2,000 resolutions in a single three-year term [140]. To manage this volume, the city uses digital platforms: the Resolution Ordinance Tracking System (ROTS) is used for encoding and routing documents, while the Legislative Information Support System Program (LISSP) lets council members track items and read ordinances on their devices [31]. However, while these systems help with logistics, the actual checking of content for "dueling provisions" is still done manually. Furthermore, scanned PDFs often have OCR noise, which makes keyword searches fail [115].

2.3 Philippine Digital Legislative Archives

Building a functional Semantic Conflict Detection system requires access to comprehensive and machine-readable legal corpora. In the Philippines, the digital landscape of legislative data is highly fragmented, split between national statutory repositories and localized municipal archives. This structural division presents significant technical hurdles for statutory data ingestion. Before deep logical inference can occur, automated auditing pipelines must first secure reliable data streams from these disparate legal environments. Consequently, understanding the specific characteristics of these sources informs the design of statutory retrieval and preprocessing methodologies.

2.3.1 National Sources

National legal repositories represent the authoritative apex of the Philippine legislative hierarchy, housing laws that universally preempt local ordinances. These databases are maintained by centralized government agencies and contain highly formalized texts such as Republic Acts, Commonwealth Acts, and Batas Pambansa. They serve as the primary knowledge base for external legal context, providing the authoritative baseline rules against which municipal drafts are evaluated. Because national legislation undergoes a rigorous, standardized drafting process, the vocabulary within these repositories is generally consistent and adheres to strict legal drafting standards. Accessing these national databases provides computational systems with the necessary macroscopic view of state-level mandates and prohibitions.

Official Gazette. The Official Gazette serves as the mandated repository for Philippine statutes, rendering it the official journal of the country for legislative text [73]. It acts as the primary publisher for all newly enacted national laws, executive orders, and administrative issuances, as mandated by decree of Republic Act No. 453 [143] and Commonwealth Act No. 638 [131]. Because it represents the final, legally binding version of a statute, any computational system evaluating municipal preemption must anchor its baseline rules to this repository. This archive effectively acts as the definitive historical ledger for state-level mandates, preserving the exact phrasing approved by the legislative and executive branches. Consequently, it remains the

most critical target for establishing an authoritative statutory knowledge base.

Open Congress API. To programmatically map the complex hierarchy of national laws, developers utilize the Open Congress API provided by the House of Representatives’ Legislative Information System (LEGIS) to access structured, machine-readable metadata regarding legislative routing and amendments [85]. This interface efficiently tracks a statute’s formal lifecycle, charting its progression from initial committee filing to final executive approval. By providing standardized output formats, the API allows computational pipelines to establish relational dependencies between different legislative acts without manually parsing unstructured text.

The LawPhil Project. The Lawphil Project functions as a massive, privately maintained database of statutory records and jurisprudential decisions [8]. Unlike official government channels, this repository provides a comprehensive, pre-digitized plaintext corpus of historical and contemporary Philippine law. This accessibility makes it mathematically ideal for the unsupervised pre-training of foundational language models tailored to the local legal domain. By ingesting this vast corpus of clean text, NLP architectures can learn the highly specific syntactic structures and deontic operators inherent in Philippine jurisprudence. Utilizing this database effectively establishes the core vocabulary required for advanced Transformer models to process specialized statutory drafting [228].

2.3.2 Local Sources

In contrast to the centralized national databases, local legal repositories are maintained independently by individual Local Government Units (LGUs). These archives house municipal ordinances, resolutions, and executive orders that govern specific geographic jurisdictions. The legislative output at this tier is entirely decentralized, resulting in highly variable archiving practices depending on the technological capacity of the specific city council. These local databases form the immediate target corpus for municipal conflict detection, containing the actual regulatory constraints imposed on city residents. Consequently, they represent a high-volume, dynamic data environment that continuously expands as the local council holds regular drafting sessions.

Davao City Sangguniang Panlungsod. The locality’s primary corpus consists of the Davao City Sangguniang Panlungsod (City Council) website, which serves as the official digital repository for the city council’s legislative outputs [32]. This database acts as the primary historical record for all municipal regulations, resolutions, and zoning policies enacted within the city’s jurisdiction. It represents a dynamic and continuously expanding text environment that directly reflects the decentralized nature of local lawmaking. The texts housed here encapsulate the specific regulatory constraints and policy directions of the city government, providing the exact ordinance text required to evaluate new drafts for logical consistency against the existing legislative framework.

Legislative Information Support System. To manage the sheer volume of local legislation, the city introduced the Legislative Information Support System Program (LISSP) as its primary administrative backbone [31]. This platform serves as a

digital tracking system designed to maintain metadata regarding ordinance routing, committee assignments, and plenary approval statuses. According to official reports, the system is openly available to the public, allowing citizens and stakeholders to monitor ordinances and resolutions in real-time [141]. By digitizing the legislative workflow, LISSP effectively functions as a centralized database of ordinances, providing the structured tracking and contextual data necessary for evaluating the lifecycle of local laws. This public transparency and administrative organization address the interoperability challenges identified in recent regional e-governance assessments [36].

2.3.3 Data Extraction Challenges

While the aforementioned repositories contain the necessary legal knowledge, extracting the needed data for computational analysis presents severe technical challenges. Scraping and formatting these diverse sources exposes the pipeline to significant data degradation and structural inconsistencies.

Optical Character Recognition. A preliminary survey of the Official Gazette and the Davao City SP archive reveals a significant barrier to automated auditing: the lack of native, machine-readable text. Both repositories predominantly host scanned, image-based PDFs rather than "searchable" or "selectable" digital documents. During manual inspection, these files were observed to contain varying degrees of visual noise, including slanted scans, faded ink, and compression artifacts. As Lopresti [115] documents, such "noisy" visual data often results in severe Optical Character Recognition (OCR) degradation, where standard extraction tools produce broken alphanumeric strings and nonsensical tokens. These observed irregularities suggest that a direct pipeline would suffer from semantic fragmentation, requiring an additional, high-precision OCR layer and extensive text-cleaning heuristics to ensure the data is viable for program analysis.

Archival Fragmentation. Another critical challenge is archival fragmentation, a state where repositories suffer from systemic gaps in their digital records. This is a significant problem for high-fidelity auditing because the absence of even a single related ordinance can lead to "false negatives" in Conflict Detection, where a system declares a draft valid simply because it cannot see the contradicting record. One primary example is the Davao City Sangguniang Panlungsod public archive, which features intermittently missing ordinance files and unorganized chronological listings that actively hinder systematic data collection [168]. Furthermore, even relatively complete national databases like the Lawphil Project can lack specific Republic Acts or historical amendments. As Westermann et al. [79] observe, legal databases are frequently "unstructured" and "fragmented," forcing developers to move beyond simple scraping toward complex data engineering to ensure a "complete" dataset. This requires developing specialized, fault-tolerant scripts to account for missing files and broken links, significantly complicating the data-gathering phase required to build a reliable local knowledge base.

Database Accessibility. A significant challenge in statutory data acquisition is the dis-

crepancy between the theoretical and actual accessibility of municipal databases. For example, while administrative legal information systems like Davao City’s LISSP are promoted in institutional reports as tools for public transparency [31, 141], empirical verification reveals strict authentication barriers requiring authorized credentials. Technical inspection reveals that municipal legislative portals often remain computationally isolated behind authentication interfaces, restricting automated programmatic extraction by public scrapers. Institutional correspondence with municipal system administrators confirms that such digital repositories are frequently not open in a programmatic sense, requiring administrative clearance and manual provisioning to access full-text enactments. As Conchas and Salapa [36] observe in similar administrative systems, isolated administrative data often exists as “digitally orphaned” artifacts, where metadata about an enactment is disconnected from its full-text searchable content. This structural disconnection necessitates specialized ingestion pipelines capable of reconciling sparse public metadata with offline full-text documents gathered from alternative archives.

Textual Scarcity. Perhaps the most significant challenge in developing an automated legal auditing system is the pervasive systematic textual scarcity within the Philippine legal informatics landscape. Despite the existence of authoritative sources like the Official Gazette and the Davao City SP website, their web architectures remain inherently hostile to modern programmatic extraction, characterized by unindexed entries and “deep-linked” PDFs that cause automated scrapers to fail. Even community-driven initiatives or secondary repositories do not offer a single, straightforward solution. For instance, while the Open Congress API provides structured metadata for legislative routing, it lacks the full-text bodies of the laws themselves. Conversely, while the Lawphil Project offers a more consistent link format and a vast plaintext corpus, it is not an exhaustive archive and contains occasional missing Republic Acts or amendments. As Shao et al. [175] observe, such fragmentation prevents the construction of a unified, high-fidelity knowledge base from a single origin. Consequently, researchers are forced into a series of technical “roundabouts” and compromises—engineering a patchwork pipeline that pulls metadata from one source and text from another—to circumvent the absence of a centralized, machine-readable repository for both national and local legislation.

2.4 Exact and Symbolic Methods

Before the adoption of neural networks, computational legal analysis relied heavily on deterministic systems to process text. Deterministic systems operate on absolute rules, meaning that a specific input will invariably produce the exact same output without any probabilistic guessing. Traditional methods for searching and evaluating legal information assumed that legislative ideas could be cleanly mapped either to specific vocabulary strings or to rigid, hand-coded logic formulas. While highly transparent and easy to debug, these foundational approaches frequently fail when forced to detect hidden conflicts in complex, unstructured legal texts. The core issue is that human language is inherently messy, and rigid systems break down when faced with the ambiguous phrasing typical of municipal

ordinances.

2.4.1 Boolean Information Retrieval

To understand why traditional legal databases struggle with semantic conflict detection, we first need to look at how they fundamentally process text at the machine level. Boolean Information Retrieval (BIR) represents the foundational search architecture for almost all commercial law databases currently in use. Instead of reading a document like a human, BIR treats every file simply as a “bag of words,” stripping away all grammar and sentence structure. It relies on an inverted index, which is essentially a massive lookup table that tracks exactly which specific words appear in which files across the entire database. Because it only registers the presence or absence of characters, the system is completely blind to the actual context or legal intent behind those words.

At its core, the retrieval mechanism in BIR is strictly based on set theory and formal Boolean logic. A user constructs a query using logical operators like **AND** ($\wedge$), **OR** ($\vee$), and **NOT** ($\neg$) to create intersecting sets of documents. For example, a search for a specific ordinance regarding explosives might be formulated mathematically as:

$$Q = (\text{fireworks} \vee \text{pyrotechnics}) \wedge \neg \text{exempt}$$

The system then retrieves documents by calculating the exact mathematical intersection of these sets, pulling only the files that satisfy every condition simultaneously. If a document contains the specific string of characters requested, it is retrieved and scored as a 1; if it misses even a single required term, it is completely ignored and scored as a 0. It evaluates the raw presence of strings, entirely ignoring the conceptual meaning or legal intent behind them [17].

2.4.1.1 Challenges with BIR

The primary limitation of BIR in legal auditing is its absolute, uncompromising reliance on exact string matching. This rigidity creates a severe “vocabulary mismatch” problem when processing natural language. Human language naturally features synonymy, where entirely different words possess the exact same meaning, and polysemy, where the identical word takes on drastically different meanings depending on the context. In statutory drafting, one author might use “prohibited” while another uses “unlawful,” referring to the exact same legal concept. Because BIR operates on character-level matching rather than conceptual mapping, it fundamentally cannot connect these two terms, leading to massive blind spots in the search results.

A landmark empirical evaluation of these systems demonstrated this vocabulary gap in a highly practical setting. Legal researchers conducting queries using standard BIR systems confidently estimated they were finding over 75 percent of the relevant documents for a given case. They experienced an illusion of effectiveness because the documents they did retrieve looked highly accurate, masking the massive volume of relevant files they missed. In reality, the system retrieved less than 20 percent of the actual relevant files contained within the database [17]. The researchers missed the remaining 80 percent

simply because the authors of those hidden documents used slightly different, completely valid phrasing that didn’t perfectly align with the strict search query [206].

In a local government context like Davao City, this specific margin of error is administratively unacceptable. Missing even one existing ordinance during an ex-ante review due to a simple vocabulary mismatch could directly result in the passage of an invalid, conflicting law. Furthermore, legislative vocabulary inherently evolves over time as drafting conventions change. An ordinance passed in the 1990s and a modern draft written in 2026 might strictly regulate the exact same municipal subject but share absolutely zero common keywords. Recent studies emphasize that resolving this requires the implementation of “Transformer” models, which abandon exact matching entirely in favor of continuous vector spaces that recognize when different vocabulary points to the same underlying concept [99].

2.4.2 Symbolic AI

Because BIR fundamentally fails to capture underlying meaning, early artificial intelligence researchers attempted to automate legal reasoning by directly modeling the rules of law themselves. This approach, widely known as Symbolic AI or “Good Old-Fashioned AI” (GOFAI), operates on the premise that human cognition can be replicated by manipulating formal symbols [208]. It moves beyond simple word matching by teaching computers actual legal logic through explicit, human-coded “IF-THEN” formulas. Instead of searching a database for raw text strings, Symbolic AI requires human engineers to manually translate the constraints of a statute into formal logic structures. This theoretically allows the computer to process the actual regulatory mechanics of a law rather than just reading its surface vocabulary.

To execute this logic, these systems specifically utilize propositional or first-order logic combined with deontic operators. Deontic logic is a highly specialized branch of mathematical logic that deals specifically with obligations, permissions, and prohibitions. For example, a system might represent a local zoning prohibition mathematically as:

$$\forall x(\text{Business}(x) \wedge \text{ResidentialZone}(x) \rightarrow \text{Prohibited}(x))$$

In this formula, the universal quantifier ($\forall$) asserts that for any entity x , if it is classified as a business ($\wedge$) and located in a residential zone, it is strictly prohibited ($\rightarrow$). By converting ordinances into these rigid formulas, a system can theoretically use deductive reasoning to spot a conflict. If Rule A dictates $\mathcal{O}(X)$ (an absolute obligation to do X) and Rule B dictates $\mathcal{F}(X)$ (an absolute prohibition against doing X), the system successfully flags a formal logical contradiction ($\perp$) [12].

2.4.2.1 Challenges with Symbolic AI

While Symbolic AI provides perfectly transparent and mathematically sound reasoning, its absolute reliance on manual programming makes it practically impossible to scale for real-world governance. The primary architectural barrier is the classical “knowledge acquisition barrier” in expert systems: human domain experts and software engineers

must manually read, interpret, and translate thousands of unstructured city ordinances into rigid computer rules. This manual translation process is unacceptably slow, extremely expensive, and highly prone to human error during the coding phase [119]. Furthermore, every time the city council amends an existing law, the underlying symbolic formula must be manually rewritten, making the system incredibly difficult to maintain over time.

Beyond the logistical nightmare of manual coding, the rigid 1-and-0 architecture of Symbolic AI fundamentally clashes with the “open texture” of natural legal language. Laws frequently rely on subjective, open-ended standards—such as requiring a citizen to exercise “reasonable care” or negotiate in “good faith.” These inherently subjective human concepts cannot be easily defined within a strict, binary rule-based formula because their definitions change depending on the specific context [11]. While these symbolic systems are highly precise in tightly controlled, hypothetical environments, they immediately shatter when exposed to the real world. Ultimately, they lack the computational flexibility required to handle the thousands of messy, unorganized, and ambiguously worded documents actually found in local government archives [229].

2.5 Graph-Based Architectures

Because exact word matching can be unreliable, some researchers use “graph-based” systems. Instead of looking at laws as just lists of words, these systems treat them like points in a large web. By looking at how laws quote each other and relate to the same topics, the computer can better understand their meaning.

2.5.1 Legal Knowledge Graphs and Graph Neural Networks

Legal Knowledge Graphs (LKGs) are a way to model how different laws depend on each other. Unlike a simple list, these graphs can use Graph Neural Networks (GNNs) to find hidden links and clashes. Researchers have created frameworks like GNN-ERE to automatically build these legal webs from messy text, which works much better than just searching for keywords [121]. Studies also show that these networks can find relevant court cases even if they do not use the exact same words [80].

The advantage of GNNs is that they can share information across the entire web structure. By treating laws as points in a network, the computer can learn the “purpose” of a law—for example, telling the difference between a rule and a procedure [16]. These systems can also perform “multi-hop reasoning.” This means they can find “indirect conflicts” where two laws do not quote each other directly but still clash because of their relationship with a third law [223]. This is a big step up from old search engines that can only look at one thing at a time.

2.5.1.1 Challenges in GNNs

While Graph Neural Networks are powerful, they only work if the web of connections is high-quality. A GNN needs a clearly built index or list of links to follow.

Dependency on Clear Connections. GNNs are limited because they need clear links between laws to "see" the meaning. As researchers have pointed out, creating these links by hand is very slow and expensive [88]. In a city with thousands of ordinances, it is almost impossible for people to keep this list of links up to date. If the system fails to find a link between two conflicting laws at the start, the GNN will be "blind" to that problem.

Disconnection in Digital Archives. Even though systems like the Legislative Information Support System Program (LISSP) have helped move Davao City's archives online, a gap remains for graph-based AI. Most digital repositories in local government are designed for storing and viewing files rather than linking ideas together. Recent studies on Philippine digital governance show that local ordinances often act as "digitally orphaned" texts; they are available online but lack the machine-readable links needed for an AI to jump from one law to another [36]. This makes the digital archive a collection of separate records rather than a connected network, making it hard for GNNs to see how they overlap.

Weakness Against Messy Text and OCR Noise. Building these legal webs automatically is also very difficult because of how local government data looks. AI frameworks usually assume the text is "clean" and easy to read [121]. However, scanned PDFs in Philippine archives often have "OCR noise" (typos from scanning), which confuses the computer [115]. When there are typos or local "Taglish" words, the computer creates broken or "hallucinated" links. This makes the whole web unreliable for finding actual legal clashes.

2.5.2 Abstract Meaning Representation

While GNNs look at links between documents, Abstract Meaning Representation (AMR) looks at the logic *inside* a single sentence. AMR is a graph system that focuses on the "who is doing what to whom" logic. Researchers argue that this provides a more stable way to understand laws than just looking at grammar [211]. By mapping out the logic, AMR can find conflicts even if laws use different wording (like "The city bans X" vs "X is banned by the city").

AMR is also useful for identifying which rules are "obligations" (must do) and which are "prohibitions" (cannot do) [199]. Unlike old keyword searches, AMR understands the role of each word. This helps find subtle clashes, such as a local law allowing "business use" when a national law forbids "non-residential activity." New legal datasets have proven that AMR is a good way to combine human-like logic with computer power [211].

2.5.2.1 Challenges in AMR

Even though AMR is very precise, using it for Philippine city ordinances is difficult. It relies on very strict rules that don't always match the "messy" way local laws are written.

Limited to Single Sentences. The biggest flaw of AMR is that it only looks at one sentence at a time [199]. However, legal conflicts in city ordinances rarely happen

in the same sentence; they are usually found in different sections or paragraphs. For example, a ban might be in Section 2, while an exception is buried in Section 5. Because AMR cannot “connect the dots” between different sentences easily, it is often “blind” to these types of conflicts.

Difficulty with Complex Legal Language. AMR relies on a static or fixed dictionary of meanings to work. The long, “run-on” sentences often found in Philippine laws are too complex for most AMR tools. For example, in RA 7160 within Section 378, such sentence can be found as follows:

”When a loss of government property occurs while the same is in transit or is caused by fire, theft, force majeure, or other casualty, the officer accountable therefor or having custody thereof shall immediately notify the provincial or city auditor concerned within thirty (30) days from the date the loss occurred or for such longer period as the provincial, city or municipal auditor, as the case may be, may in the particular case allow, and he shall present his application for relief, with the available evidence in support thereof.”

Such a sentence causes the AI often fails or produces the wrong logic map, as studies show that as sentences get longer and more confusing, AMR becomes much less accurate [201].

Problems with Local Words and Typographical Errors. AMR is also very sensitive to typographical errors and unknown words. When the computer runs into a scanning error from an old PDF, or a local word that isn’t in its dictionary, it fails to map the logic correctly. Research shows that this “lexical noise” often results in broken logic maps that lose the actual meaning of the law [205]. Without a tool specifically made for Philippine Legal English, relying on AMR for Davao City’s archives would be unreliable.

2.6 Deep Learning and Transformer Models

The application of Natural Language Processing (NLP) in the legal domain has shifted from rigid, rule-based systems and basic word-matching toward Deep Learning [96]. Legacy algorithms treat legal texts as simple character strings, which frequently fail to capture the dense and interconnected nature of statutory drafting. Deep Learning, conversely, utilizes neural networks to map out the actual semantic meaning of legal language. This allows systems to process complex syntax and perform the deductive reasoning required to spot conflicts. This section breaks down the Transformer architecture, explaining its underlying mechanics, computational challenges, and why it remains the most practical framework for auditing unstructured municipal archives.

2.6.1 Transformer Models and Self-Attention Mechanism

A major breakthrough was the Transformer architecture, introduced by Vaswani et al. [207] in 2017. Prior models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed text sequentially. This step-by-step process-

ing was computationally slow and made it difficult for the model to retain information from earlier in a long document. Transformers bypass this sequential processing limit by relying on a Self-Attention mechanism.

Self-Attention allows a model to analyze all words in a sequence simultaneously to determine their contextual relationships. This enables a specific term to connect with any other word in the document, regardless of the physical distance between them. Because it processes data in parallel, Transformers are significantly faster to train, directly enabling the creation of Large Language Models (LLMs). Modern iterations have introduced refinements like Disentangled Attention to better parse complex syntactic structures, alongside Windowed Attention to process long-context legal documents efficiently. However, while Generative LLMs like ChatGPT excel at text summarization, they are prone to ‘hallucinations’—often inventing plausible but entirely fake legal citations [46]. To ensure legal validity, a system must be discriminative: specifically classifying whether statements exhibit a Conflict or Entailment, rather than simply generating new text.

2.6.2 Challenges in Transformer Models

Despite these architectural advantages, deploying Transformer models within the hardware limitations of Philippine Local Government Units (LGUs) introduces specific computational hurdles.

Quadratic Memory Scaling and Context Limits. The Transformers’ core processing engine scales quadratically ($O(n^2)$) with sequence length, requiring significantly more memory and compute as input text grows [207]. Because of this quadratic complexity, standard models like BERT are generally capped at reading 512 tokens at a time [49]. Philippine ordinances are frequently verbose, often spreading rules and exceptions across multiple pages. To manage this, developers use a **Sliding Window** technique. This involves taking a fixed-size “window” (e.g., 512 words) and moving it across the document with a specific **Stride** [7]. For example, if the window moves 256 words at a time, the windows will overlap, ensuring that the end of one chunk is also the start of the next to try and keep context together [83]. However, this is like reading a long book through a small slot; if a general rule is on Page 1 and its specific exception is on Page 5, the AI can never see them at the same time [172]. This physically cuts the “semantic link” between related clauses, creating logical blind spots where the system might flag a conflict simply because it cannot see the full picture [172].

Morphological Noise and Domain Adaptation. Transformers require massive corpora of clean text for proper pre-training. While specialized models like Legal-BERT [25] perform well for US or European law, there is no equivalent foundation model trained specifically on Philippine local legislation. Applying Western-trained tokenizers to local government archives—which are saturated with uncorrected OCR errors, archaic colonial phrasing, and administrative ‘Taglish’—causes severe tokenization fragmentation. The model fails to recognize these localized terms and shatters them into meaningless sub-tokens, effectively ruining its ability to comprehend the law.

Edge Deployment and Hardware Constraints. Executing deep logical checks across thousands of document pairs demands significant computing power, typically requiring expensive GPU resources [151]. Most municipal governments simply lack this caliber of server infrastructure. Consequently, deploying these heavy models on standard LGU workstations requires aggressive optimization and decoupled pipeline architectures to ensure the system does not freeze during inference.

2.6.3 Transformer Models as the Ideal Method

Notwithstanding these hardware limitations, Transformer models remain the most viable approach for Semantic Conflict Detection in local ordinances. General-domain language models typically struggle with legal text because common words frequently take on entirely different legal definitions. For instance, "party" refers to a litigant rather than a celebration, and "suit" signifies a legal action rather than clothing. Greco et al. [74] emphasize that this high degree of polysemy is exactly why domain-specific models are required. While older static systems assign a single, fixed meaning to a word [74], a Transformer dynamically adjusts its reading of a word based on the surrounding sentence context [25]. Chalkidis et al. [26] empirically showed that this contextual awareness is what allows a model to distinguish between amending a law and violating it—a distinction that simple keyword searches invariably miss.

The transition to Deep Learning necessitates explicitly moving away from legacy methodologies. Boolean keyword searches fundamentally fail because legislative drafting vocabulary constantly evolves over the decades. Conversely, manually mapping thousands of unlinked Republic Acts and local ordinances into deterministic graph structures creates an intractable administrative burden. As Li et al. [110] observed, hand-crafting these legal knowledge graphs simply does not scale for volatile, unstructured municipal records. Ultimately, Transformers represent the ideal method: they possess the capacity to map the deep, latent semantic relationships in unstructured legal text without requiring an intractable amount of manual data entry.

2.7 Transformer Pipeline

The appeal of an end-to-end, single-stage neural architecture is its operational simplicity: funneling raw text directly into final outputs without intermediate steps. For general Natural Language Processing (NLP) tasks like sentiment analysis, a single Transformer model is usually enough. However, legal informatics, specifically Semantic Conflict Detection, requires a level of deductive precision that a single monolithic model structurally cannot provide. Forcing one architecture to act simultaneously as a high-volume search engine and a strict logical judge creates impossible mathematical trade-offs.

2.7.1 Challenges in Single-Stage Pipeline

Deploying a single Transformer model to scan decades of uncodified municipal ordinances introduces severe architectural friction. The system inevitably compromises either its

recall (missing conflicting laws) or its precision (flagging false positives). This limitation changes depending on the specific type of single-stage model used.

Generative LLM Constraints. A common but flawed approach feeds a document pair directly into a Generative Large Language Model (Decoder-only architectures like GPT or LLaMA) and prompts it to declare a conflict. These models are autoregressive text predictors, not discriminative logical classifiers. As Linna [113] notes, applying them to strict statutory auditing invites ‘hallucinations’—the fabrication of plausible but legally fake citations. Furthermore, their massive parameter counts make offline Edge Deployment on standard Local Government Unit (LGU) hardware impossible without aggressive quantization. Dettmers et al. [47] demonstrated that this quantization process severely damages zero-shot reasoning fidelity in specialized domains.

Bi-Encoders and their Limitations. A Bi-Encoder, also known as a “Dual-Tower” or “Two-Tower” model, is an AI architecture designed for high-speed searching [151]. It works by processing the user’s search query and the documents in the archive completely separately from each other. The model turns every ordinance into a mathematical coordinate (a vector) and stores them in a database [94]. Because it does not have to read the documents at the same time a person is searching, it can compare a query against more than 25,000 national statutes in just a few milliseconds [29]. This makes it the perfect tool for the “Coarse” stage of the pipeline, where the goal is to narrow down the entire legal archive to just a few relevant provisions [66].

Current municipal search systems predominantly rely on Dense Retrievers (Bi-Encoders). While highly efficient for filtering millions of documents, Thakur et al. [197] explain that Bi-Encoders suffer from information loss during vector compression because they collapse the semantic meaning of an entire document into a single, fixed-size vector. This architecture optimizes for topical similarity rather than logical consistency. Consequently, a Bi-Encoder has a critical logical blindspot: it will retrieve an ordinance “allowing fireworks” and one “banning fireworks” and score them as highly relevant because their vocabulary distributions are nearly identical. Kim et al. [99] observed that Bi-Encoders fundamentally lack the computational architecture to detect the logical contradiction operating beneath the shared syntax.

Overcoming this blindspot requires a secondary phase using a Cross-Encoder architecture. Instead of independently compressing documents into isolated vectors, a Cross-Encoder computes token-level attention across both documents simultaneously ([CLS] Ordinance A [SEP] Ordinance B). This allows the model’s attention heads to directly compare negators (e.g., “banning” vs. “allowing”) in their immediate contexts. Nogueira and Cho [135] demonstrated that this cross-attention mechanism shifts the task from basic topical retrieval to rigorous Natural Language Inference (NLI), enabling the precise Semantic Conflict Detection required for legal texts.

Quadratic Memory Constraints If a single-stage Bi-Encoder lacks logical depth, the

theoretical alternative is running a single-stage Cross-Encoder across the entire database. However, this approach is computationally impossible on standard office workstations. Because the Self-Attention matrix scales quadratically ($O(n^2)$) with sequence length, Beltagy et al. [13] highlight that cross-encoding a target ordinance against the 25,432 national statutes simultaneously instantly exceeds available graphics memory. The model would be forced to split the database into tiny, fragmented sliding windows, severing the critical semantic links between general rules and remote exceptions. This memory constraint demonstrates that isolating a small candidate shortlist through coarse retrieval is essential before running deep sequence classification.

2.7.2 Single-Stage vs. Dual-Stage Pipeline

Given these hardware and logical constraints, the literature discourages single-stage monoliths for legal analysis. Expanding to a dual-stage pipeline represents a structural necessity for processing high-volume, unstructured text. Several technical considerations justify this architectural split:

Retrieve-Then-Entail Architecture. Legal auditing requires a shift from topical similarity to logical inference. A Retrieve-Then-Entail pipeline serves as a Coarse-to-Fine guardrail: an efficient retriever isolates candidates, and a deep-reasoning NLI Cross-Encoder identifies specific policy violations. Williams et al. [219] indicate that this approach reduces manual review time by up to 70 percent while maintaining conceptual coherence.

Hardware Feasibility. A dual-stage approach segregates the computational heavy lifting. Stage one relies on Information Retrieval via pre-computed vector indexes that use minimal active memory. This offloads the intense $O(n^2)$ matrix multiplications of the Cross-Encoder exclusively to the top-K retrieved pairs. Khattab and Zaharia [98] demonstrated that this allows advanced NLP to run locally on standard servers without requiring enterprise GPU clusters.

Pragmatic Solution to Compression. Splitting the pipeline lets each model specialize. The first stage accepts the trade-off of vector compression to maximize search speed. As Humeau et al. [87] observed, the second stage avoids vector pooling entirely, reading raw, uncompressed tokens across both texts to achieve high-precision logic.

2.7.3 Competition on Legal Information Extraction and Entailment

The necessity of architectural decoupling is highlighted by over a decade of results from the Competition on Legal Information Extraction and Entailment (COLIEE), moving beyond mere theoretical preference. As Goebel et al. [66] detail, COLIEE serves as the premier international proving ground for legal NLP, challenging research teams to solve complex jurisprudential problems using standardized datasets. In particular, the competition structurally reinforces the separation of search and logic by bifurcating statutory

analysis into distinct, sequential tasks, which includes two relevant tasks:

Statutory Information Retrieval. Systems must scan a vast civil code corpus to retrieve all candidate provisions relevant to a specific legal query.

Statutory Textual Entailment. Systems must perform rigorous logical deduction to determine whether the retrieved provisions logically entail or contradict the given query [6, 66].

Year after year, architectures deployed by winning teams consistently reject single-stage monoliths in favor of pipelined approaches. The consensus among top-performing systems is that Information Retrieval requires high-recall dense retrievers (e.g., BM25 paired with Bi-Encoders) to handle corpus volume, while the entailment task strongly demands deep-reasoning Cross-Encoders. While early benchmarks in COLIEE 2021 established this separation [6], modern iterations through COLIEE 2025 [67] and the newly published COLIEE 2026 proceedings [68] confirm that this decoupled architecture remains the gold standard. For statutory retrieval (Task 3), top-performing teams such as NOWJ [133] demonstrated that hybrid dense-sparse retrieval achieves near-perfect candidate recall, while statutory textual entailment (Task 4) systems like Team DU [5] and Tran and Nguyen [200] reached over 95% accuracy by isolating granular reasoning from broad corpus retrieval. This sustained global validation indicates that separating Information Retrieval from Natural Language Inference is the established operational blueprint for resolving semantic conflict across statutory corpora.

2.7.4 Pipeline Synthesis

Addressing context limits, logical blindspots, and morphological noise simultaneously is computationally prohibitive for a single monolithic model operating on standard computing hardware. Consequently, contemporary computational law literature converges on a dual-stage Coarse-to-Fine pipeline. Rather than relying on speculative monolithic designs, Reimers and Gurevych [151] note that this decoupled Retrieve-Then-Entail approach represents the empirical consensus in modern legal informatics.

The mathematical realities of Transformer scaling dictate this functional separation. Foundational work by Karpukhin et al. [94] established that Dense Retrievers must be independently optimized for Maximum Inner Product Search (MIPS) to handle corpus-level volume effectively. However, because Bi-Encoders lack the cross-attention necessary for deductive reasoning, they must feed into a secondary mechanism. Louis and Spanakis [117] validated this architectural split specifically for statutory law, proving that utilizing a high-capacity vector retriever purely for filtering, followed by a localized Cross-Encoder for Semantic Conflict Detection, dramatically reduces false positives compared to single-stage retrieval systems.

Operating within this rigid competitive framework, Althammer et al. [6] demonstrated that in legal entailment tasks, delegating the reading phase to a disentangled Cross-Encoder prevents the semantic collapse inherent in long-document vectorization. Prior studies by Shao et al. [175] consistently corroborate this, showing that isolating the computational heavy-lifting of document filtering from the granular scrutiny of log-

ical evaluation successfully bypasses the hardware constraints that cause single-model architectures to fail. Synthesizing these specific architectural components provides a rigorously tested blueprint: Stage One (Information Retrieval) absorbs the corpus volume and context length, while Stage Two (Textual Entailment) resolves the logical and morphological noise. This deliberate division of labor ensures semantic consistency across the unstructured archives of Philippine local legislation.

2.8 Information Retrieval

Information Retrieval (IR) is the process of extracting relevant documents from massive, unstructured datasets based on a specific query. In legal informatics, IR acts as the foundational filtering mechanism. It serves as the “coarse” stage of a dual-pipeline system, isolating candidate statutes from thousands of irrelevant municipal records before any intensive logical analysis takes place.

2.8.1 Nature of ‘Relevance’ in Retrieval

To understand how computational systems extract documents, we must first examine the human cognitive process that defines relevance in the physical world. In cognitive science, relevance is not a static property but a dynamic relationship between information and a specific task. According to Relevance Theory, humans naturally filter information by seeking the greatest cognitive benefit for the least mental effort [182]. When a legal researcher checks for conflicts, they do not simply count matching words; they instinctively evaluate the “intent” and “applicability” of a rule based on their background knowledge of the legal system. This human-centric view suggests that a document is relevant only if it provides a new piece of information that changes the researcher’s understanding of the legal landscape [170].

Beyond the cognitive baseline, the theoretical definition of relevance in legal information retrieval is anchored in the philosophical requirements of a functional legal system. According to the jurisprudence established by Lon Fuller in *The Morality of Law*, a primary requirement for the internal morality of law is consistency; a system that issues contradictory rules fails to be a legal system at all [62]. Within this philosophical framework, relevance is not merely about finding documents that talk about the same subject matter, but about identifying provisions that share a “normative overlap.” This means that a retrieved document is relevant if it occupies the same jurisdictional or behavioral space as a draft municipal ordinance, potentially creating a situation where a citizen cannot obey one law without violating another. Grounding the retrieval stage in this quest for consistency treats relevance as the systematic identification of potential normative collisions and threats to the rule of law [62].

However, traditional computational methods struggle to replicate this human judgment because they rely on a rigid, mathematical definition of relevance. In standard deterministic systems, relevance is treated as a binary mathematical property based entirely on set intersection [17]. If we denote a candidate document as D and a user query as Q , this lexical relevance is formally evaluated as a binary function $R(D, Q) \in \{0, 1\}$. Under

this strict lexical definition, a document is deemed fully relevant (i.e. 1) if and only if it contains the exact sequence of characters requested by the user [206]. Consequently, this mechanism establishes a purely topical baseline, effectively grouping documents together simply because they share a surface-level vocabulary [17]. While this mathematical simplicity allows for rapid indexing across massive databases, it completely strips away the legal context required for complex legislative analysis.

This stark discrepancy between superficial lexical matching and actual legal consequence represents a major stumbling block for legacy search architectures. In statutory law, two documents discussing the exact same topic might be lexically relevant to one another, but they are only *legally* relevant if one provision actively modifies, permits, restricts, or preempts the other [206]. For instance, a retrieved document might use identical vocabulary but apply to a completely different jurisdiction or regulatory scope, rendering it completely useless for an ex-ante conflict check. Conversely, a document that governs the exact same civic behavior but uses entirely different phrasing—such as substituting “motor vehicle” for “automobile”—will be scored as mathematically irrelevant (i.e. 0) by a strict Boolean system [17]. Therefore, to accurately audit local ordinances against superior law, statutory retrieval systems must move beyond binary, keyword-driven matching in favor of dense semantic representations capable of capturing actual jurisprudential relationships [99].

To resolve these limitations, modern computational pipelines fundamentally redefine relevance spatially rather than lexically. Instead of scanning a query as a raw string of text, dense retrieval architectures utilize Bi-Encoder models to compress entire legal provisions into high-dimensional mathematical coordinates, commonly referred to as dense vectors [151]. Under this advanced semantic framework, relevance is no longer determined by counting shared characters, but by calculating the geometric distance—such as cosine similarity—between these coordinate points [99]. If a newly drafted local ordinance and an existing national statute govern the same underlying civic behavior, the neural network learns to map their respective vectors closely together in the mathematical space, completely bypassing the vocabulary mismatch problem [220]. Consequently, the retrieval mechanism evaluates relevance as a continuous proximity score rather than a rigid binary output, successfully surfacing conceptually intertwined laws that basic keyword searches otherwise miss.

2.8.2 Long-Context Information Retrieval

Relying on standard BERT-based architectures introduces an artificial constraint on legal analysis: the 512-token limit. Because standard Self-Attention mechanisms scale quadratically with sequence length, processing long texts requires severe truncation. To get around this, researchers like Chalkidis et al. [27] routinely use Sliding Window techniques to segment documents into overlapping chunks. However, applying this to Philippine ordinances—which are historically wordy and often bury exceptions deep within 1,000-token provisions—is structurally destructive. It severs the substantive semantic link between a primary rule and its distant exception.

Consequently, the model operates with a logical blindspot. Beltagy et al. [13] note

that this causes logical hallucinations, where a rule might be flagged as an absolute prohibition simply because its exempting clause was pushed into a separate, unconnected processing window. Furthermore, these systems must navigate the trade-off between Sparse Retrieval (e.g., BM25) for exact keyword matching, and Dense Retrieval for capturing conceptual meaning even without direct word overlap.

Resolving this fragmentation means moving beyond short sliding-window chunking where possible. The architectural solution requires an initial retrieval mechanism capable of native long-context retention. Recent Transformer models incorporate advanced positional encoding, such as Rotary Positional Embeddings (RoPE), alongside sparse or optimized attention matrices. Warner et al. [216] demonstrated that these upgrades extend context windows up to 8,192 tokens without triggering memory failures. Deploying this class of architecture allows the system to read entire local ordinances as cohesive semantic units, preserving the law’s internal structure and eliminating the Sliding Window fallacy.

2.8.3 IR Benchmarks for Candidate Selection

Selecting effective retrieval architectures from the extensive landscape of open-weight models requires objective empirical benchmarking. In contemporary legal information retrieval, candidate models are evaluated against standardized empirical leaderboards. While these testing frameworks mostly evaluate highly curated, Global North datasets, their rigorous zero-shot and clustering parameters provide a mathematically sound baseline to assess model capacity for handling complex statutory text.

Benchmarking Information Retrieval (BEIR) The Benchmarking Information Retrieval (BEIR) framework evaluates a Dense Retriever’s capacity for zero-shot retrieval across diverse and specialized datasets [197]. Zero-shot retrieval measures a model’s ability to operate in a specific domain without prior fine-tuning. Because Philippine local legislation completely lacks annotated training sets, a candidate model must inherently possess robust zero-shot generalization. Thakur et al. [197] establish that a dominant position on the BEIR leaderboard is a critical proxy for this. It indicates the model’s underlying capacity to map novel, localized legal vocabulary, including archaic syntax and uncoded administrative terms, into a coherent vector space without previous exposure.

Massive Text Embedding Benchmark (MTEB) Expanding on BEIR, Muennighoff et al. [128] introduced the Massive Text Embedding Benchmark (MTEB), which serves as the global standard for ranking embedding systems across over 130 datasets and multiple languages. In information retrieval literature, candidate architectures are frequently filtered using MTEB’s “Retrieval” and “Clustering” benchmarks. Models excelling on these specific leaderboards demonstrate a proven capacity to group overlapping concepts and retrieve relevant texts based purely on semantic proximity.

COLIEE Task 3 (Statute Law Retrieval) The annual Competition on Legal Information Extraction and Entailment (COLIEE) organizes the premier international benchmark for computational law [67, 68]. Specifically, Task 3 evaluates statute law

retrieval, challenging computational systems to extract the exact legal provisions from an extensive statutory corpus that are relevant to a given legal query or case scenario. Success on COLIEE Task 3 requires balancing high-throughput filtering with precise semantic discrimination, serving as the benchmark standard from which leading statutory retrieval architectures—such as the dense augmentation methods of AIIR Lab [220] and the multi-stage ranking pipelines of Team JNLP [66]—are established and validated.

2.8.4 Candidate Model Architectures for IR

Surveying leading open-weight architectures documented across recent information retrieval literature and standardized benchmarks (such as MTEB and BEIR) reveals eight prominent dense bi-encoder models relevant to statutory candidate discovery, organized across four primary inductive categories:

Distilled Edge Baselines. Reimers and Gurevych [151] established the Sentence-BERT framework, mapping text into fixed-dimensional dense vector spaces optimized for cosine proximity. The `all-MiniLM-L6-v2` model compresses a 6-layer BERT architecture down to 22.7 million parameters through knowledge distillation, producing 384-dimensional dense vectors. Despite its ultra-compact footprint, empirical benchmarks demonstrate that it retains strong semantic clustering capabilities with sub-two-millisecond per-query latency on consumer CPUs. Complementing this, Xiao et al. [222] introduced `bge-small-en-v1.5` (33.4M parameters), which optimizes contrastive representation learning for high rank sensitivity on the Massive Text Embedding Benchmark (MTEB). In public-sector edge deployment, these distilled bi-encoders provide high-throughput candidate filtering without requiring dedicated GPU accelerators.

General MTEB Leaders. Scaling bi-encoder capacity to standard 110M-parameter base dimensions enables more expressive semantic representations. Xiao et al. [222] developed `bge-base-en-v1.5` (109M parameters, 768 dimensions), utilizing retrofitted contrastive pretraining and instruction-tuned query prefixes to excel across multi-domain retrieval leaderboards. Similarly, Wang et al. [214] introduced `e5-base-v2` (109M parameters), trained on over one billion weakly-supervised text pairs via contrastive objectives, demonstrating high generalization on unseen technical and legal search tasks. Furthermore, Song et al. [180] developed `all-mpnet-base-v2` (109M parameters), combining masked language modeling with permuted language pretraining to eliminate positional dependence and generate highly stable semantic embeddings.

Legal-Domain Adapted Encoders. Chalkidis et al. [25] demonstrated that pre-training bidirectional encoders on specialized legal repositories enables dense representations to capture statutory taxonomy and regulatory jargon that general-domain embedders conflate. Mean-pooled sentence embeddings derived from `legal-bert-base-uncased` (110M parameters) map statutory provisions into vector spaces that reflect legislative hierarchy and administrative codification rather than colloquial word associations.

Modern Long-Context and Multilingual Encoders. Addressing the quadratic attention complexity of traditional 512-token encoders, Warner et al. [217] developed **ModernBERT-base** (149M parameters), which integrates FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activations, and sequence unpadding to natively process up to 8,192 tokens. In statutory retrieval, this allows direct encoding of verbose multi-paragraph enactments without destructive sliding-window chunking. Expanding to the multilingual domain, Chen et al. [29] introduced **bge-m3** (568M parameters, 1,024 dimensions), supporting 8,192-token contexts and multi-functionality (dense, sparse, and multi-vector scoring). In statutory consistency checking, its multi-vector representations provide resilience across code-switched municipal records and bilingual administrative issuances.

2.8.5 Retrieval Architectures, Filtering Strategies, and Hybrid Fusion

In computational legal informatics, isolating candidate enactments from extensive statutory collections requires understanding the theoretical foundations, failure modes, and complementary trade-offs of diverse retrieval strategies. Recent findings from the COLIEE benchmarks and legislative consistency literature identify five core methodological dimensions governing coarse statutory search:

Probabilistic Lexical Retrieval (Okapi BM25). The foundational baseline for document retrieval in computational law is the Okapi BM25 model, formulated by Robertson and Zaragoza [158] within the probabilistic relevance framework. BM25 scores the relevance of a statutory document D to a query Q by evaluating non-linear term frequency saturation and document length normalization:

$$S_{\text{BM25}}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

where $f(q, D)$ denotes term frequency, $|D|/\text{avgdl}$ normalizes for provision length, and parameters $k_1 = 1.5$ and $b = 0.75$ govern term saturation and length penalization. In statutory retrieval (such as COLIEE Task 3), BM25 serves as an indispensable baseline (as demonstrated by Team AIIR [220]) because legislative drafting routinely depends on exact statutory section citations, formal administrative acronyms, and verbatim penal monetary amounts. However, as established by Blair and Maron [17] and Manning et al. [122], exact lexical matching suffers from the vocabulary mismatch problem: when local ordinances paraphrase statutory commands or enact indirect prohibitions using alternative civic vocabulary, BM25 fails to retrieve governing laws.

Dense Vector Semantic Retrieval. To overcome lexical mismatch and synonymy limits, dense retrieval maps provisions into continuous vector spaces using Siamese Bi-Encoder architectures [151]. Rather than evaluating token overlap, dense retrievers score relevance via cosine proximity in latent semantic space:

$$S_{\text{dense}}(D, Q) = \frac{\mathbf{e}_Q \cdot \mathbf{e}_D}{\|\mathbf{e}_Q\| \|\mathbf{e}_D\|}$$

where $\mathbf{e}_Q$ and $\mathbf{e}_D$ represent pooled contextual embeddings generated by models such as `all-MiniLM-L6-v2`. In statutory preemption analysis, dense representations capture conceptual legal intent, successfully retrieving governing statutes even when local drafts employ entirely disparate phrasing.

The Domain Filtering Dilemma: Hard Pruning vs. Soft Priors. In heterogeneous statutory databases spanning tens of thousands of enactments, researchers frequently investigate topic classification or domain partitioning to reduce search space complexity. However, empirical literature highlights the severe danger of *hard categorical filtering* in legal domains. Foundational statutory instruments—such as the Philippine Local Government Code of 1991 (RA 7160) or comprehensive civil and penal codes—function as omnibus legislation that simultaneously govern municipal administration, taxation, public safety, and environmental sanitation. Hard filtering by a single predicted regulatory category discards the governing omnibus code whenever a query touches an adjacent regulatory area, causing catastrophic candidate recall collapse. In contrast, information retrieval theory establishes *soft domain priors* [158]. Instead of binary candidate exclusion, soft priors apply a calibrated multiplicative score boost (e.g., +20%) to candidate documents matching the predicted topic or latent semantic cluster, preserving access to multi-domain statutory codes while gently elevating topically congruent provisions.

Score-Level Linear Combination vs. Rank-Level Fusion. Because lexical and semantic retrievers exhibit orthogonal failure modes, leading computational law frameworks combine both streams. In the literature, two primary fusion approaches predominate:

- *Score-Level Linear Combination (The JNLP Approach)*: Championed by Team JNLP in the COLIEE competitions [66], weighted linear interpolation combines normalized sparse and dense scores:

$$S_{\text{hybrid}}(D, Q) = \alpha \cdot \hat{S}_{\text{dense}}(D, Q) + (1 - \alpha) \cdot \hat{S}_{\text{BM25}}(D, Q)$$

where $\alpha \in [0, 1]$ balances semantic and lexical bias. However, because BM25 scores are unbounded while cosine similarities span $[-1, 1]$, linear combination requires min-max normalization, which can introduce instability across heterogeneous query lengths.

- *Rank-Level Reciprocal Rank Fusion (RRF)*: To eliminate score normalization artifacts, Cormack, Clarke, and Buettcher [38] formulated Reciprocal Rank Fusion:

$$\text{RRF}(D) = \sum_{m \in \mathcal{M}} \frac{1}{k_{\text{rrf}} + \text{rank}_m(D)}$$

where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ and $k_{\text{rrf}} = 60$. Because RRF operates strictly on ordinal ranking positions, it provides scale invariance, eliminates distribution sensitivity, and ensures that provisions performing well across both lexical and semantic modalities are prioritized.

Hierarchical Statutory Priority Safeguard. Anchored in the constitutional doctrine of legislative supremacy and the hierarchy of legal norms [97, 71], statutory sys-

tems operate under strict formal ordering. In national digital archives, subordinate administrative issuances (such as executive orders, administrative circulars, and departmental memos) frequently constitute nearly half of all indexed records. Because administrative issuances share broad governmental vocabulary with municipal drafts, unconstrained retrieval often allows routine circulars to crowd out superior national statutes. Literature in legal informatics emphasizes structural safeguards that prioritize primary legislative acts (Republic Acts) over subordinate executive issuances, shielding governing statutes from administrative noise and significantly improving Mean Reciprocal Rank (MRR).

2.9 Natural Language Processing

Natural Language Processing (NLP) provides the computational techniques needed to analyze and represent text at various linguistic levels, aiming for human-like comprehension [112]. As Jurafsky and Martin [92] explain, it acts as the bridge between human language and machine logic, allowing algorithms to parse and interpret unstructured text. In the legal field, NLP shifts document auditing from slow manual reviews to scalable, automated systems. However, applying standard NLP to law requires more than basic grammar checks; it demands systems that can navigate dense sentence structures, deontic operators (obligations, permissions, prohibitions), and the highly specific vocabulary found in statutory law.

Early Legal NLP relied heavily on Symbolic Artificial Intelligence (AI) [96]. These systems used manually built ontologies and strict Boolean logic to extract legal information. However, because municipal drafting language changes constantly over decades, these exact-match frameworks proved far too brittle. Contemporary NLP architectures solve this by moving to continuous vector representations. By mapping textual provisions into high-dimensional semantic spaces, modern Transformers bypass superficial vocabulary mismatches to mathematically quantify the deeper concepts connecting different legal terms. This allows the system to act as an active diagnostic engine that structurally maps legal conflicts, rather than just a passive search index.

This section breaks down how NLP operates within legislative analysis. It covers the systemic challenges of processing localized legal text, the structural need for Natural Language Inference (NLI), the benchmarks used for model selection, and the specific architectures built for these deductive tasks.

2.9.1 Challenges in NLP

Deploying NLP frameworks against municipal archives quickly exposes the mechanical limits of standard Language Models. These architectures must simultaneously overcome the geographic bias built into their pre-training data and the mathematical flaws of relying purely on lexical overlap.

The Global North Bias There is a noticeable lack of research addressing the noisy, low-resource reality of local government data in the Global South. As Ranathunga

et al. [149] point out, the vast majority of Legal NLP benchmarks, like LexGLUE [27] and CaseHOLD, are trained exclusively on pristine, digitally native Western legal text. These highly optimized models degrade drastically when applied to Philippine Local Government Unit (LGU) archives. These datasets are filled with scanned PDFs suffering from Optical Character Recognition (OCR) errors, archaic American colonial phrasing, untranslated Spanish legal terms, and local "Taglish." Rust et al. [162] demonstrated that standard WordPiece tokenizers shatter these unrecognized words into meaningless sub-tokens, destroying the model's ability to infer legal intent. Fixing this localized morphological noise requires abandoning low-capacity tokenizers. Instead, developers must deploy models equipped with massive Byte-Pair Encoding (BPE) vocabularies (e.g., 128,000+ tokens) that can absorb linguistic quirks without triggering Out-of-Vocabulary (OOV) errors. Furthermore, He et al. [81] note that utilizing models with Disentangled Attention separates the actual content of a corrupted word from its noisy positional data. This allows the architecture to track logic accurately even when the text is heavily degraded by OCR artifacts.

Shortcomings of NLP in Lexical Similarity While early NLP systems excel at finding topical overlap, they remain structurally deficient for Semantic Conflict Detection. Traditional NLP heavily prioritizes Lexical Similarity—measuring the surface-level token intersection between documents using algorithms like TF-IDF or shallow word embeddings. Bowman et al. [18] explain that pure lexical similarity is fundamentally unsuited for detecting legal contradictions because lexical metrics are entirely blind to deontic logic and syntactic negation. For example, an ordinance stating, "The city permits the sale of fireworks," and a later one asserting, "The city strictly prohibits the sale of fireworks," share near-perfect Lexical Similarity scores. A standard NLP similarity model will cluster them as highly related, completely missing the absolute policy collision happening beneath the vocabulary.

2.9.2 From NLP to Natural Language Inference

Historically, checking legislative consistency relied on Symbolic NLP (hand-written rules) and, later, Statistical NLP. In Statistical NLP, models learned probabilistic weights for specific inputs, like Term Frequency-Inverse Document Frequency (TF-IDF) or N-grams [96]. These models depended heavily on human-engineered features to define grammar rules and categories. While they allowed for soft, probabilistic decisions, they consistently struggled with the deep lexical and semantic ambiguities inherent in legal drafting.

The field largely transitioned to Neural NLP models in the mid-2010s [146]. Unlike statistical systems that treat features as separate entities, neural networks operate end-to-end, mapping raw text directly to output predictions through hidden layers. These architectures capture deeper linguistic patterns than traditional statistical methods by learning complex relationships during optimization. However, while machine learning excels at finding patterns, it can overly mimic its training data, making domain-specific pre-training essential for high performance in rigorous legal tasks.

Within modern neural approaches, Natural Language Inference (NLI) has emerged

as the definitive benchmark for evaluating a model’s capacity for actual reasoning. Yang et al. [224] define NLI as determining the logical relationship between two text sequences: a “premise” and a “hypothesis.” The model must classify whether the hypothesis is true (Entailment), false (Contradiction), or unrelated (Neutral) based solely on the premise. This specific task exposes the core flaw of earlier statistical NLP, which mistakenly equated high word overlap with logical entailment [224]. Modern neural approaches fix this by capturing deep compositional semantics. As Le Hong et al. [108] point out, this allows modern models to resolve complex linguistic phenomena—like coreferences and modality—that rigid rule-based systems historically failed to grasp.

2.9.3 NLI Benchmarks

Identifying effective Cross-Encoder architectures for the inference stage requires isolating models with robust, pre-trained deductive logic. In legal natural language processing, research literature relies on established Natural Language Inference (NLI) leaderboards to benchmark candidate models. While these benchmarks primarily evaluate Western jurisprudence, Chalkidis et al. [27] note that a model’s baseline performance on these standardized tests serves as an informative indicator of its capacity to parse dense statutory syntax.

GLUE and SuperGLUE Wang et al. introduced the General Language Understanding Evaluation (GLUE) [213] and its more rigorous successor, SuperGLUE [212], as foundational baselines for model reasoning. These frameworks evaluate a Transformer’s ability to handle diverse linguistic challenges, such as Recognizing Textual Entailment (RTE) and resolving complex coreferences. While a high SuperGLUE score confirms baseline cognitive capacity, it does not independently guarantee success in the legal domain due to the severe vocabulary shifts present in statutory drafting.

LexGLUE To evaluate specialized legal logic, legal NLP research relies on the Legal General Language Understanding Evaluation (LexGLUE) benchmark. Chalkidis et al. [27] designed LexGLUE specifically to test a model’s ability to navigate “legalese,” where common words assume highly context-dependent meanings (e.g., “executed” denoting a signed contract rather than a penal death). High performance on LexGLUE entailment tasks indicates that an architecture possesses the foundational deductive capacity required to navigate complex statutory drafting.

Hugging Face NLI and Zero-Shot Leaderboards While traditional NLP benchmarks evaluate general reading comprehension, contemporary Hugging Face NLI leaderboards evaluate models specifically on diverse, adversarial textual entailment tasks, aggregating MultiNLI (MNLI), FEVER, Adversarial NLI (ANLI R1–R3) [134], and WANLI [114]. Benchmarking candidate models across these suites tests their resilience against spurious lexical heuristics, trick premise-hypothesis alignments, and negation traps. High-performing architectures on these leaderboards—such as the pre-aligned DeBERTa-v3 variants curated by Lauer et al. [107]—demonstrate strong zero-shot deductive capacity, serving as leading candidate foundations for statutory conflict detection.

COLIEE Task 4 (Statutory Legal Textual Entailment) Within legal informatics, the definitive international competition benchmark for statutory inference is Task 4 of the Competition on Legal Information Extraction and Entailment (COLIEE) [147, 67, 68]. In Task 4, systems are presented with a statutory article (the premise) and a natural language legal query (the hypothesis), requiring binary classification to determine whether the query is legally entailed by or contradicts the governing statute. Evaluating architectures against COLIEE Task 4 methodologies provides direct empirical evidence of a model’s capacity to navigate statutory conditions, exceptions, and deontic modalities.

2.9.4 Candidate Model Architectures for NLI

In computational law and natural language inference, evaluating deontic relationships across statutory provisions demands specialized inductive biases. While early neural NLI research relied on general-domain masked language models, contemporary literature demonstrates that statutory preemption and normative collision detection require architectures tailored to dense legalese, long-context boundaries, disentangled logical negation, and computational efficiency [27, 81, 217]. Across recent benchmarks—such as LexGLUE, the Hugging Face NLI leaderboards, and the international COLIEE competitions—researchers converge on six primary architectural and domain families of candidate sequence encoders:

Legal-Domain Pretrained Encoders. Standard language models trained exclusively on general corpora (such as Wikipedia and BookCorpus) frequently fail to capture the specialized semantics of legal drafting, where everyday terms carry specialized juridical meanings. To address this vocabulary and stylistic divergence, researchers develop domain-adapted encoders initialized with custom legal vocabularies or continuously pretrained on expansive legal repositories. Chalkidis et al. [25] introduced **LEGAL-BERT**, pretrained from scratch on diverse English legislation, court judgments, and contract corpora, demonstrating superior term sensitivity across legal classification tasks. Henderson et al. [82] expanded this domain pretraining approach with **PoL-BERT** (Pile of Law), pretraining encoders on a 256 GB corpus encompassing administrative regulations, federal registers, municipal codes, and state statutory provisions. Complementary efforts by Paul et al. [142] (**InLegalBERT**) and Zheng et al. [228] (**CaseLaw-BERT**) investigate the transferability of common-law statutory and litigation pretraining, while Xiao et al. [221] (**Lawformer**) investigate domain-adapted long-document architectures. In the literature, this family provides a critical testbed for whether extensive pretraining on statutory and regulatory text provides intrinsic protection against legal misinterpretation.

Pre-Trained NLI and Cross-Encoder Reasoners. A pervasive vulnerability of unadapted sequence encoders when evaluated on deontic tasks is *affirmative bias*—the empirical tendency of language models to predict Entailment or Neutral when reading complex statutory clauses that actually contain subtle prohibitions or conflicting conditions [5, 200]. In foundational natural language processing, Devlin et al. [49] established **bert-base-uncased** (110M parameters), introducing bidirectional masked language modeling (MLM) and next sentence prediction (NSP).

As the canonical baseline architecture across standard NLP benchmarks, standard BERT provides the primary general-purpose foundation against which domain adaptation and architectural improvements are measured. To overcome the heuristic reliance on superficial lexical overlap, contemporary literature explores models pre-aligned on multi-task natural language inference datasets (such as MNLI, FEVER, ANLI, and WANLI). Laurer et al. [107] demonstrate that initializing sequence classifiers from warm NLI checkpoints equips models with robust directional contradiction sensitivity before domain adaptation. At the architectural level, He et al. [81] developed **DeBERTa-v3**, which introduces Disentangled Attention—representing word content and relative spatial positions with separate vectors—and replaces standard masked language modeling with an adversarial replaced token generator. In parallel, Clark et al. [33] developed **ELECTRA**, utilizing Replaced Token Detection (RTD) to optimize contextual representations across all input positions. These architectures excel at distinguishing subtle deontic shifts (such as isolating an absolute prohibition from a conditional authorization) that standard unidirectional or masked encoders routinely miss.

Modern Long-Context Encoders. Statutory drafting is inherently verbose, characterized by complex preamble clauses, extensive definitions, and multi-sentence penal qualifications. Traditional transformer encoders are constrained by quadratic self-attention complexity, imposing a rigid 512-token ceiling that necessitates destructive document chunking or sliding window heuristics [27]. Early long-context architectures attempted to resolve this constraint using localized or sparse attention approximations, such as the sliding-window attention in **Longformer** [13] and the block-sparse random attention in **BigBird** [226]. However, empirical legal informatics evaluations note that sparse attention patterns can dilute or sever semantic dependencies between distant provisos and governing commands. To overcome these limitations, modern architectures integrate architectural optimizations that support extended contexts natively. Warner et al. [217] introduced **ModernBERT**, an 8,192-token encoder incorporating FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activation functions, and sequence unpadding. Complementing this, Nussbaum et al. [137] developed **Nomic-BERT** to extend context retention. These modern long-context designs enable joint encoding of complete multi-sentence statutory provisions without sparse approximation artifacts.

Native Cross-Encoder Rerankers. Unlike bi-encoders that process queries and documents independently into pooled vector coordinates, cross-encoders execute full bidirectional self-attention across the concatenated pair simultaneously. In retrieval and inference benchmarks, joint cross-attention captures deep inter-token interactions that vector similarity cannot represent. In COLIEE statute retrieval benchmarks, researchers deploy multilingual cross-encoder rerankers, notably **BGE-Reranker-v2-m3** [222], to score pairwise relevance across multi-thousand token contexts. Similarly, sequence classifiers adapted from MS-MARCO reranking frameworks employ deep cross-attentive scoring to evaluate pair compatibility. In literature, native cross-encoders represent the conceptual bridge between Stage 1 candidate filtering and Stage 2 logical validation.

Multilingual and Cross-Lingual Encoders. In post-colonial statutory jurisdictions like the Philippines, statutory archives exhibit unique sociolinguistic complexities. While statutory enactments are drafted in legal English, municipal ordinances and administrative records frequently feature localized terminology, regional administrative phrasing, Latin legal maxims, and English-Tagalog code-switching. Conneau et al. [37] introduced **XLM-RoBERTa**, trained across 100 languages, demonstrating strong cross-lingual transfer on semantic benchmarks. He et al. [81] extended disentangled attention to the multilingual domain with **mDeBERTa-v3**, providing resilience to mixed-language syntax. Furthermore, Cruz and Cheng [41] developed Tagalog-adapted transformers (**roberta-tagalog-base**) to capture Philippine syntactic structures. In statutory consistency checking, multilingual encoders evaluate whether localized linguistic variances can be processed without catastrophic comprehension loss.

Distilled and Compact Edge Baselines. Deploying deep learning in public-sector municipal environments requires strict adherence to available edge computing infrastructure. Because local government units typically operate without high-performance computing clusters or enterprise cloud APIs, running billion-parameter language models is computationally unfeasible. To establish realistic performance baselines under constrained hardware, the literature explores knowledge distillation and parameter sharing. Sanh et al. [169] introduced **DistilBERT**, which compresses transformer depth while retaining over 95% of language comprehension. Lan et al. [103] introduced **ALBERT**, utilizing cross-layer parameter sharing to reduce memory footprint down to tens of megabytes. In dense retrieval and inference, Reimers and Gurevych [151] established **all-MiniLM-L6-v2** as an ultra-compact 22-million parameter baseline. Evaluating distilled edge models allows researchers to quantify the exact accuracy penalty incurred when prioritizing sub-second CPU latency over multi-layer neural capacity.

2.10 Ground Truth

In supervised machine learning, a Ground Truth (or Gold Standard) dataset provides the correct, human-verified labels used to train and test models. During training, the neural network uses this dataset as an answer key. It constantly compares its own predictions against these true labels, calculating the difference (or loss) and adjusting its internal weights to make fewer mistakes over time [69]. Without a highly accurate set of target labels, the model will essentially learn the wrong rules, leading to a phenomenon where it memorizes flawed data instead of learning actual reasoning patterns.

Beyond just teaching the model, the ground truth is the only objective way to measure success. Researchers typically split this data into training, validation, and testing sets. The test set acts as the final exam, kept completely hidden from the model until the very end. Evaluating the model against this unseen ground truth allows researchers to calculate standard performance metrics like precision, recall, and F1-score. Without this baseline, it is mathematically impossible to prove that a model generalizes well to new data or works as intended in a real-world scenario [19].

2.10.1 Ground Truth in Natural Language Inference

In Natural Language Inference (NLI), ground truth datasets define the logical relationship between two pieces of text. Annotators read a source text (the Premise) and a claim (the Hypothesis), then categorize their relationship into one of three distinct labels. If the premise guarantees the hypothesis is true, it is labeled as Entailment. If they contradict each other, it is labeled as Contradiction. If there is not enough information to prove either, the pair is labeled Neutral [224].

Most traditional NLI datasets, such as the widely used SNLI (Stanford Natural Language Inference) corpus, are built using crowdsourced workers who write simple variations of everyday sentences [19]. While this approach is cheap and easily generates hundreds of thousands of training pairs, the resulting datasets are incredibly basic and often embed unintended annotation artifacts [78]. They usually feature short, single-sentence premises with obvious logical connections. Training a model exclusively on these simple datasets leaves it entirely unprepared for the dense, multi-layered logic required in specialized fields.

2.10.2 Annotation Procedures and Reliability Metrics

Turning raw text into a useful NLI dataset requires strict guidelines to minimize human bias and prevent annotators from relying on subjective legal interpretations. In standard NLI, guidelines dictate simple factual agreement. However, legal conflict detection requires evaluating normative relationships based on deontic logic. Drawing from the frameworks established by Agotnes et al. [3] and Li et al. [111], the annotation guidelines strictly instruct experts to map the text into obligations (must do), permissions (may do), and prohibitions (cannot do). A "Contradiction" is only flagged if the local ordinance mandates an action the national statute prohibits, or prohibits an action the national statute explicitly permits. If a local ordinance merely adds a regulatory step (e.g., a local processing fee) to a national permit without banning the core activity, the guidelines strictly dictate a "Neutral" classification. This rigid deontic framework is designed to ensure that the dataset measures actual legal preemption rather than just topical friction.

To evaluate the effectiveness of these strict guidelines, researchers must utilize Inter-Rater Reliability (IRR) metrics [125] rather than mathematically flawed percentage agreements. Raw percentage agreement fails to account for statistical probability, where in a multiple-choice classification task, there is an inherent likelihood that annotators will occasionally select the same label by pure random guessing.

The two primary statistical formulas used to subtract this random chance are Cohen's Kappa [34] and Fleiss' Kappa [59]. Cohen's Kappa evaluates agreement between exactly two raters using the following formula:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o represents the relative observed agreement between the two individual raters, and p_e represents the hypothetical probability of chance agreement.

When a dataset requires a multi-expert consensus—such as utilizing three or more domain experts to establish a majority rule—Fleiss’ Kappa becomes the mandatory statistical metric. Fleiss generalized Cohen’s formula to accommodate a fixed, larger number of multiple raters:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

In this generalized equation, $\bar{P}$ represents the mean observed agreement across all pairs of raters, while $\bar{P}_e$ represents the expected chance agreement across the entire pool. The resulting κ score is interpreted using the universally accepted statistical matrix established by Landis and Koch [105]:

Table 2.1: Interpretation of Fleiss’ Kappa (κ) Scores

Kappa (κ) Score	Interpretation
< 0.00	Poor agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

While objective tasks routinely achieve κ scores above 0.90, applying this expectation to the legal domain is mathematically unrealistic. Bowman et al. [19], in their foundational study for the Stanford Natural Language Inference (SNLI) corpus, established that evaluating semantic relationships is inherently subjective. Their benchmark study hit an approximate performance ceiling with an overall Fleiss’ Kappa of 0.70. They explicitly demonstrated that achieving near-perfect agreement (> 0.80) in textual inference usually indicates overly simplistic or fabricated test data.

2.10.2.1 Mitigating Sub-Threshold Reliability

While achieving a ”Substantial” Inter-Rater Reliability score ($\kappa > 0.61$) validates the clarity of the annotation guidelines, failing to meet this threshold does not inherently invalidate the dataset or halt the experimental pipeline. Klebanov and Beigman [100] establish that a sub-threshold κ score typically indicates systematic ambiguity within the text itself—categorized as ”hard” instances—rather than incompetence by the annotators. Their research demonstrates that it is still entirely possible to extract a reliable, low-noise subset from the instances where annotators do achieve consensus.

When researchers encounter this sub-threshold reliability, the mitigation strategy depends entirely on the size of the dataset. For large-scale studies, including the disaster-tweet classification by Ablazo [1], the standard procedure is data omission. Beigman et al. [100] explicitly advocate for filtering out difficult, borderline cases from the gold standard, arguing that evaluating algorithms on inherently ambiguous text introduces

statistical noise akin to random coin-flips. In these massive datasets, items that fail to reach majority consensus are simply discarded because the volume of remaining "easy" data easily absorbs the loss.

However, applying this standard omission strategy to a specialized, few-shot legal dataset mathematically destroys the test set's Minimum Detectable Effect. Furthermore, in municipal legislation, these "hard" cases often represent the exact jurisdictional friction the logic engine needs to learn. To preserve dataset volume without introducing benchmarking noise, researchers in specialized domains employ Expert Adjudication rather than omission. Gao et al. [65] establish that in these scenarios, a hierarchical adjudication procedure is used where a senior domain expert reviews the conflicting annotations. The senior expert resolves the ambiguity based on governing jurisprudential principles and manually establishes the Gold Standard label, treating the initial low κ metric as a measure of text difficulty rather than a rigid experimental blocker.

2.10.3 Ground Truth in the Legal Domain

When applying NLI to the legal domain, the standard approach of comparing short, isolated sentences quickly breaks down. Statutory phrasing is notoriously complex, filled with conditional clauses, exceptions, and domain-specific vocabulary. Consequently, a hypothesis in a legal context rarely aligns perfectly with a single sentence in the premise. Instead, proving an entailment or contradiction often requires the model to connect evidence scattered across multiple paragraphs or pages.

To address this, researchers have developed specialized legal datasets that reflect real-world document structures. For instance, the Competition on Legal Information Extraction/Entailment (COLIEE) dataset constructs its ground truth by pairing civil code statutes with legal bar exam queries, requiring models to identify specific legal provisions before determining entailment [147]. Similarly, the ContractNLI dataset by Koreeda & Manning [101] engineers its gold standard around non-disclosure agreements. Rather than giving the model pre-cut sentence pairs, ContractNLI forces the system to evaluate hypotheses against entire legal documents, requiring annotators to both classify the logical relationship and extract the specific supporting text span.

Furthermore, unlike general-domain datasets crowdsourced through platforms like Amazon Mechanical Turk, legal ground truth relies heavily on domain experts. Constructing datasets like CaseHOLD [228], which evaluates legal holding statements, requires law students and legal professionals to ensure the labels accurately reflect actual jurisprudence. The push towards comprehensive legal benchmarks, such as LexGLUE [27], demonstrates that evaluating legal NLP models demands rigorously structured, expert-annotated data rather than automated or amateur labeling.

This difficulty is magnified when dealing with local or municipal legislation. While national or federal laws generally follow standardized drafting conventions, municipal texts are highly heterogeneous, exhibiting significant variations in terminology, structure, and overall organization across different jurisdictions [21]. Because of this localized variance, municipal legal documents have historically been neglected in NLP research, hampered primarily by a severe lack of annotated ground truth datasets [22]. Standard legal cor-

pora fail to capture the specific legislative templates, localized governance terminology, or idiosyncratic cross-referencing styles used in municipal laws, such as Philippine local ordinances. Without existing resources that reflect these structural and linguistic complexities, evaluating models on local legislation requires building highly specialized, expert-annotated datasets from the ground up.

A critical architectural decision in dataset construction is determining the exact granularity of the extracted legal premise. While the hypothesis (the local draft) is evaluated as a multi-sentence span to capture context, the premise (the national statute) must be strictly isolated at the sentence level. In Philippine statutory drafting, a single paragraph frequently contains a general rule, multiple conditional modifiers, and distinct exceptions. Holzenberger et al. [84] demonstrate that feeding an entire statutory paragraph into an NLI model as a premise dilutes the model’s attention mechanism, causing it to lose track of specific variable constraints. By forcing human annotators to isolate the national rule down to its exact sentence boundary, the resulting Ground Truth prevents the candidate logic engines from getting distracted by surrounding, non-governing text, ensuring the classification mathematically targets the exact point of semantic collision.

The difficulty of dataset construction is magnified by the scarcity of legal domain experts available for annotation. As Zhong et al. [229] explicitly note, the complex terminology required for legal texts makes manual labeling highly time-consuming and difficult to scale, as it strictly requires domain experts rather than crowdsourced workers. Because constructing massive datasets for localized municipal law is practically unfeasible, researchers must rely on constrained, highly curated Gold Standards. To compensate for this limited data volume, the current NLP standard mandates treating legal conflict detection as a few-shot learning scenario. Gao et al. [64] validate this approach, proving that transformer models can be successfully fine-tuned on specialized datasets containing merely hundreds of examples, provided the evaluation framework rigorously controls for overfitting.

2.10.4 Multi-Sentence Context and Cross-Document Limits

Traditional Natural Language Inference (NLI) evaluates text using a strict 1-to-1 pairing, comparing a single premise sentence directly against a single hypothesis sentence. However, this highly granular approach fundamentally breaks down in the legal domain due to two specific edge cases. First, intra-document context: a primary rule in one section of an ordinance is frequently modified or negated by an exception buried several paragraphs later (the “1v2” or multi-sentence problem). Second, cross-document dependencies: a contradiction may only emerge when synthesizing provisions from two entirely separate statutes simultaneously.

To address the first edge case, modern legal frameworks utilize Asymmetric Span-Level Pairing. Yin et al. [225] critique traditional sentence-level datasets, proving that capturing contextual exceptions requires evaluating multi-sentence blocks rather than isolated lines. As demonstrated by Koreeda and Manning in the ContractNLI dataset [101], the premise remains a single, focused legal rule, while the corresponding hypothesis is extracted as a multi-sentence chunk. This asymmetric structure allows the model’s

attention mechanism to accurately evaluate a primary rule against a distant exception contained within the same local provision.

While asymmetric spans successfully capture intra-document context, resolving the second edge case—cross-document multi-hop reasoning—remains computationally prohibitive. Identifying semantic preemption that spans multiple independent laws currently represents the bleeding edge of NLP research and is not yet fully explored for local governance. Recent frameworks have attempted to solve this, but they introduce severe architectural overhead. For instance, Lu et al. [118] address cross-document retrieval by programmatically mapping Entity Graphs to mine evidence paths between laws, a process that is highly resource-intensive and unscalable for unstructured LGU archives. Alternatively, Shen et al. [176] proposed the HopWeaver framework, which relies on massive generative Large Language Models (LLMs) to synthesize complex reasoning paths across documents.

Both of these state-of-the-art solutions immediately violate the strict hardware constraints of consumer-grade LGU workstations. Consequently, practical, edge-deployable systems must intentionally bound their theoretical scope to single-premise, asymmetric span evaluations, deliberately avoiding the quadratic memory scaling required for multi-document fusion.

2.10.5 Train/Test Partitions for Small Datasets

When evaluating NLI models on constrained datasets, researchers often default to k -fold cross-validation to maximize testing coverage. However, Vabalas et al. [203] mathematically demonstrated that applying cross-validation to datasets under 1,000 items introduces severe optimistic bias, causing models to perform significantly worse in real-world deployment than their test scores suggest. To rigorously control for overfitting in few-shot legal scenarios, contemporary evaluation frameworks must abandon k -fold approaches. Instead, they require strict, static Train/Validation/Test partitions (e.g., 70/15/15) to ensure the evaluation metrics remain entirely untainted by training exposure.

Beyond merely preventing optimistic bias, utilizing a static partition allows for rigorous validation monitoring during the fine-tuning phase. In datasets with limited volume, neural networks are highly susceptible to memorizing the training data rather than learning underlying logical patterns. Goodfellow et al. [69] establish that permanently isolating a validation set allows architectures to employ early stopping criteria, halting the training process the moment the model’s accuracy on unseen data begins to degrade. Furthermore, as Zhang et al. [227] note regarding few-sample fine-tuning, this static separation ensures strict document-level boundaries, preventing clauses from a specific municipal draft from accidentally bleeding across the training and testing sets to artificially inflate performance metrics.

2.10.6 Small Test Sets and Minimum Detectable Effect

A standard critique of static splits on small datasets is that the resulting test set lacks the statistical power to reliably compare candidate architectures. However, Card et al. [23] es-

tablished that a test set’s validity is fundamentally determined by its Minimum Detectable Effect (MDE) rather than arbitrary size thresholds. The MDE dictates that if the expected accuracy gap between two models is small, a massive test set is required to establish that observed performance differences exceed random chance. Conversely, because fine-tuning foundational models on a highly novel, localized municipal domain yields massive margins of performance improvement, the required MDE drops significantly. Consequently, even a constrained test set of approximately 50 pairs is mathematically sufficient to definitively detect the optimal logic engine without sacrificing statistical significance [64].

The practical application of MDE fundamentally shifts how model performance is evaluated in specialized domains. In general NLP research, massive test sets are required because researchers are often evaluating models against benchmarks like GLUE to prove marginal, incremental gains of one percent to two percent over previous state-of-the-art architectures [213]. However, applying a base foundation model—which possesses no prior knowledge of Philippine local governance for example—to a highly specialized task generally results in baseline performance barely above random chance [25]. Zheng et al. [228] demonstrate that in the legal domain, fine-tuning foundational architectures on expertly curated pairs triggers a massive, double-digit jump in classification accuracy compared to zero-shot baselines. Because this expected performance delta is substantial, legal NLP frameworks do not require thousands of annotated examples to demonstrate statistically valid improvements, proving that an evaluation set of approximately 50 curated pairs possesses sufficient statistical power to mathematically capture this distinct leap in model reasoning capacity.

2.11 Explainable Artificial Intelligence

In legal informatics, a “black box” prediction is fundamentally unacceptable, regardless of nominal classification accuracy. Automated systems that flag normative conflicts between statutory enactments must provide verifiable reasoning (*ratio decidendi*) behind their classifications to genuinely assist legislative analysis [53]. Without transparent justification, legal practitioners and legislative staff cannot trust, audit, or formally defend algorithmic determinations. Consequently, contemporary legal NLP literature emphasizes Explainable AI (XAI) not as an optional secondary layer, but as an intrinsic structural requirement of high-stakes legal decision support, prioritizing faithful mathematical traces over ungrounded generative summaries.

2.11.1 Intrinsic Explainability Mechanisms

Within the explainability literature, researchers distinguish between Post-Hoc Surrogate Models and Intrinsic Explainability Mechanisms. Surrogate methods, such as Local Interpretable Model-agnostic Explanations (LIME) [155] and SHapley Additive exPlanations (SHAP) [120], attempt to approximate the behavior of complex deep networks by fitting a simplified, interpretable linear model around individual predictions. However, in text classification benchmarks, these external explainers are computationally demanding and inherently prone to approximation errors, as they explain a surrogate rather than the underlying network itself [160]. In contrast, intrinsic approaches extract internal model

representations—such as self-attention weights and contextualized hidden states—directly from the transformer layers during the forward pass. Because native attention extraction reflects the mathematical weights computed during inference without fitting secondary approximations, it offers an intrinsically faithful and computationally lightweight mechanism for structural transparency.

While utilizing attention weights as a proxy for feature importance has sparked significant debate, empirical findings demonstrate its utility for targeted textual inspection. Jain and Wallace [91] argued that raw attention distributions do not always correlate consistently with gradient-based feature importance. However, Wiegrefe and Pinter [218] provided critical counter-evidence, demonstrating that attention mechanisms provide faithful representations of internal prioritization under rigorous diagnostic testing. In statutory conflict detection and deontic reasoning, extracting token-level attention spans allows reviewers to identify the specific conflicting provisions (for example, isolating an “absolute prohibition” in a superior statute against an “unauthorized permit” in a local draft), establishing a direct visual link between textual evidence and classification decisions without incurring the computational overhead of auxiliary surrogate models.

Complementing structural attention heatmaps in modern explainability frameworks is Confidence Calibration. Deep learning models, particularly multi-layer transformers, frequently exhibit severe overconfidence in erroneous predictions when evaluated on out-of-domain or specialized text. As Guo et al. [76] demonstrated, modern neural networks are fundamentally miscalibrated, such that raw softmax probabilities rarely reflect the true empirical likelihood of correctness. In high-stakes regulatory auditing, relying on uncalibrated probabilities produces misleading risk assessments. To restore probabilistic reliability, calibration literature advocates post-hoc temperature scaling applied to pre-softmax logits. Rescaling logits by a learned temperature parameter (τ) before softmax normalization aligns predicted confidence with observed empirical accuracy, enabling decision support systems to distinguish between definitive contradictions and borderline ambiguities with sound statistical justification.

2.11.2 Edge Deployment Mechanisms

In edge and on-premise computing environments, operational feasibility requires strict adherence to memory and latency boundaries. Post-hoc explainability methods operate by repeatedly perturbing input tokens, requiring hundreds or thousands of forward passes per document to approximate Shapley values or linear boundary slopes [155, 120]. Empirical evaluations by Kumar et al. [102] confirm that permutation-based explainers inflate per-document inference latency by several orders of magnitude, rendering them computationally prohibitive for lengthy legislative texts. Furthermore, Wang et al. [215] demonstrate that running complex post-hoc explainers on edge workstations severely degrades throughput, making interactive review impractical. In resource-constrained deployment settings, intrinsic extraction of native attention weights and calibrated logits provides a computationally viable pathway to transparency without requiring specialized high-performance computing clusters.

2.11.3 Metrics for Explanation Quality

Evaluating explanation quality in natural language processing requires distinguishing between two fundamental evaluation criteria: Plausibility and Faithfulness. Plausibility reflects how convincing, fluent, and intuitive an explanation appears to a human reader, whereas Faithfulness measures the degree to which an explanation accurately reflects the internal reasoning operations employed by the model to generate its prediction [90, 53]. Jacovi and Goldberg [90] demonstrate that optimizing explanations purely for plausibility frequently produces deceptive explanations that mask internal reasoning failures and biases. In legal domains, legal scholarship emphasizes that faithfulness must take precedence, as an explanation that sounds plausible but misrepresents statutory text can lead to catastrophic compliance failures.

This trade-off is particularly evident in generative explainability approaches that deploy auto-regressive large language models to draft plain-language explanations of legal predictions. While generative summaries achieve high human plausibility scores, empirical legal informatics studies demonstrate that generative models frequently suffer from legal hallucinations, fabricating non-existent statutory citations or altering legal interpretations during text generation [113, 95]. Because generated natural language explanations are decoupled from the classifier’s internal mathematical representations, they remain inherently unfaithful [90]. Consequently, legal informatics literature increasingly favors extractive and attention-grounded visualizations that display the exact mathematical spans influencing classification, avoiding generative distortions.

2.11.4 XAI within Multi-Stage NLP Architectures

In multi-stage NLP pipelines, literature on Explainable AI emphasizes situating interpretability mechanisms at the fine-grained inference stage rather than the coarse retrieval phase. In retrieve-then-entail systems, the initial retrieval phase functions as a high-throughput candidate selector across extensive document repositories. As Anand et al. [7] observe, computing dense feature attributions during coarse document filtering provides negligible analytical utility while incurring prohibitive computational latency. In contrast, the subsequent cross-encoder inference stage conducts fine-grained semantic entailment across a constrained candidate set, making it the mathematically appropriate locus for interpretability analysis.

At this second stage, legal NLP frameworks utilize token-level saliency and attention visualizations to bridge model internals with human inspection [209]. By mapping cross-attention weights and calibrated class probabilities back to corresponding spans in the premise and hypothesis texts, systems provide structural verification of detected contradictions without introducing generative paraphrasing risks. This dual mechanism of intrinsic attention mapping and post-hoc confidence calibration allows legislative researchers to trace classification outputs directly back to source statutory phrases.

2.12 Theoretical Framework

As established in legal informatics literature, the Retrieve-then-Entail architecture represents the leading theoretical framework for statutory conflict detection and legislative consistency checking. Originating in the international benchmarks of the Competition on Legal Information Extraction and Entailment (COLIEE 2025 and 2026) [67, 68], this framework integrates two complementary theoretical pillars: a scalable Information Retrieval (IR) architecture to isolate relevant statutory context from expansive corpora, followed by a fine-grained Natural Language Inference (NLI) architecture for rapid deontic conflict detection. Specifically, this theoretical framework synthesizes the dense statutory retrieval methodology of Team AIIR Lab [220, 106], the multi-stage ensemble retrieval principles of Team JNLP [67, 66], and the encoder-based textual entailment framework of Team KIS [93].

2.12.1 Statute Law Retrieval by Team AIIR

In statute law retrieval, dense bi-encoder architectures offer an effective theoretical foundation for bridging the semantic gap between newly drafted provisions and expansive statutory corpora (e.g., 25,000+ legislative documents) without relying on computationally demanding generative text decoding or manual legal taxonomy engineering [220, 106]. Drawing from the empirical methodology developed by the AIIR Lab in recent COLIEE benchmarks, this retrieval pipeline comprises four theoretical components: Domain-Adapted Data Augmentation, Semantic Vectorization, Corpus Pre-computation, and Nearest-Neighbor Distance Evaluation.

Data Augmentation and Domain Adaptation. By training on synthetically augmented legal datasets, dense retrieval models learn to recognize identical underlying statutory concepts regardless of whether they are expressed in archaic legal vocabulary or modern drafting syntax, substantially enhancing semantic robustness against lexical divergence.

Semantic Vectorization via Bi-Encoder. The central component of Phase 1 is a Bi-Encoder architecture (such as `all-mpnet-base-v2`), which projects statutory text into a dense, continuous vector space. Unlike cross-encoders, which must process query-document pairs jointly and are computationally prohibitive across thousands of statutory enactments, bi-encoders encode queries and corpus documents independently. During contrastive fine-tuning, the embedding space is optimized such that vectors representing logically relevant statutory pairs exhibit high geometric proximity, while irrelevant pairs are separated under distance metrics such as cosine similarity.

Corpus Pre-computation and Vector Indexing. To satisfy the operational requirement of querying tens of thousands of statutory documents efficiently, dense retrieval pipelines rely on offline pre-computation. Prior to query execution, the fine-tuned bi-encoder processes the statutory corpus, transforming individual statutory provisions into static, dense vector representations stored within an indexed vector database. Because corpus vectorization is decoupled from live query execution,

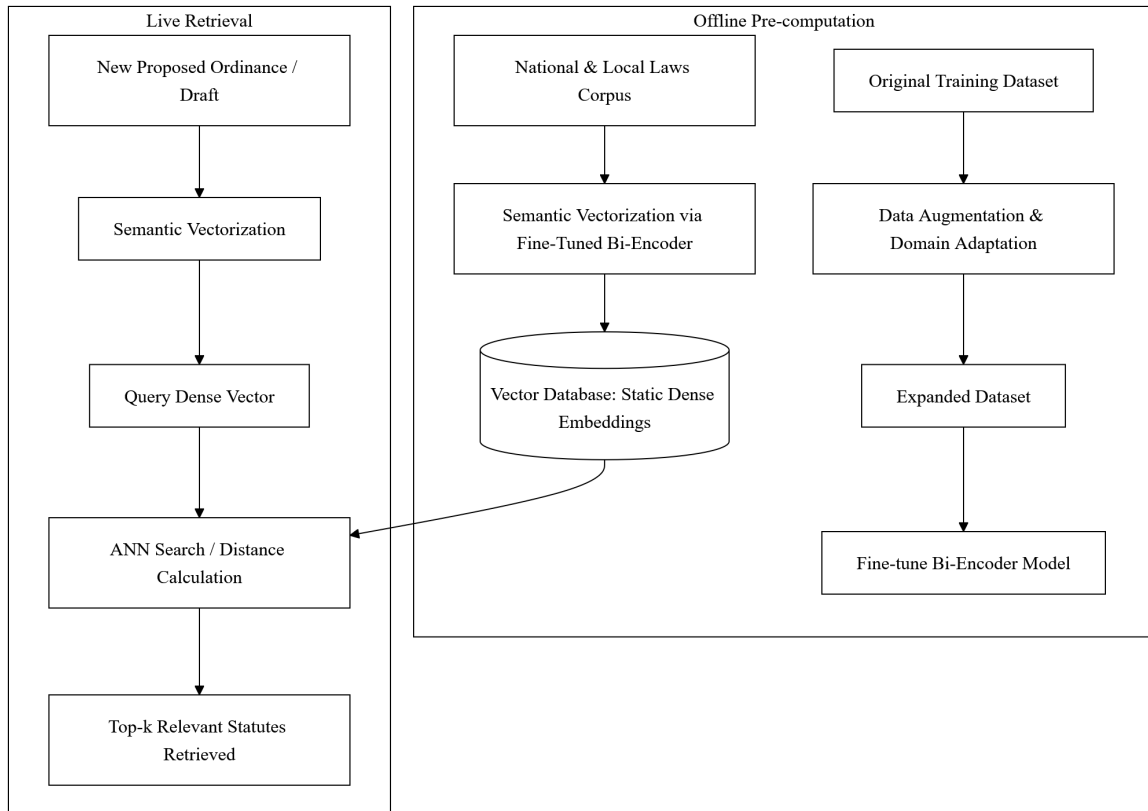


Figure 2.1: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from AIIR Lab’s retrieval methodology.

dense retrieval scales effectively across extensive statutory archives without requiring real-time GPU re-encoding.

Live Query Vectorization and Proximity Scoring. During query evaluation, the bi-encoder projects candidate draft provisions into the shared mathematical embedding space. Approximate nearest neighbor (ANN) search algorithms subsequently calculate geometric proximity (e.g., cosine similarity) between the query vector and pre-indexed statutory embeddings. Because this process computes continuous geometric distances rather than autoregressive tokens, high-throughput retrieval surfaces the top- k most relevant statutory candidates in sub-second latency, providing high candidate recall to prime downstream natural language inference.

2.12.2 Statute Law Retrieval by Team JNLP

Complementing pure bi-encoder retrieval, literature in computational law emphasizes multi-stage retrieval pipelines to balance candidate recall against ranking precision. Exemplified by Team JNLP’s high-performing architecture in COLIEE benchmarks [67, 66], this multi-stage pipeline progressively filters statutory search spaces across three structured phases: Pre-Retrieval Candidate Pruning, Model Fine-Tuning, and Result Ensembling.

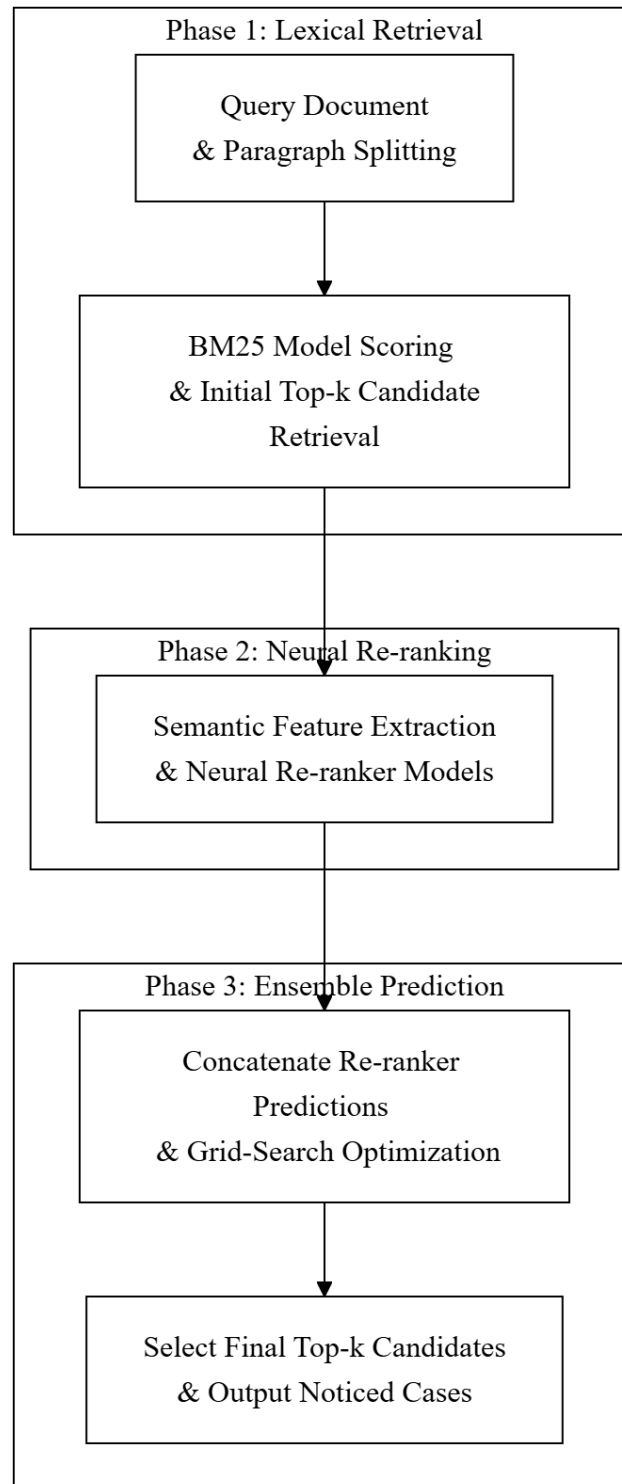


Figure 2.2: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from JNLP's retrieval methodology.

2.12.2.1 Pre-Retrieval

The primary objective of the initial phase is to drastically reduce the expansive search space while maintaining high recall across governing provisions, achieved through a multi-stage candidate pruning procedure.

Distance-Based Heuristic Filtering. Dense embeddings for statutory queries and corpus provisions are extracted using high-capacity multilingual representations (e.g., BGE-M3). Rather than relying solely on raw cosine similarity, the methodology calculates multi-dimensional distance metrics (such as L1 distance bins), training a gradient-boosted decision tree to predict relevance and counter severe positive-negative sample imbalance.

Passage Re-ranking and Pruning. The candidate pool is subsequently refined to eliminate non-relevant documents. A specialized neural passage re-ranker (such as RankLLaMA) evaluates candidate texts and prunes lower-ranked provisions, successfully narrowing the candidate set to top-ranked candidates while preserving recall.

2.12.2.2 Model Fine-Tuning

Once the candidate space is constrained, the framework applies targeted natural language models to perform fine-grained relevance scoring.

Parameter-Efficient Model Adaptation. Because fine-tuning large models across full parameter sets is computationally intensive, legal NLP frameworks employ Parameter-Efficient Fine-Tuning (PEFT) techniques, such as LoRA and QLoRA. These techniques enable foundational language models to adapt to specialized statutory semantics with minimal memory overhead.

Class Imbalance Mitigation. Because statutory candidate sets exhibit severe class imbalance (where only a small minority of retrieved provisions contain true preemption conflicts), sampling strategies and asymmetric loss functions are applied during training to preserve sensitivity to positive conflict signals.

2.12.2.3 Result Ensemble

To maximize retrieval robustness and mitigate individual model variance, multi-stage retrieval pipelines aggregate relevance scores across distinct retrieval architectures.

Weighted Score Aggregation. Rather than relying on uniform voting, relevance predictions are aggregated through weighted linear combinations of calibrated probabilities outputted by complementary lexical and neural models.

Hyperparameter Optimization. The weighting coefficients assigned to individual scoring components are optimized via systematic hyperparameter search (e.g., Optuna), ensuring balanced contributions between lexical keyword precision and dense semantic recall.

2.12.3 Statute Law Textual Entailment by Team KIS

In the second phase of legal retrieve-then-entail architectures, the theoretical objective transitions from broad candidate discovery to fine-grained deontic reasoning. To ensure high computational throughput and deterministic classification, modern legal NLP frameworks deploy non-generative cross-encoder architectures, exemplified by the ModernBERT-based rationale extraction methodology developed by Team KIS [93].

This entailment framework is structured around four theoretical components:

Unified Input Construction. While traditional transformer encoders are constrained by short sequence limits (e.g., 512 tokens), requiring aggressive document truncation, modern architectural designs support extended context windows up to 8,192 tokens. Under this formulation, retrieved statutory context (the premise) and candidate draft provisions (the hypothesis) are concatenated into a unified sequence. Strategic token boundaries clearly demarcate statutory premises from draft hypotheses.

Token-Level Problem Formulation. To evaluate targeted normative conflicts within candidate provisions, the framework structures input sequences with dedicated segment tokens (e.g., <sep>). This formats multi-clause statutory text into an aligned representation suitable for direct cross-attention.

Simultaneous Multi-Target Classification. The unified sequence is processed by an efficient bidirectional encoder. Because the architecture operates purely as a sequence classifier, it eliminates the computational latency and hallucination vulnerabilities inherent in autoregressive text generation. Instead, the model evaluates segment separation tokens to generate class probabilities, determining whether candidate provisions logically entail, contradict, or remain neutral relative to governing law.

Probability Calibration and Ensembling. To maximize predictive reliability and mitigate single-model bias, the framework incorporates probability aggregation across diverse model checkpoints or hyperparameter variations. Averaged or calibrated posterior probabilities exceeding decision thresholds indicate detected normative conflict with formal statistical backing.

2.12.4 Theoretical Synthesis

This synthesis integrates the foundational theoretical models reviewed throughout this chapter into a unified framework for ex-ante legislative conflict detection. While each individual methodology addresses an isolated phase of legal document processing, their conceptual integration establishes a principled retrieve-then-entail framework:

Dense Retrieval Methodology. The theoretical foundation for statutory candidate discovery draws from the dense retrieval methodology established by the **AIIR Lab** [220, 106]. Their approach demonstrates that targeted domain augmentation mitigates OCR noise in archival legal text, while continuous semantic vectorization enables high-throughput candidate identification across large statutory collections without relying on rigid keyword indices.

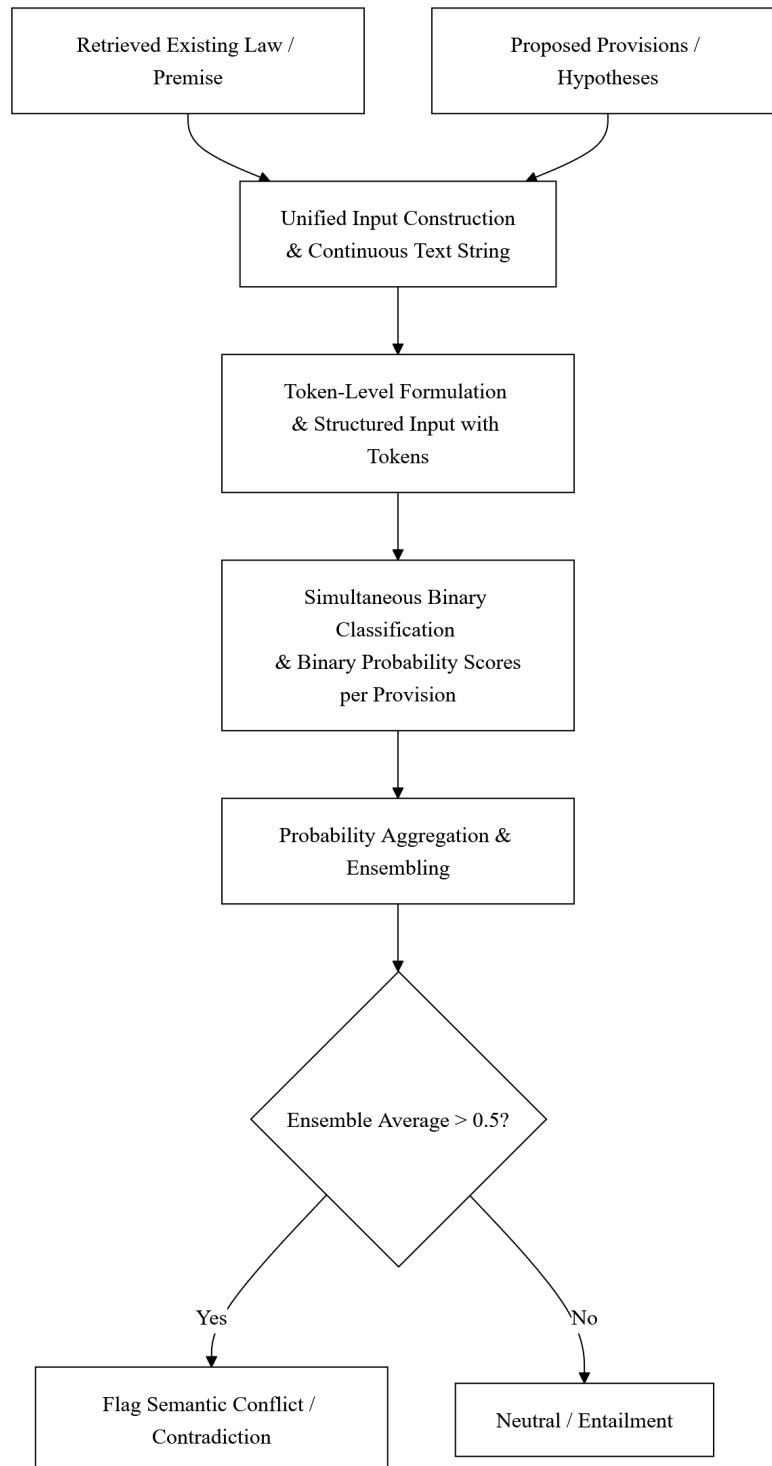


Figure 2.3: Phase 2 - Statutory Textual Entailment (NLI Pipeline). Adapted from Team KIS's ModernBERT rationale extraction methodology.

Ensemble Learning Strategy. To balance candidate recall against precision, the framework incorporates the multi-stage ranking principles demonstrated by **Team JNLP** [67, 66]. This methodology utilizes lexical filtering (BM25) to prune extraneous documents, followed by neural scoring and rank fusion to isolate authoritative candidate enactments on standard computing hardware.

Textual Entailment. For granular deontic conflict evaluation, the framework adopts the **Team KIS framework** for textual entailment [93]. Their methodology formats premise-hypothesis pairs into a continuous sequence evaluated via lightweight cross-encoders, utilizing token-level separation boundaries to classify logical relationships without generative hallucination risks.

Synthesis and Benchmark Consensus. Synthesizing these foundational frameworks establishes an end-to-end theoretical foundation for automated preemption auditing. The retrieval stage narrows extensive statutory search spaces down to high-probability candidate provisions, while the textual entailment stage conducts deterministic normative conflict checks. The latest findings from the COLIEE 2026 proceedings [68, 133] reinforce this architectural consensus, validating that pipelined retrieval and textual entailment provide the most reliable foundation for automated legal consistency analysis.

Chapter 3

Methodology

This chapter explains the methodology used to build and evaluate the Coarse-to-Fine Semantic Conflict Detection System. It describes how Philippine legal texts are collected and cleaned, and how the two-stage Retrieve-then-Entail architecture is organized. The chapter also details the creation of the ground truth dataset and the selection of candidate models. Finally, it outlines the evaluation metrics and hardware constraints, which are chosen to match the computing capacity of an average Local Government Unit (LGU).

3.1 Conceptual Framework

This section maps how data flows through the proposed system. The architecture follows an Input-Process-Output (IPO) design organized into a two-stage “Coarse-to-Fine” retrieve-then-entail pipeline (Figure 3.1). By decoupling broad statutory candidate retrieval from compute-intensive textual entailment classification, the framework overcomes the high computational demands and memory limits that traditionally impede local legal NLP deployment. The workflow is divided into two operational phases: Phase 1 (Offline Foundation and Optimization) and Phase 2 (Live Online Retrieve-then-Entail Pipeline).

3.1.1 Phase 1: Offline Foundation and Optimization

This preliminary phase establishes the empirical foundation of the system by compiling the statutory corpus, building pre-computed retrieval indexes, and selecting candidate transformer logic engines through a structured benchmark screening process.

During dual-corpus ingestion and pre-computation, the system compiles a comprehensive dual-jurisdiction statutory archive combining the Philippine national statutory collection (Republic Acts, Executive Orders, Presidential Decrees, Acts, Batas Pambansa, and Commonwealth Acts) with Davao City municipal legislation (historical Landmark Ordinances and digital LISSP municipal records). The compiled corpus undergoes legal text normalization, optical character recognition (OCR) noise mitigation on historical scans, and structural section segmentation into granular, self-contained regulatory units (~ 200 – 300 tokens). Cleaned statutory provisions are then indexed offline: dense seman-

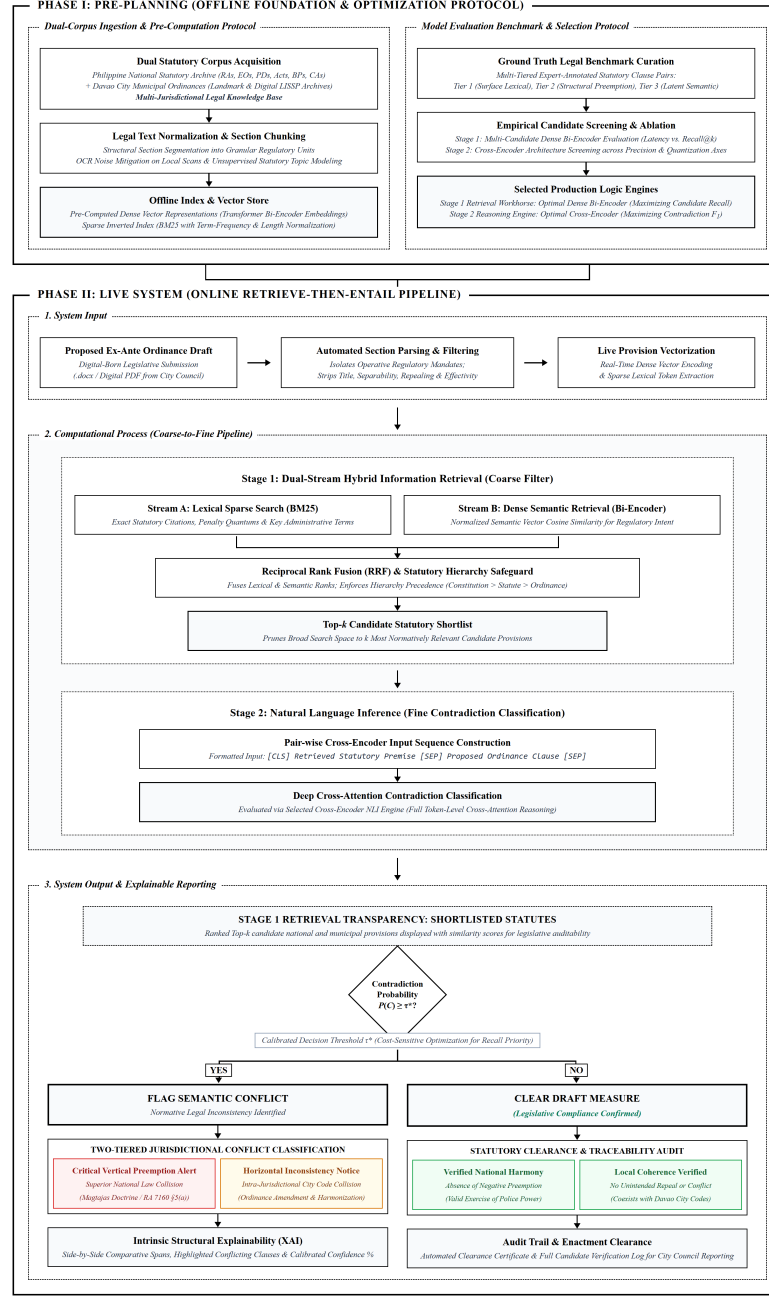


Figure 3.1: Detailed Information Flow of the Coarse-to-Fine Semantic Conflict Detection Architecture.

tic representations are pre-computed using a lightweight transformer bi-encoder, while sparse inverted lexical indexes are constructed using BM25 ($k_1 = 1.5, b = 0.75$) alongside unsupervised statutory topic modeling.

Following corpus indexing, model benchmarking and selection evaluates candidate neural architectures against a curated Ground Truth Legal Benchmark stratified across three granular difficulty tiers: Tier 1 (Surface Lexical), Tier 2 (Structural Preemption), and Tier 3 (Latent Semantic). A multi-stage empirical screening and ablation process eval-

uates candidate bi-encoders by balancing inference latency against candidate Recall@ k , alongside candidate cross-encoders evaluated across precision, latency, and quantization axes. Based on these empirical evaluations, the highest-performing models are selected as the production engines: an optimal dense bi-encoder for Stage 1 candidate retrieval, and an optimal fine-tuned cross-encoder for Stage 2 contradiction reasoning.

3.1.2 Phase 2: Live System Workflow

The live system executes the automated conflict detection pipeline when a newly drafted municipal measure is submitted for pre-enactment review, operating through an Input-Process-Output structure.

The workflow begins with system input ingestion, where a legislative researcher uploads a newly proposed ex-ante ordinance draft. Because draft measures originate within local government offices as digital-born documents (.docx or digital PDF), they are free from optical scanning noise. The ingestion pipeline segments the draft into individual substantive provisions, filtering out non-regulatory procedural boilerplate such as the Title, Separability, Repealing, and Effectivity clauses. The isolated substantive provisions are then vectorized in real time via the selected dense bi-encoder alongside sparse lexical token extraction.

The computational process core then executes two complementary stages: hybrid candidate retrieval followed by natural language inference. In Stage 1 (Dual-Stream Hybrid Information Retrieval), the system narrows the comprehensive statutory search space down to a focused candidate pool. Stream A executes lexical sparse search via BM25 to match exact statutory citations, numerical penalty amounts, and administrative agency terms, while Stream B executes dense semantic search via the selected bi-encoder, measuring cosine similarity between the draft provision and statutory embeddings to capture abstract regulatory intent. The ranked outputs are combined using Reciprocal Rank Fusion (RRF) integrated with a Statutory Hierarchy Safeguard prioritizing higher-ranking national statutes over municipal provisions, extracting the Top- k candidate provisions without overburdening local LGU compute resources.

In Stage 2 (Natural Language Inference), each retrieved statutory candidate is paired with the draft ordinance provision into a single cross-encoder sequence:

$$\mathbf{x} = [\text{CLS}] \text{ Statutory Premise } [\text{SEP}] \text{ Ordinance Hypothesis } [\text{SEP}] \quad (3.1)$$

The unified sequence is processed by the selected fine-tuned cross-encoder NLI engine, which performs full inter-token cross-attention to classify the directional logical relationship across Entailment, Neutral, or Contradiction, producing a calibrated contradiction probability $P(\text{Contradiction})$.

Finally, system output and explainable reporting presents an audit trail displaying the Top- k shortlisted candidate statutes for human oversight before evaluating contradiction probability against a calibrated decision threshold τ^* , optimized for recall priority via F_2 -score weighting. When the contradiction probability meets or exceeds the threshold ($P(\text{Contradiction}) \geq \tau^*$), the system flags a normative legal inconsistency and routes

it through a two-tiered jurisdictional classification: Critical Vertical Preemption Alerts signaling collisions with superior national statutes under the *Magtajas* doctrine, and Horizontal Inconsistency Notices highlighting collisions or uncoordinated overlaps with prior Davao City municipal ordinances. To ensure decision-support transparency, the system renders intrinsic structural explainability featuring side-by-side comparative spans, highlighted conflicting phrases, and calibrated confidence percentages. Conversely, when contradiction probabilities remain below the calibrated threshold ($P(\text{Contradiction}) < \tau^*$), the system issues a formal legislative compliance confirmation, accompanied by a statutory clearance audit confirming both national statutory harmony and local municipal code coherence.

3.2 Research Design

This study uses an applied research design to build a practical software tool for local governance. The researchers use an experimental systems development approach to solve the delays and difficulties of manual ordinance review through automated detection. This setup connects principles from legal informatics and natural language processing to address a real local need. By focusing on practical transformer-based models, the design connects theoretical computer science with everyday government office work. The experiment directly tests whether a retrieve-then-entail pipeline can operate reliably in a local legislative setting.

The experimental setup tests the retrieval and classification modules separately to measure search speed and accuracy independently. Building the system step-by-step allows the researchers to adjust the information retrieval and natural language inference stages using local legal texts. This engineering process ensures that the finished tool accounts for the writing styles found in both city ordinances and national statutes. In this way, applied research methods help turn modern NLP concepts into a usable tool for the Davao City Sangguniang Panlungsod. As a result, the findings establish a clear benchmark for automating pre-enactment legal checks on local legislation.

3.3 Corpus Acquisition and Exploratory Analysis

Building the legal database requires collecting text from different public repositories so that the system includes as many relevant laws as possible. These sources provide the national statutes and city ordinances needed to check draft legislation for consistency. To keep the language consistent and make vector processing simpler, the system gathers only the English-language versions of all legal texts. This choice matches Philippine statutory rules under Section 20, Chapter 5, Book I of Executive Order No. 292 (Administrative Code of 1987) [153], which explicitly mandates that in the interpretation of laws promulgated in official languages, the English text shall control unless otherwise specifically provided. National statutes and local ordinances are gathered using separate automated scripts designed for the format of each source.

3.3.1 Corpus Composition

The Lawphil Project (Arellano Law Foundation) serves as the primary digitized extraction source for national statutory text because its documents are provided in clean HTML, follow a consistent layout, and offer broad historical coverage [8]. Automated scraping scripts browse through chronological and alphabetical indexes to collect Republic Acts, Batas Pambansa, Commonwealth Acts, Acts, Executive Orders, and Presidential Decrees. Using a custom web extraction pipeline built by the proponents, statutory extraction was conducted as of **March 2026**, collecting 25,432 unique legal records enacted between 1900 and 2026. The Official Gazette of the Republic of the Philippines [73] and the Chan Robles Virtual Law Library were used as authoritative secondary cross-references to verify enactment dates, chronological numbering, and historical amendment status. All collected national laws are converted into standardized JSON records with metadata, such as dates of approval, titles, and section numbers. Listing 3.1 illustrates the web scraping and provision extraction logic implemented to retrieve and structure the statutory archive. As illustrated in Figure 3.2, the Lawphil repository presents statutory text in clean, unstyled HTML typography with a predictable hierarchical URL schema (e.g., `/statutes/repacts/ra2018/ra_11035.2018.html` for Republic Act No. 11035). This standardized structure separates introductory legislative headers from operative provisions and eliminates the need for dynamic JavaScript browser rendering, facilitating rapid, fault-tolerant web scraping.

```

1  def extract_statutory_corpus(base_url, start_year=1900, end_year=2026):
2      """
3      Extracts, sanitizes, and structures national statutes from the online archive.
4      """
5      statute_records = []
6
7      for year in range(start_year, end_year + 1):
8          year_index = fetch_page(f"{base_url}/year/{year}")
9          statute_links = parse_index_links(year_index)
10
11         for link in statute_links:
12             # Fetch raw web page and parse HTML DOM tree
13             html_content = fetch_page_with_retry(link, delay_seconds=1.0)
14             soup = BeautifulSoup(html_content, "html.parser")
15
16             # Extract statutory metadata and clean document body
17             metadata = extract_header_metadata(soup)
18             body_text = clean_document_body(soup)
19
20             # Segment statute into discrete section-level provisions
21             sections = segment_into_sections(body_text)
22
23             record = {
24                 "statute_id": metadata.id,
25                 "title": metadata.title,
26                 "instrument": metadata.type,
27                 "date_enacted": metadata.date,
28                 "sections": sections
29             }
30             statute_records.append(record)
31
32     return statute_records
33

```

Listing 3.1: Code Snippet of the Statutory Corpus Acquisition and Provision Extraction Pipeline



Figure 3.2: Sample Statutory Enactment Webpage from the Lawphil Project Repository (Republic Act No. 11035, *Balik Scientist Act*). The repository features a streamlined, clean HTML document layout without intrusive styling or dynamic client-side rendering. Furthermore, its transparent and predictable URL hierarchy (`/statutes/repacts/ra2018/ra_11035_2018.html`) systematically organizes enactments by instrument type, legislative year, and statutory numbering, enabling automated, fault-tolerant corpus harvesting.

Collecting national laws from Lawphil helps avoid the disorganization common to public legal websites. Although the Official Gazette and congressional websites also host statutory records, their web pages frequently change layout and often provide older laws only as image-only PDF scans. In contrast, Lawphil keeps a standard HTML structure across all historical eras, clearly separating introductory preambles from operative legal sections. Scraping from this repository ensured uniform text formatting across the national collection (over 25,000 enactments), preventing formatting differences from skewing vector embeddings during Stage 1 retrieval. The resulting collection provides a clean, consistent text foundation for semantic analysis and conflict detection.

To capture the complete statutory search space, the national collection is paired with Davao City municipal legislation gathered across two complementary channels:

1. **Davao City SP Public Web Portal:** Provides selective historical “Landmark Ordinances” and ad-hoc municipal uploads spanning approximately fifteen years, capturing prominent regulatory codes in public safety, zoning, and health [168].
2. **Legislative Information Support System Program (LISSP):** Acquired directly through formal administrative coordination with the Sangguniang Panlungsod Secretariat and Information Technology division, who officially granted the proponents access to the LISSP documents via a shared Google Drive repository containing digitized enactments primarily from the 19th and 20th City Councils between 2021 and 2025 [31].

Together, these two channels yielded an archive of exactly 1,664 municipal enactments spanning from 1950 to 2025. These 1,664 documents represent the complete accessible digital corpus gathered within the proponents’ institutional capabilities. In practice, the true cumulative historical total of ordinances enacted since Davao City was chartered under Commonwealth Act No. 51 in 1936 remains unrecorded on the internet and unknown to municipal authorities. In institutional consultations, Sangguniang Panlungsod personnel confirmed that an official cumulative ledger does not exist because local legislation is numbered sequentially per calendar year and council term rather than in an unbroken cumulative sequence. Furthermore, the vast majority of historical enactments, particularly those prior to the 2021 LISSP digital initiative, exist exclusively as physical paper documents preserved in archive rooms and filing cabinets at the Sangguniang Panlungsod building on San Pedro Street, requiring manual page-by-page physical inventorying to tally. When combined with the 25,432 national statutes, the total statutory search space comprises **27,096 legal enactments** ($N_{\text{sections}} = 176,421$).

3.3.2 Corpus Biases

A close look at the collected national laws reveals a large imbalance across legal topics. Over 40% of the national statutes consist of routine administrative actions, such as converting local schools, renaming roads, adjusting municipal boundaries, or granting broadcast franchises. In contrast, major regulatory areas that interact with city powers, such as public health, land zoning, and local taxation, make up less than 15% of the total national collection. Searching through this raw distribution without filtering would cause search algorithms to return irrelevant administrative laws instead of actual regulatory rules. To keep search results useful, the system must separate these routine administrative laws from substantive regulatory provisions.

The municipal archive also shows an uneven time distribution and gaps in older records. For ordinances passed before 2021, city record-keeping mainly focused on digitizing major “Landmark Ordinances,” leaving ordinary regulations from earlier decades in physical paper storage [168]. As a result, the pre-2021 local records represent a selective collection of prominent laws rather than a complete legislative record. While post-2021 records from the digital LISSP database are much more complete, some numerical gaps remain due to missing committee drafts and unindexed attachments. These temporal biases mean that the available municipal archive contains substantially more recent ordinances than older enactments.

The physical state of the scanned city ordinances adds both visual and linguistic noise. Unlike national laws extracted from clean HTML repositories or digital-born text exported directly from word processors, the raw municipal records consist entirely of scanned physical paper PDFs. Older ordinances scanned at low resolution suffer from faded print, ink smudges, and crooked pages, as illustrated in Figure 3.3. The paper copies also contain non-text items such as government stamps, official signatures, and handwritten notes that confuse sentence splitting tools. At the language level, local ordinances sometimes include regional terms and administrative Taglish in their introductory sections [54]. These informal drafting habits differ noticeably from the formal, standardized English found in national laws.

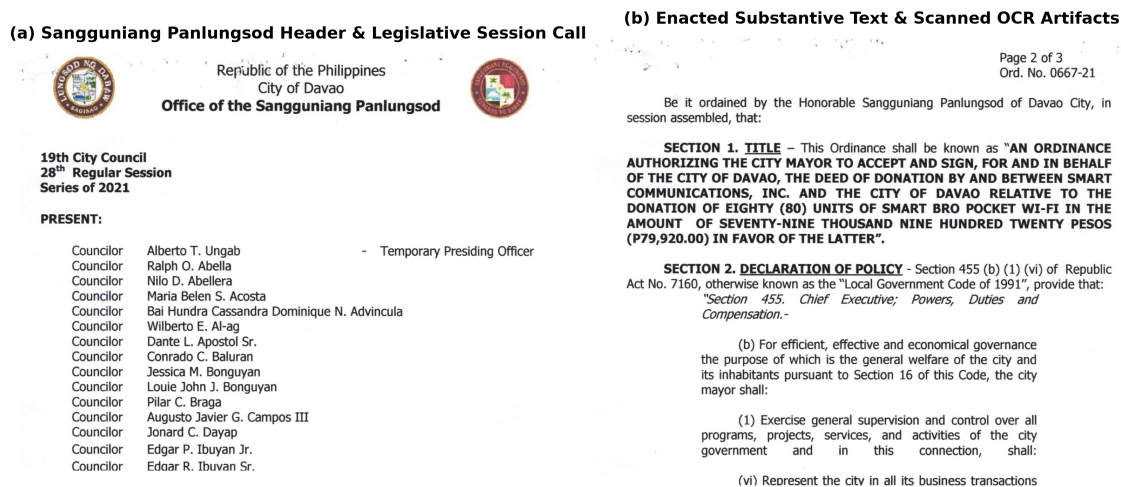


Figure 3.3: Sample Archival Document from the Davao City Sangguniang Panlungsod Corpus Illustrating Physical Scanning Artifacts and Optical Character Recognition (OCR) Noise. Panel (a) illustrates the official municipal session heading, seal, and city council metadata. Panel (b) shows the enacted substantive provisions exhibiting real OCR scanner noise, character substitutions (e.g., “0667-2L” for 2021, “TITTE” for Title), and document skew.

In real-world legislation, direct contradictions between laws are very rare. Law-makers generally try to write rules that work alongside existing higher laws rather than directly opposing them [62]. As a result, actual legal conflicts make up less than 1% of random pairs formed between municipal ordinances and national statutes. Taking random pairs of text from the unedited collections would lead to an overwhelming majority of unrelated pairs. A machine learning model trained on such unbalanced data could score 99% nominal accuracy simply by guessing “Neutral” every time, while remaining unable to detect real legal conflicts.

3.3.3 Philippine National Statutory Corpus Analysis

Across the five main types of legislation with sequential numbers, the collected corpus includes 19,683 statutes out of an estimated 20,246 historical laws, achieving a coverage rate of 97.22%. When combined with the 5,749 Executive Orders from fifteen presidential

administrations, the compiled national archive holds 25,432 out of about 26,451 total historical legal documents, reaching a national recall rate of 96.15%. All official legislative totals reflect the chronological numbering sequence established by Congress and presidential records as of **March 2026**, where sequential Republic Acts reached Republic Act No. 12,313. Table 3.1 provides a breakdown of the national corpus, comparing the collected documents with official legislative totals for each statutory type. Figure 3.4 shows the share of each legal instrument in the archive, illustrating the large volume of executive orders alongside congressional acts.

A key methodological decision in compiling this database is the explicit inclusion of **Executive Orders (EOs)** alongside congressional legislation. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III of Executive Order No. 292 [153], Executive Orders are presidential acts providing rules of a general character in execution of statutory mandates. Unlike Supreme Court decisions that settle private adversarial controversies under *stare decisis*, Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force of law. Modern Executive Orders establish binding nationwide regulatory mandates, such as Executive Order No. 26 (s. 2017) mandating nationwide public smoking bans, that legally preempt inconsistent local ordinances. Excluding Executive Orders would create an artificial regulatory gap, allowing local ordinances to contradict operative presidential mandates.

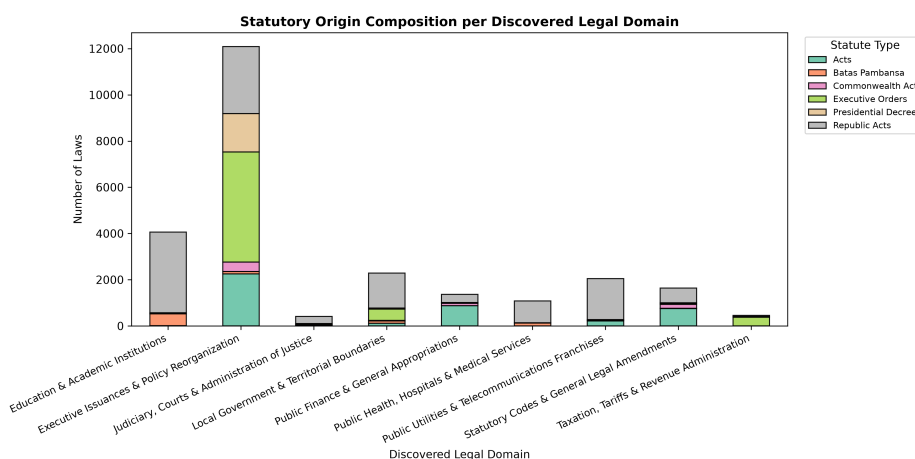


Figure 3.4: Composition of the Philippine National Statutory Corpus by Legislative Instrument ($N = 25,432$).

The coverage rates confirm that the national legal archive is nearly complete across all historical governance eras. Commonwealth Acts show full 100.00% coverage across all 733 enactments, while American-era Acts (99.58%) and Batas Pambansa (99.66%) miss only 18 and 3 enactments, respectively. For Republic Acts, the 11,963 collected records cover 97.16% of the 12,313 official total, with missing entries mainly being local bills or recent laws still being indexed. For Presidential Decrees, the 1,844 gathered records represent 90.57% of the 2,036 numbered decrees from the Martial Law period, where uncollected entries were unpublished directives until the Supreme Court required full publication in *Tañada v. Tuvera* [185]. For Executive Orders, the 5,749 collected records

Table 3.1: Comparative Statutory Coverage and Composition of the Philippine National Legal Archive ($N = 25,432$)

Statutory ID	Instrument	Corpus	Official Total	Coverage Rate	Corpus Share	Historical Era and Governing Authority
01	Republic Acts (RA)	11,963	12,313	97.16%	47.04%	Post-war and Fifth Republic legislation (1946–present, up to RA 12313 as of March 2026)
02	Executive Orders (EO)	5,749	6,205*	92.65%	22.61%	Presidential administrative issuances (1901–2026, across 15 administrations)
03	Acts	4,257	4,275	99.58%	16.74%	American Insular Government enactments (1900–1935)
04	Presidential Decrees (PD)	1,844	2,036	90.57%	7.25%	Martial law legislative decrees enacted by executive fiat (1972–1986)
05	Batas Pambansa (BP)	886	889	99.66%	3.48%	Unicameral parliamentary legislation under the 1973 Constitution
06	Commonwealth Acts (CA)	733	733	100.00%	2.88%	Transitional governance enactments under the 1935 Constitution
Sequential Subtotal		19,683	20,246	97.22%	77.39%	All numbered legislative enactments of Congress
Total Archive		25,432	26,451	96.15%	100.00%	Comprehensive corpus extracted from the Lawphil Project

*Historical total for Executive Orders represents the aggregate sum of maximum numbered issuances across all fifteen presidential administrations (1935–2026) and colonial governorships (1901–1935) as recorded in the Official Gazette and Lawphil archives as of March 2026.

represent 92.65% of the 6,205 numbered orders issued across fifteen presidential terms, offering an extensive record of administrative policy.

This institutional breakdown also explains why executive and administrative orders account for nearly half (48.01%) of the subsequent topic groups. Executive Orders (22.61%) and Presidential Decrees (7.25%) together make up 7,593 documents, representing nearly one-third of the entire national archive. When combined with colonial-era civil administration Acts and agency reorganizations, administrative directives form the largest single category of Philippine legal text. In language processing, this large volume of administrative text poses a challenge during search, as queries can easily match routine bureaucratic language rather than municipal regulations. This computational reality demonstrates why automated topic grouping is necessary to keep local ordinance checks focused on relevant laws.

To trace how the six statutory instruments evolved across Philippine jurisprudence, the proponents categorized statutory enactments across six chronological eras. Each historical era established distinct legislative authorities, ranging from colonial commissions to modern constitutional congresses. Figure 3.5 charts the chronological succession and aggregate volume of statutory instruments from 1900 through 2026. The distribution

reflects the shifting primacy of legislative acts, presidential decrees, and administrative executive orders across political transitions. Examining these enactment waves illustrates how historical governance structures directly shaped the composition of the national legal corpus.

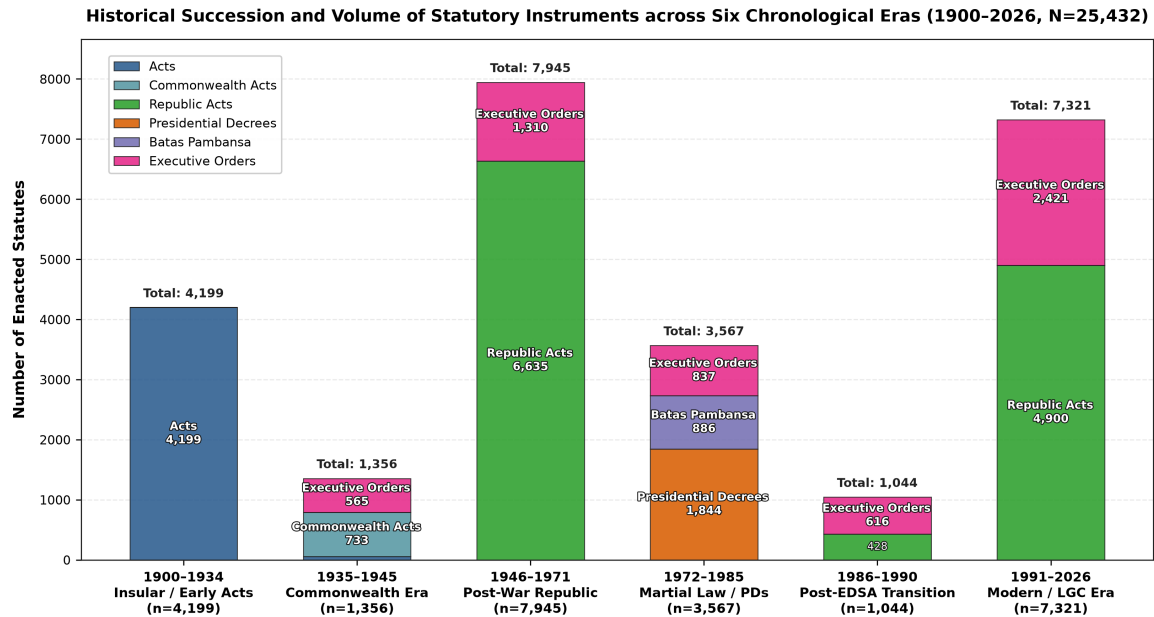


Figure 3.5: Historical Succession and Volume of Statutory Instruments across Six Chronological Eras (1900–2026, $N = 25,432$).

The temporal breakdown reveals that Philippine lawmaking expanded during periods of national administrative reorganization. Early colonial governance relied entirely on 4,199 foundational Acts during the Insular period (1900–1934). Centralized presidential lawmaking predominated during the Martial Law period through 1,844 Presidential Decrees and 886 Batas Pambansa parliamentary enactments (1972–1985). In the modern era governed by Republic Act No. 7160 (1991–2026), statutory production stabilized into a dual stream of 4,900 congressional Republic Acts and 2,421 executive policy orders. Because vocabulary standards shifted markedly across these legislative transitions, search models require chronological awareness to avoid misinterpreting historical statutory phrasing.

During preliminary exploratory research, the proponents conducted initial machine learning experiments on the national statutory archive and identified severe dataset imbalances. In the national collection of 25,432 statutes, routine executive issuances and administrative reorganizations comprise nearly half of all historical enactments, while specialized regulatory domains remain relatively small. Searching for a local regulatory ordinance across this raw, unfiltered collection causes severe search drift because dense retrieval models readily match repetitive administrative phrasing across unrelated legal areas. To resolve these dataset quirks, the proponents formalized a two-stage topic modeling architecture that organizes the statutory archive into cohesive, well-separated subject areas. This structured categorization prevents routine administrative directives from

crowding the candidate shortlist during coarse retrieval, establishing a balanced foundation for subsequent conflict checks.

The topic classification pipeline groups documents automatically while executing efficiently on the Primary Desktop Host. Statutory texts are first stripped of HTML tags and filtered against a curated list of 49 legal stopwords, such as “hereby” and “enacted,” which appear ubiquitously without distinguishing legal subjects. Cleaned texts are converted into term frequency-inverse document frequency (TF-IDF) feature vectors across a 25,000-term vocabulary, rewarding distinctive statutory terminology over frequent administrative vocabulary [122]. To maintain computational efficiency while preserving conceptual cohesion, the pipeline applies Truncated Singular Value Decomposition (SVD) to project the sparse feature vectors into 64 latent semantic dimensions [43]. Following the latent semantic modeling principles of Landauer et al. [104], this 64-dimensional subspace captures core semantic associations while eliminating lexical noise, followed by standard L_2 vector normalization.

Partitioning statutory texts directly into a small set of broad categories risks conflating heterogeneous legislative subjects. To prevent premature cluster mixing, the normalized 64-dimensional vectors are initially partitioned into $K = 28$ micro-clusters using MiniBatch K -Means clustering [174]. Following the over-clustering strategy recommended by Grootendorst [75] and Aggarwal [2], this over-complete partition isolates narrow recurring enactments, such as franchise charters or school conversions, into tight homogenous groups. The pipeline then adapts class-based TF-IDF (c -TF-IDF) from Grootendorst [75] to identify salient keywords by treating all text within a micro-cluster as a unified document:

$$c\text{-TF-IDF}_{i,c} = \frac{|t_{i,c}|}{|c|} \times \log \left(1 + \frac{A}{\sum_k |t_{i,k}|} \right)$$

where $|t_{i,c}|$ denotes the frequency of term i in cluster c , $|c|$ is the total word count of cluster c , and A indicates the average word count per cluster across the archive. This cluster-level weighting elevates terminology that specifically characterizes a legislative group while penalizing ubiquitous statutory phrasing.

To align mathematical clusters with operative jurisprudence, the extracted c -TF-IDF profiles are mapped against regulatory competencies defined in the Local Government Code of 1991 [71]. Micro-clusters sharing congruent regulatory functions are merged into consolidated macro domains, requiring each domain to contain at least 1.0% of the corpus ($N \geq 254$ statutes) to ensure statistical stability [2]. Outlier micro-clusters falling below this volume threshold are reassigned to their nearest semantic centroid, producing balanced clusters suitable for coarse indexing:

$$\mathbf{C}_m = \frac{\sum_{i \in \mathcal{D}_m} \mathbf{z}_i}{\|\sum_{i \in \mathcal{D}_m} \mathbf{z}_i\|_2}, \quad P(m \mid \mathbf{z}_i) = \frac{\exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_m))}{\sum_j \exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_j))}$$

where $\mathbf{z}_i$ represents the normalized 64-dimensional SVD vector of document i , $\mathbf{C}_m$ denotes the normalized centroid of macro domain m , and $\tau = 6.0$ functions as the inverse temperature scaling parameter. Adapting temperature scaling principles from Guo et al. [77], the parameter $\tau = 6.0$ prevents uncalibrated softmax dispersion across cosine distances,

ensuring that dominant domain probabilities remain sharply separated. Executing this consolidation across all 25,432 national statutes organizes the historical collection into eight Consolidated Macro Legal Domains, as detailed in Table 3.2 and illustrated in Figure 3.6.

Table 3.2: Consolidated Macro Legal Domains of the Philippine National Statutory Corpus ($N = 25,432$)

ID	Consolidated Macro Domain	Statutes	Share	Top Salient $cTF-IDF$ Keywords
00	Executive Issuances & Policy Reorganization	12,210	48.01%	executive, commission, committee, policy, code
01	Education & Academic Institutions	3,492	13.73%	high school, elementary school, campus, college
02	Local Government & Territorial Boundaries	2,134	8.39%	barangay, municipality, boundary, sitio, territorial
03	Public Utilities & Telecom Franchises	2,041	8.03%	franchise, broadcast, telecommunications, radio
04	Public Health, Hospitals & Medical Services	1,890	7.43%	hospital, bed capacity, medical center, infirmary
05	Statutory Codes & General Legal Amendments	1,873	7.36%	administrative code, revised, amend, article, penalty
06	Public Finance & General Appropriations	1,335	5.25%	appropriation, treasury, funds, expenditures
07	Taxation, Tariffs & Revenue Administration	457	1.80%	tariff, customs code, internal revenue, excise
Total Consolidated National Corpus		25,432	100.00%	All Enacted Statutes (1900–2026)

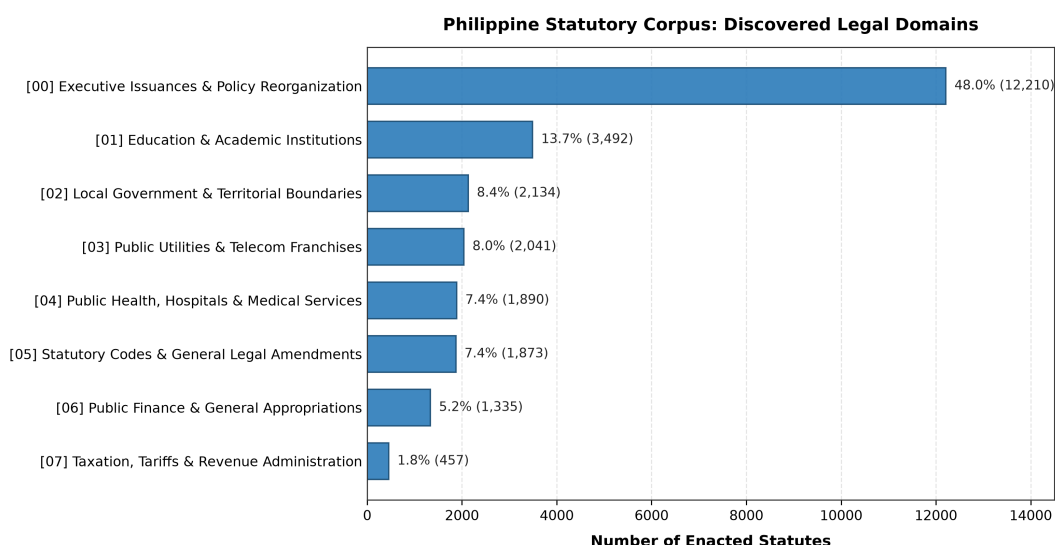


Figure 3.6: Distribution of the 25,432 Philippine National Statutes across the Consolidated Macro Legal Domains.

The topic distribution reflects the historical patterns of Philippine lawmaking. Executive issuances, administrative orders, and presidential decrees make up nearly half of

the entire statutory collection, reflecting centralized authority during past political transitions. Common municipal subjects, including school infrastructure (13.73%) and public utility franchises (8.03%), form large distinct clusters with repetitive drafting patterns. In contrast, key regulatory topics like taxation (1.80%) and public health (7.43%) occupy smaller, highly focused portions of the database. Separating these regulatory clusters keeps routine administrative orders from crowding the candidate list during Stage 1 retrieval.

To examine the geometric separation of these legal topics, the proponents projected the 64-dimensional SVD document vectors into a two-dimensional latent semantic landscape. Dimensionality reduction maps document similarities onto coordinate axes, allowing direct visual inspection of cluster cohesion. Figure 3.7 illustrates this topic landscape across a representative stratified sample ($N = 3,500$), highlighting the core clusters and their spatial distributions. Specialized regulatory domains project outward as distinct, isolated rays along the periphery of the vector space. Enactments establishing academic institutions and telecommunications franchises form tight, cohesive rays with minimal dispersion.

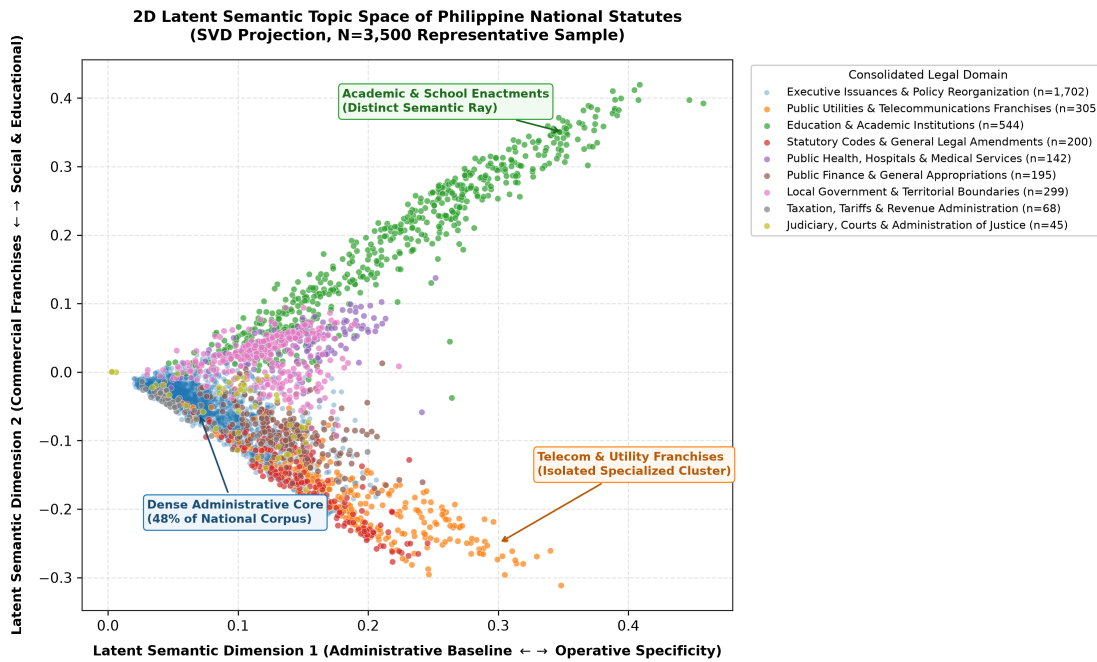


Figure 3.7: Two-Dimensional Latent Semantic Topic Space of the Philippine National Statutory Corpus (SVD Projection, $N = 3,500$ Representative Sample).

In contrast, administrative issuances and executive orders cluster densely near the origin, comprising almost half of the entire archive. This dense central mass explains why unconstrained semantic searches frequently drift toward routine bureaucratic orders unless restrained by hierarchical safeguards. The coordinate axes reflect the two primary orthogonal directions of vocabulary variation discovered through singular value decomposition. The horizontal axis measures operative statutory specificity against general administrative phrasing, where higher positive values indicate detailed technical mandates rather

than procedural orders. Conversely, the vertical axis captures functional divergence between social infrastructure such as public schools and commercial infrastructure such as telecommunications franchises.

To confirm that the mathematical clusters reflect substantive legal categories rather than shared administrative boilerplate, the proponents examined the most salient vocabulary terms discovered by the model. Words that appear frequently across all documents fail to differentiate legal topics, requiring class-based weightings to isolate subject matter. Figure 3.8 displays the top class-based TF-IDF ($c\text{-}TF\text{-}IDF$) keywords across all eight consolidated macro domains. The vocabulary profiles confirm that each macro domain is defined by operative legal terminology unique to its statutory field. Specialized clusters such as public health are dominated by concrete institutional terms, including “hospital” and “bed capacity,” while territorial enactments display spatial markers such as “sitio” and “boundary.”

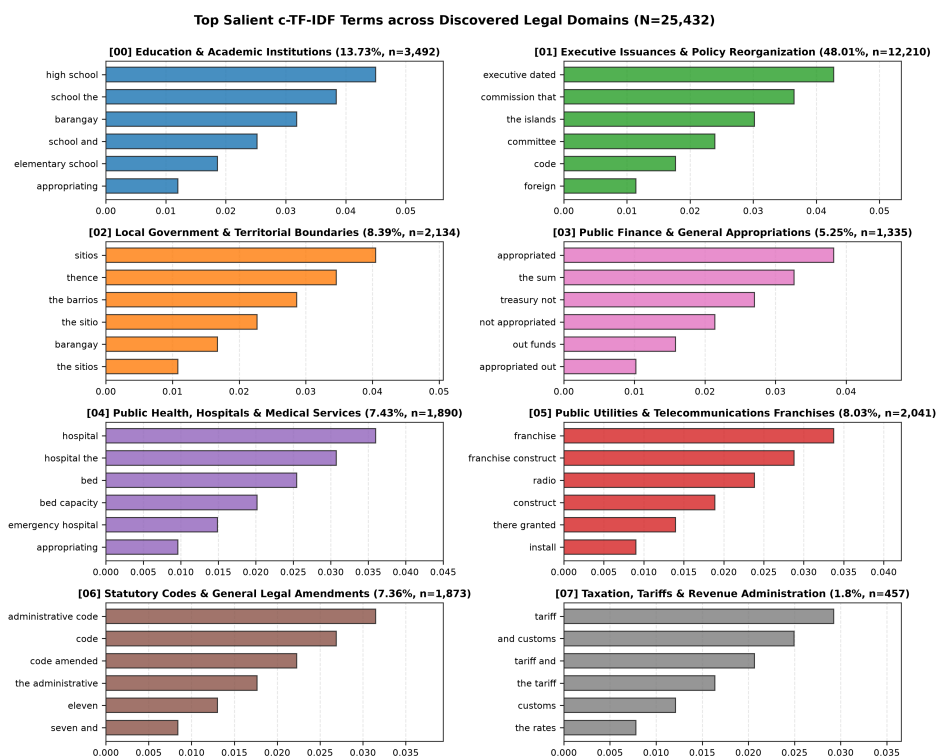


Figure 3.8: Top Salient $c\text{-}TF\text{-}IDF$ Keyword Profiles across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

To examine how individual lawmaking authorities contribute to these eight subject areas, the proponents mapped the six statutory instruments across the consolidated macro domains. In Philippine legislative history, different legal instruments serve distinct constitutional functions, ranging from broad congressional acts to presidential executive directives. Figure 3.9 presents this cross-allocation matrix, detailing both absolute enactment counts and percentage distributions for each statutory source. This cross-tabulation reveals an explicit division of legal responsibility across Philippine statutory instruments. Analyzing this allocation confirms that regulatory subjects concentrate within specific

legislative formats rather than distributing evenly across the archive.

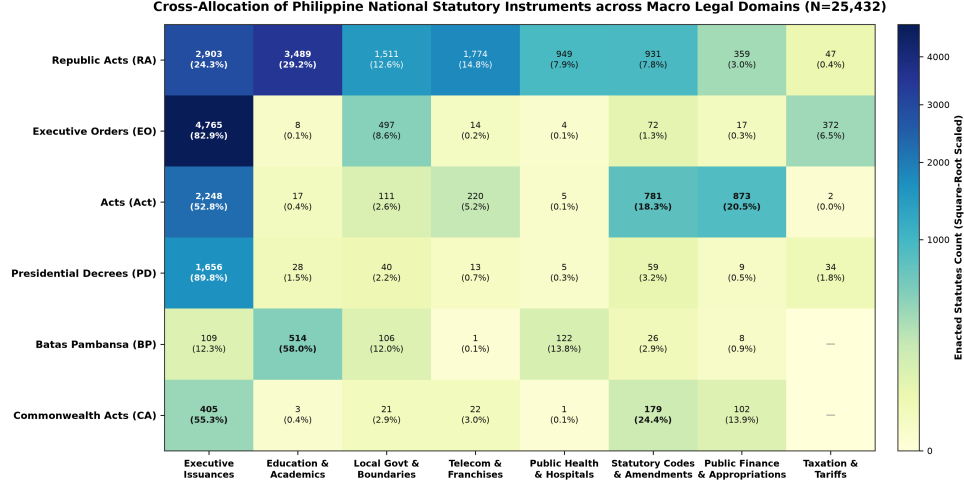


Figure 3.9: Cross-Allocation Matrix of the Six National Statutory Instruments across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

The cross-allocation matrix reveals that congressional enactments and executive orders govern entirely separate legal spheres. Republic Acts account for nearly all specialized regulatory measures, containing 99.9% of academic enactments and 86.9% of telecommunications franchises. In contrast, Executive Orders concentrate heavily within administrative policy (82.9%) and presidential tariff adjustments (81.4%). Because municipal ordinances regulate local community behavior rather than internal executive operations, matching draft local clauses against executive orders produces high rates of irrelevant candidate retrievals. These empirical disparities validate using statutory source indicators and domain filters during Stage 1 retrieval to keep candidate shortlists focused on genuine regulatory mandates.

To confirm that the topic pipeline discovered authentic legal boundaries rather than grouping surface vocabulary, the proponents evaluated the model using a stratified 80/20 train/test split. The 80/20 proportion follows the standard Pareto partition for large-scale legal collections exceeding 20,000 documents [2], withholding 20% of the corpus (5,087 statutes) to test generalizability on unseen enactments while training the model on the remaining 80% (20,345 statutes). In statistical testing, an out-of-sample holdout of 5,087 legal texts provides high statistical power with a margin of error below $\pm 1.4\%$, making repetitive cross-validation computationally redundant as vocabulary frequencies and SVD components stabilize asymptotically at this volume. Stratified sampling preserves the natural historical proportions of the eight macro domains, ensuring that smaller regulatory categories like taxation (1.80%, $N = 457$) maintain adequate representation in the test set ($N_{\text{test}} \approx 91$) without class starvation. Table 3.3 presents the out-of-sample performance scores, default random baselines, literature targets, and empirical verdicts across all evaluation criteria.

The supervised benchmark results confirm that the discovered macro domains generalize reliably to unseen legislation. In an eight-domain classification setting, the uni-

Table 3.3: Empirical Holdout Validation Metrics, Baseline Comparisons, and Literature Targets for the Topic Modeler ($N_{\text{train}} = 20,345$, $N_{\text{test}} = 5,087$)

Validation Metric / Benchmark Test	Observed	Default Baseline	Target Range	Empirical Verdict
Out-of-Sample Supervised Accuracy	97.37%	12.50%	$\geq 85.0\%$	Strong Generalization
Out-of-Sample Macro F1-Score	96.26%	12.50%	$\geq 80.0\%$	Balanced Discrimination
Mean Test Cosine Confidence	0.5508	0.1250	> 0.4000	Strong Dominant Probability
High-Confidence Predictions (≥ 0.50)	56.97%	$< 20.0\%$	$> 50.0\%$	Clear Majority Cluster
Holdout Cosine Silhouette Score	0.1602	≤ 0.0000	0.1000–0.2000	Solid Text Separation
Holdout Davies-Bouldin Index	2.4496	> 4.0000	2.0000–3.0000	Compact Cluster Scatter
Explicit Statutory Title Probes (<i>Unambiguous Indicator Subsets</i> , $N = 1,259$)				
Telecom Franchise Pattern	97.8%	12.50%	$> 90.0\%$	High Precision ($N = 360$)
Hospital Bed Capacity Pattern	96.6%	12.50%	$> 90.0\%$	High Precision ($N = 203$)
School Infrastructure Pattern	92.2%	12.50%	$> 90.0\%$	High Precision ($N = 645$)
Judiciary & Court Pattern	72.5%	12.50%	$> 70.0\%$	Acceptable ($N = 51$, embedded laws)

form proportional chance criterion established by Morrison [127] yields a default random baseline accuracy of only 12.50% (1/8). In contrast, the validation classifier reached an out-of-sample accuracy of 97.37% and a Macro F1-Score of 96.26%, surpassing the 85.0% target threshold typically expected for reliable document categorization [2]. To provide an external sanity check beyond learned cluster labels, the evaluation tested four explicit statutory title probes ($N = 1,259$ total) where legislative intent is unmistakable from the statutory catchline. The model achieved 97.8% precision on telecommunications franchises ($N = 360$), 96.6% on hospital bed capacity acts ($N = 203$), and 92.2% on school infrastructure laws ($N = 645$).

The probabilistic confidence metrics evaluate how decisively the model assigns unseen statutes to their respective legal domains. Across all 5,087 holdout statutes, the model achieved a Mean Test Cosine Confidence of 0.5508, exceeding the typical 0.4000 operational target and demonstrating that the top assigned category captured dominant probability mass. Furthermore, 56.97% of all unseen statutes were classified with high certainty (≥ 0.50 probability), well above the majority target threshold of 50.0%. The holdout evaluation yielded a Cosine Silhouette Score of 0.1602 and a Davies-Bouldin Index of 2.4496, well within the target range formulated by Davies and Bouldin [42]. These geometric measurements prove that the topic model establishes stable, mathematically separated legal boundaries before documents are indexed for coarse retrieval.

3.3.4 Davao City Municipal Legislative Corpus Analysis

To evaluate local legislative patterns, the proponents conducted an empirical census of all 1,664 digitized Davao City municipal ordinances. Unlike the national archive which spans over a century of centralized governance, the municipal collection documents municipal self-governance across four distinct legislative eras. As illustrated in Figure 3.10(a), the archive contains 50 foundational enactments from the Pre-LGC Era (1950–1991), 184 landmark regulatory codes from the Post-LGC Historical Era (1992–2015), and 145 enactments from the Pre-Pandemic Era (2016–2020). Statutory volume concentrates heavily within the modern digital LISSP database, contributing 1,285 enactments (77.2%) between 2021 and 2025. This temporal surge reflects the establishment of standardized electronic archiving by the Sangguniang Panlungsod, providing complete records of contemporary municipal lawmaking.

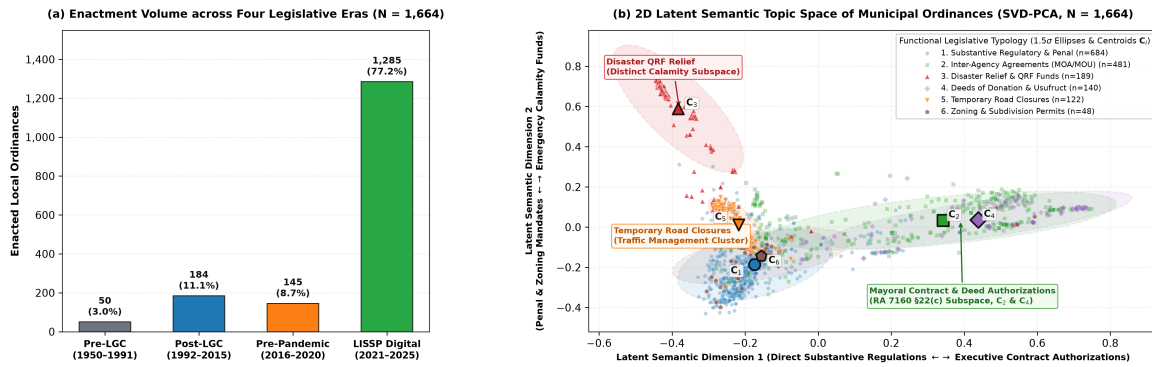


Figure 3.10: Temporal Distribution across Four Legislative Eras and Two-Dimensional Latent Semantic Topic Space of the Davao City Municipal Legislative Corpus ($N = 1,664$). Panel (a) illustrates digitized ordinance counts across historical eras, highlighting the digital archiving expansion between 2021 and 2025. Panel (b) depicts the two-dimensional latent semantic space generated through 64-component Truncated SVD, featuring 1.5 σ covariance confidence ellipses and geometric cluster centroids (C_1 – C_6) across the six discovered functional typologies.

Applying Truncated SVD to the municipal corpus revealed that city legislation divides into six functional archetypes grounded in the Local Government Code of 1991 [71]. To eliminate procedural noise before modeling, the pipeline extracts substantive operative text while setting aside routine session attendance headers. Word vectors were constructed across an 8,000-feature vocabulary and projected into 64 latent semantic dimensions, capturing 50.65% of the total vocabulary variance. Figure 3.10(b) displays the two-dimensional latent semantic topic space for the municipal collection. Emergency Quick Response Fund (QRF) calamity grants form an isolated, tight cluster in the upper-left quadrant (C_3), while temporary road closures project into a compact traffic management cluster (C_5). Inter-agency agreements (C_2) and deeds of donation (C_4) span an elongated horizontal subspace along the positive horizontal axis, corresponding to executive contract authorizations under Section 22(c) of Republic Act No. 7160. Direct police power enactments (C_1) and zoning permits (C_6) anchor the lower-left quadrant, governing local community conduct.

A critical methodological comparison between the national statutory topic landscape (Figure 3.7) and the municipal semantic space (Figure 3.10(b)) is the degree of visual dispersion. In the national corpus, over 3,500 cookie-cutter school conversions and 2,000 public utility franchises collapse into sharp, isolated radial rays due to identical statutory drafting templates. In contrast, local municipal legislation is intrinsically cross-functional under Republic Act No. 7160. Under Sections 16, 21, 22(c), 455, and 458 of the Local Government Code, a single substantive regulatory ordinance routinely incorporates budgetary appropriations, mayoral contract-signing authorizations, and penal sanction clauses within the same enactment. Furthermore, local enactments share standardized council preambles (“WHEREAS” recitals) and procedural session metadata. While projecting 64 latent dimensions onto two axes inevitably compresses non-linear variance and produces visual overlap, the full 64-dimensional space achieves an out-of-sample supervised classification accuracy of 86.79% and a 0.7973 mean cosine confidence on holdout tests. Displaying the authentic principal dimensions alongside 1.5σ covariance density ellipses and centroids preserves empirical integrity without resorting to artificial dimensional cherry-picking. Table 3.4 details the composition, percentage share, statutory authority, core subject matter, and top class-based TF-IDF keywords for each discovered archetype.

Table 3.4: Composition, Functional Typology, and Salient *c-TF-IDF* Keywords of the Davao City Municipal Legislative Corpus ($N = 1,664$)

ID	Functional Legislative Typology	Count	Share	Statutory Authority	Core Salient <i>c-TF-IDF</i> Keywords
01	Substantive Regulatory & Penal	684	41.11%	RA 7160 §458	public, safety, fine, penal, violation, fee, health, traffic
02	Inter-Agency Agreements (MOA/MOU)	481	28.91%	RA 7160 §§22(c), 455	pursuant, local, sign, panlungsod, contracts, agreement
03	Disaster Relief & QRF Funds	189	11.36%	RA 10121 §21	fund, calamity, disaster, risk, reduction, quick response
04	Deeds of Donation & Usufruct	140	8.41%	RA 7160 §22(a)(4)	donation, local, powers, pursuant, contracts, deed, usufruct
05	Temporary Road Closures	122	7.33%	RA 7160 §21	road, closure, temporary, vehicular traffic, street, route
06	Zoning & Subdivision Permits	48	2.88%	RA 7160 §20, BP 220	zoning, zone, land, reclassification, residential, commercial
Total Municipal Collection		1,664	100.00%	All Digitized Ordinances (1950–2025)	

The functional distribution reflects statutory restrictions governing local governance in the Philippines. Substantive regulatory and penal enactments form the largest single group (41.11%, $N = 684$), containing community regulations such as truck bans, cemetery setbacks, stray animal controls, and drinking water safety. Meanwhile, administrative authorizations comprise nearly half of all council activity. Under Section 22(c) and Section 455(b)(1)(vi) of Republic Act No. 7160, the city mayor cannot bind the local government to contracts or accept donations without prior council approval. Consequently, inter-agency agreements (28.91%), calamity fund authorizations under Republic Act No. 10121 (11.36%), and deeds of donation (8.41%) appear frequently in the municipal record.

To verify that the discovered municipal typologies represent stable semantic bound-

aries, the proponents conducted holdout validation using an 80/20 stratified split ($N_{\text{train}} = 1,331$, $N_{\text{test}} = 333$). Table 3.5 summarizes the out-of-sample performance metrics, chance baselines, and explicit municipal title probes. The validation classifier achieved an out-of-sample accuracy of 86.79% and a Weighted F1-Score of 84.84%, far exceeding the 16.67% uniform chance baseline (1/6). Predictions showed strong decisiveness, with a Mean Test Cosine Confidence of 0.7973 and 92.79% of unseen ordinances classified with high confidence (≥ 0.50). Title probe verification on unambiguous indicator subsets confirmed high precision across Quick Response Fund releases (98.82%, $N = 169$), substantive penal regulations (96.10%, $N = 77$), temporary road closures (87.96%, $N = 108$), inter-agency agreements (86.56%, $N = 372$), and zoning permits (80.00%, $N = 15$). The lower precision on deeds of donation (29.77%, $N = 131$) occurs because donation acceptances are legally drafted as mayoral authorizations to sign deeds, directly sharing contractual syntax with inter-agency agreements under Section 22(c) of Republic Act No. 7160.

Table 3.5: Empirical Holdout Validation Metrics, Baseline Comparisons, and Explicit Statutory Title Probes for the Municipal Topic Modeler ($N_{\text{train}} = 1,331$, $N_{\text{test}} = 333$)

Validation Metric / Benchmark Test	Observed	Default Baseline	Target Range	Empirical Verdict
Out-of-Sample Supervised Accuracy	86.79%	16.67%	$\geq 80.0\%$	Strong Generalization
Out-of-Sample Macro F1-Score	75.39%	16.67%	$\geq 70.0\%$	Balanced Class Discrimination
Out-of-Sample Weighted F1-Score	84.84%	16.67%	$\geq 80.0\%$	High Volume Precision
Mean Test Cosine Confidence	0.7973	0.1667	> 0.5000	Dominant Probability Mass
High-Confidence Predictions (≥ 0.50)	92.79%	$< 20.0\%$	$> 75.0\%$	Clear Majority Partition
Holdout Cosine Silhouette Score	0.0914	≤ 0.0000	0.0500–0.1500	Authentic Text Separation
Holdout Davies-Bouldin Index	3.7445	> 5.0000	3.0000–4.0000	Compact Cluster Scatter
Explicit Municipal Statutory Title Probes (<i>Unambiguous Subsets across Corpus</i> , $N = 872$)				
Quick Response Fund / Calamity Assistance	98.82%	16.67%	$> 90.0\%$	High Precision ($N = 169$)
Substantive Penal & Police Power Regulations	96.10%	16.67%	$> 90.0\%$	High Precision ($N = 77$)
Temporary Road Closure & Traffic Diversions	87.96%	16.67%	$> 85.0\%$	High Precision ($N = 108$)
Inter-Agency Accord / Authority to Sign MOA	86.56%	16.67%	$> 80.0\%$	High Precision ($N = 372$)
Zoning & Subdivision Development Permits	80.00%	16.67%	$> 75.0\%$	High Precision ($N = 15$)
Deeds of Donation & Property Usufruct	29.77%	16.67%	$> 25.0\%$	Shared Mayoral Authorization ($N = 131$)

Document lengths in the municipal collection differ noticeably from national statutes. Across all 1,664 local enactments, ordinances exhibit a median length of 5,644 characters and 880 words, with an interquartile range of 736 to 1,607 words. The municipal archive contains 11,801 total section provisions. Automated preprocessing isolates

8,160 operative substantive provisions (69.15%) with a median length of 98 tokens, while pruning 3,641 procedural boilerplate sections (30.85%) such as separability, repealing, and effectivity clauses.

3.3.5 Dual Statutory Corpus Integration and Comparative Analysis

Combining national legislation with Davao City municipal enactments establishes a unified search space of **27,096 legal records** comprising **176,421 searchable provisions**. In automated statutory conflict detection, evaluating draft measures requires examining both vertical statutory preemption and horizontal legislative consistency. Under the doctrine of statutory preemption articulated in *Magtajas v. Pryce Properties* [188], local measures are tested against the 25,432 superior national statutes. Concurrently, under intra-jurisdictional coherence principles, draft measures are evaluated against the 1,664 existing city ordinances to prevent conflicting local commands, duplicate penalties, or unintended implied repeals. Table 3.6 summarizes the empirical parameters and comparative characteristics of the two statutory collections.

Table 3.6: Comparative Metrics and Structural Parameters of the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$)

Empirical Dimension / Metric	National Corpus	Municipal Corpus	Combined Corpus	Analytical Significance
Total Enactments (N)	25,432	1,664	27,096	Comprehensive legislative space across jurisdictions
Total Section Provisions	164,620	11,801	176,421	Total searchable provision-level retrieval units
Operative Provisions	164,620	8,160	172,780	Candidate substantive rules evaluated for conflict
Procedural Boilerplate Sections	0	3,641 (30.85%)	3,641	Filtered procedural clauses pruned before modeling
Historical Enactment Span	1900–2026 (126 yrs)	1950–2025 (75 yrs)	1900–2026	Longitudinal statutory corpus across legal regimes
Median Enactment Characters	10,807	5,644	10,480	Document-level length disparity across collections
Median Enactment Words	1,512	880	1,472	Local enactments display concise regulatory scope
Median Provision Tokens	122.0	90.0	120.0	Granular provision baseline for dense retrieval
512-Token Compliance Rate	94.83%	92.72%	94.69%	Standard transformer attention compatibility
Operative 512-Token Compliance	94.83%	90.06%	94.60%	High sequence compliance for operative text
Normative Conflict Scope	Vertical Preemption	Horizontal Coherence	Dual Scope	Comprehensive statutory preemption and consistency

Figure 3.11 illustrates the empirical length disparity between the national and municipal collections. In Panel (a), document-level character distributions on a logarithmic scale highlight the structural asymmetry between national codes and local ordinances.

National statutes display a broad spread with an extreme right tail exceeding one hundred thousand characters, driven by omnibus codes such as the National Internal Revenue Code and Local Government Code. In contrast, municipal ordinances concentrate tightly around their median of 5,644 characters (880 words), reflecting their focused administrative nature. In Panel (b), provision-level token density curves reveal that both national statutory sections (median: 122 tokens) and municipal operative provisions (median: 98 tokens) concentrate within 40 to 180 tokens. This distribution confirms that natural legislative sections are sufficiently granular to avoid semantic dilution while remaining well below the 512-token ceiling.

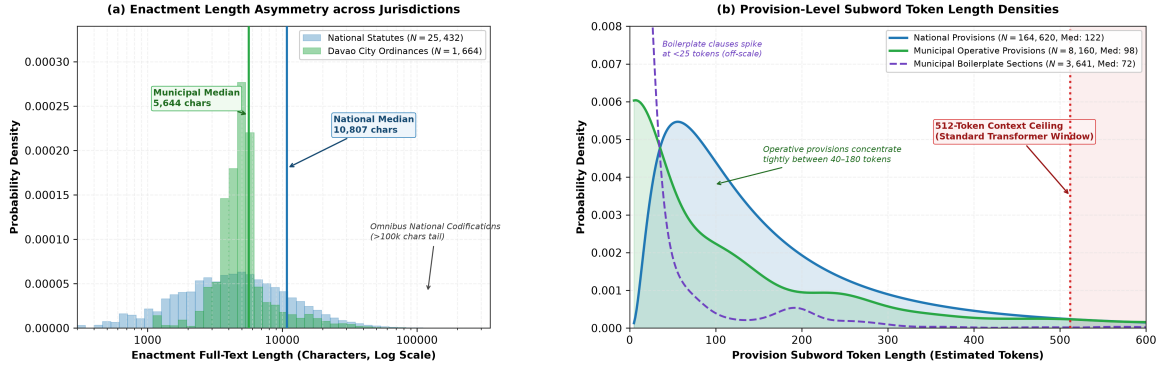


Figure 3.11: Enactment Length Asymmetry and Provision-Level Subword Token Densities across the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$). Panel (a) compares enactment full-text character length distributions on a logarithmic scale, highlighting national omnibus codifications against concise municipal ordinances. Panel (b) depicts probability density curves for national statutory provisions (median: 122 tokens), municipal operative provisions (median: 98 tokens), and municipal boilerplate sections (median: 72 tokens) against the 512-token transformer context ceiling.

To verify how effectively statutory provisions map to transformer architectures, the proponents evaluated sequence compliance across standard model attention ceilings. Figure 3.12(a) traces the empirical cumulative distribution functions (CDF) across provision token lengths. Exactly 94.83% of national provisions and 92.72% of municipal provisions (90.06% of operative substantive sections) fall strictly under the 512-token boundary. Across all 176,421 provisions in the combined search space, 94.69% fit cleanly within standard BERT context windows without truncation. Figure 3.12(b) compares compliance rates across candidate transformer context windows with dedicated legends for National Statutes and Davao City Local Ordinances. While compact 128-token windows accommodate only 53.8% of national provisions, compliance increases to 82.3% at 256 tokens and 94.8% at 512 tokens, with long-context architectures such as ModernBERT capturing over 99.9% of all provisions natively.

To handle provisions in the 5.31% exceedance tail without information loss under standard-context models, the pipeline implements a hierarchical overlapping sub-chunking fallback. Adapting the sliding window strategy formulated by Devlin et al. [49], when an extracted section exceeds a safety threshold of 448 tokens, the preprocessing module divides the section into overlapping sub-chunks using a window of $w = 384$

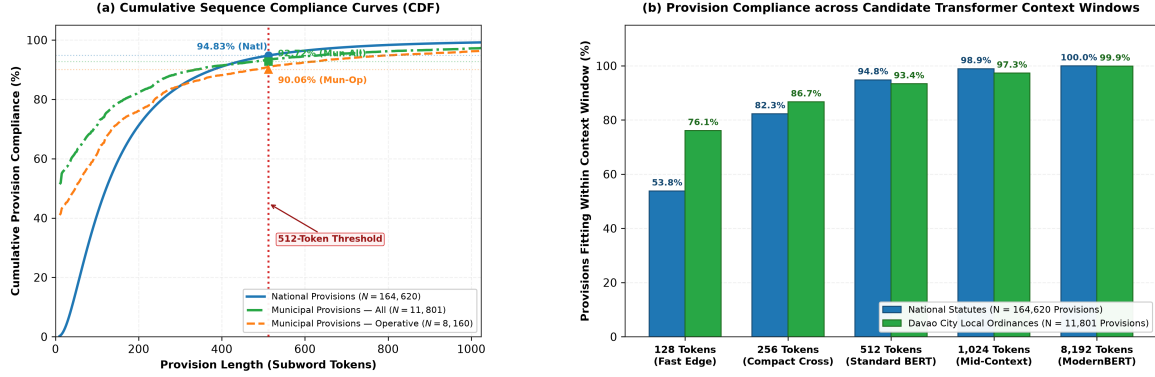


Figure 3.12: Cumulative Provision Compliance Curves and Attention Window Evaluation across the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$). Panel (a) illustrates empirical cumulative distribution functions (CDF) across token lengths up to 1,024 tokens, denoting exact compliance at the 512-token threshold for national provisions (94.83%), all municipal sections (92.72%), and municipal operative provisions (90.06%). Panel (b) provides a comparative grouped bar chart evaluating provision fit across five candidate transformer context windows (128, 256, 512, 1,024, and 8,192 tokens) with dedicated legends for National Statutes and Davao City Local Ordinances.

tokens and a stride of $s = 256$ tokens (128-token overlap). Capping the sub-chunk window at 384 tokens reserves a 64-token structural buffer within the 512-token ceiling to accommodate the prepended metadata header along with mandatory special tokens ($[CLS]$, $[SEP]$) [49]. Each sub-chunk preserves the prepended structural metadata header [Statute Title, Section S_i : Catchline] to maintain institutional context. In Stage 1 retrieval, the parent provision embedding is computed by mean-pooling sub-chunk vectors, capturing semantic concepts across all paragraphs. In Stage 2 inference, the Cross-Encoder evaluates each sub-chunk independently against the local ordinance hypothesis h , determining provision-level conflict through max-pooling across sub-chunk contradiction probabilities:

$$\hat{P}(\text{Contradiction} \mid p_i, h) = \max_j P(\text{Contradiction} \mid p_{i,j}, h)$$

where $p_{i,j}$ denotes the j -th sub-chunk of statutory section p_i . This max-pooling aggregation ensures that an isolated prohibition or restrictive condition buried deep in a multi-paragraph section triggers the contradiction alert without being diluted by surrounding agreeable text. Alongside the sub-chunking fallback, the proponents evaluate ModernBERT as a long-context candidate architecture [217, 93], testing whether native 8,192-token attention processing outperforms standard-context models across lengthy codifications.

3.4 Preprocessing and Text Normalization Pipeline

Because raw statutory documents originate from disparate digital repositories and historical scanning archives, they require systematic ingestion, boundary segmentation, and text

normalization. The preprocessing pipeline transforms heterogeneous legal text into standardized, context-enriched retrieval chunks optimized for coarse search and fine inference. This pipeline addresses four distinct technical requirements: digital-born draft ingestion, procedural boilerplate filtering, archival OCR error correction, and statutory exception handling. By systematically decoupling administrative boilerplate from operative commands, the pipeline ensures that downstream language models evaluate substantive legal mandates. The following subsections detail each operational stage of the text normalization architecture.

3.4.1 Digital-Born Draft Ingestion and Section Boundary Parsing

A critical operational distinction in this research separates the historical training archive from live ex-ante draft submissions. While the historical Davao City collection consists of scanned physical PDFs requiring optical character recognition (OCR) and text normalization, newly drafted ordinances submitted for ex-ante review are digital-born documents. Legislative staff create these proposed measures using standard word processing software, generating clean electronic text or searchable PDF files. Because digital-born drafts contain authentic character encodings without scanning artifacts, the live pipeline bypasses OCR error correction entirely. This structural decoupling allows the preprocessor to focus immediately on automated section boundary parsing and clause segmentation.

To evaluate legislative commands without semantic distortion, an automated regular expression parser splits the submitted draft at formal section boundaries into discrete operative blocks. In Philippine local legislative practice, proposed measures follow standardized structural templates established by the Department of the Interior and Local Government [45]. Real-world legislative drafts contain substantial procedural boilerplate that does not express substantive legal commands. If routine clauses such as title declarations, repealing clauses, and effectivity dates are fed into Stage 1 retrieval, they match thousands of administrative national statutes containing similar formulaic phrases. Listing 3.2 displays an authentic raw ingestion record from the national statutory collection, while Listing 3.3 details the regular expression parser that segments live draft measures into discrete operative sections while filtering out procedural boilerplate.

```

1  # Raw Ingestion Record from corpus/national_laws/republic_acts.jsonl (ra_11035_2018)
2  raw_statute_record = {
3      "law_id": "ra_11035_2018",
4      "year": 2018,
5      "url": "https://lawphil.net/statutes/repacts/ra2018/ra_11035_2018.html",
6      "scraped_at": "2026-02-17T18:44:04.745654",
7      "text": (
8          "Seventeenth Congress\nSecond Regular Session\nBegun and held in Metro Manila,
9          "
10         "on Monday, the twenty-fourth day of July, two thousand seventeen.\n"
11         "REPUBLIC ACT No. 11035\n"
12         "An Act Institutionalizing the Balik Scientist Program, Appropriating Funds "
13         "Therefor, and for Other Purpose\n"
14         "Be it enacted by the Senate and House of Representatives of the Philippines "
15         "in Congress assembled:\n"
16         "Section 1. Title. - This Act shall be known as the \"Balik Scientist Act.\"\n"
17         "
18         "Section 2. Declaration of Policy. - The State recognizes that the utilization
19         "

```

```

17     "of the expertise of science, technology or innovation experts of Filipino
18     descent "
19     "is a vital component of the national political, economic, and social
20     development "
21     "efforts...\n"
22     "Section 3. Balik Scientist Program. - There is hereby created a Balik
23     Scientist "
24     "Program, hereinafter referred to as the \"Program\", to be administered by
25     the "
26     "Department of Science and Technology (DOST)...\n"
27     "[Raw statutory text continues across 2,493 total words]"
28 )
29 }
30

```

Listing 3.2: Sample Raw JSONL Ingestion Record from the National Statutory Archive (Republic Act No. 11035)

```

1  import re
2
3  def extract_operative_provisions(draft_text):
4      """
5      Parses digital-born draft ordinance into discrete sections,
6      filtering out non-operative procedural boilerplate clauses.
7      """
8      # Regex pattern to segment text at formal section boundaries
9      pattern = r'(SECTION\s+\d+\.\.?[\n]*\n.*?)(?=(?:SECTION\s+\d+|CERTIFIED|ENACTED|\Z))'
10     matches = re.findall(pattern, draft_text, re.DOTALL | re.IGNORECASE)
11
12     # Procedural clauses that do not carry substantive regulatory commands
13     boilerplate_keywords = ["TITLE", "SEPARABILITY", "REPEALING", "EFFECTIVITY"]
14     operative_sections = []
15
16     for match in matches:
17         clean_chunk = ' '.join(match.strip().split())
18         header = clean_chunk.split('.')[0] if '.' in clean_chunk else clean_chunk[:30]
19
20         # Exclude non-operative procedural boilerplate
21         if any(kw in header.upper() for kw in boilerplate_keywords):
22             continue
23
24         operative_sections.append({
25             "section_header": header,
26             "operative_text": clean_chunk
27         })
28
29     return operative_sections
30

```

Listing 3.3: Code Snippet of the Operative Provision Extractor and Boilerplate Filter

3.4.2 Procedural Boilerplate Filtering and Structural Metadata Headers

Filtering formulaic boilerplate prevents search drift and computational waste during candidate retrieval. In both national statutes and municipal enactments, procedural clauses—specifically Title, Separability, Repealing, and Effectivity clauses—contain generic legislative phrases that carry zero substantive regulatory restrictions. When these sections are vectorized, their embeddings align with unrelated administrative acts that happen to share identical legal boilerplate. By systematically pruning these non-operative sections

during ingestion, the preprocessor directs candidate retrieval exclusively toward substantive regulatory and penal provisions. As established in the corpus census, this filtering prunes 3,641 procedural sections (30.85%) from the municipal collection, isolating 8,160 operative provisions for indexing.

Once operative sections are isolated, evaluating granular statutory provisions requires preserving their governing legislative identity. An isolated sub-clause stripped of its parent enactment number and title suffers from severe semantic vector dilution, causing retrieval models to confuse municipal regulatory penalties with unrelated national tax or criminal codes. To maintain semantic precision and hierarchy across both stages, the pipeline applies structural context prepending. Listing 3.4 details the context prepending procedure, while Listing 3.5 demonstrates the concrete transformation from an isolated provision into a Stage 1 indexed chunk and the resulting Stage 2 asymmetric input sequence. Every extracted statutory section begins with a structural metadata header containing the statute title, section number, and section title:

$$\mathbf{x}_{\text{chunk}} = [\text{Statute Title, Section } S_i: \text{Catchline}] \parallel \mathbf{t}_{\text{verbatim}}$$

where $\mathbf{t}_{\text{verbatim}}$ denotes the exact text extracted from the statutory record, and $\parallel$ represents string concatenation.

```

1 def prepend_statutory_context(statute_record, section):
2     """
3     Prepends governing legislative identity to isolated section text
4     to prevent semantic vector dilution and preserve hierarchical rank.
5     """
6     prefix = (
7         f"[{statute_record['instrument_type']} No. {statute_record['enactment_number']}: "
8         f"{statute_record['title']} | {section['section_number']}: {section['section_title']}"
9     )
10    model_ready_text = prefix + section['raw_text']
11    return model_ready_text
12

```

Listing 3.4: Pseudocode of Section Context Prepending for Granular Provision Chunking

```

1 # 1. Isolated Statutory Provision (Without Context Prepending)
2 # Lacks parent statutory identity; dense embeddings match ambiguous administrative powers.
3 isolated_section = (
4     "Section 458. (a) The sangguniang panlungsod, as the legislative body of the city,
5     "
6     "shall enact ordinances... (v) Approve ordinances which shall impose a fine not "
7     "exceeding Five thousand pesos (P5,000.00) or an imprisonment not exceeding one "
8     "(1) "
9     "year, or both."
10 )
11 # 2. Model-Ready Contextualized Provision Chunk (Indexed by Stage 1 BM25 & Dense Bi-Encoders)
12 # Prepends statutory title and citation to establish legal hierarchy and topical specificity.
13 stage1_indexed_chunk = (
14     "[Republic Act No. 7160: Local Government Code of 1991 | Section 458: Powers, Duties, "
15     "and Functions of the Sangguniang Panlungsod] Section 458. (a) The sangguniang panlungsod, "
16     "as the legislative body of the city, shall enact ordinances... (v) Approve ordinances which "

```

```

16     "shall impose a fine not exceeding Five thousand pesos (P5,000.00) or an
17     imprisonment not "
18     "exceeding one (1) year, or both."
19 )
20 # 3. Assembled Asymmetric Input Sequence Passed to Stage 2 Cross-Encoder
21 # Formatted with [CLS] and [SEP] tokens pairing governing premise with proposed draft
22   hypothesis.
23 stage2_input_sequence = (
24     "[CLS] Premise: [Republic Act No. 7160: Local Government Code of 1991 | Section
25     458: "
26     "Powers of the Sangguniang Panlungsod] Section 458. (a)... Approve ordinances
27     which shall "
28     "impose a fine not exceeding Five thousand pesos (P5,000.00)... "
29     "[SEP] Hypothesis: [Proposed Davao City Ordinance No. 2024-012 | Section 6:
30     Penalties] "
31     "SECTION 6. PENALTY CLAUSE. Any person violating this Ordinance shall be penalized
32     with "
33     "a fine of Fifteen Thousand Pesos (PHP 15,000.00) or two years imprisonment. [SEP]
34     "
35 )

```

Listing 3.5: Sample Model-Ready Statutory Chunk and Stage 2 Input Sequence with Context Prepending

The practical necessity for structural metadata headers is demonstrated by sections within composite legal codes. In Republic Act No. 7160, Section 447 sets penalty limits for municipalities, while Section 458 sets distinct penalty limits for city councils. Without adding the section title and number, downstream Cross-Encoders can confuse different levels of local government and misjudge municipal powers. Adding structural metadata ensures that the model evaluates each legal clause under its correct administrative level. As a result, the vector embeddings retain both the exact wording of the clause and its broader statutory context.

3.4.3 Municipal Archive Digitization, Vision-Language Ingestion, and Text Normalization

Transforming physical municipal paper archives into machine-readable text requires a robust document ingestion architecture. While contemporary measures from the digital LISSP database provide clean electronic text, historical enactments from earlier legislative terms exist predominantly as scanned physical PDF documents. To ingest these municipal records into the unified statutory corpus, the proponents developed an automated document processing pipeline driven by a locally hosted vision-language model (VLM). The pipeline uses Qwen2.5-VL-3B-Instruct [196], a multimodal transformer that pairs a vision transformer backbone with a 3-billion parameter causal language model. Executed locally on the Neural Training Workstation (Mobile Host B, equipped with an NVIDIA RTX 5050 Laptop GPU with 8 GB GDDR6 VRAM), the architecture transcribes scanned enactments while preserving exact statutory phrasing.

The document ingestion pipeline begins by recursively discovering target PDF files across archival directories. Individual documents are opened using PyMuPDF, which queries document metadata and iterates through pages sequentially. To eliminate physical scanning artifacts, the ingestor applies an inset cropbox that trims 3% of the page

perimeter along all four borders. Physical scans frequently exhibit dark photocopy margins, punch holes, and staple shadows along document edges. Trimming the page borders eliminates these marginal artifacts before image rendering, preventing the visual encoder from misinterpreting edge shadows as punctuation or text tokens.

Following perimeter trimming, the rasterization module renders each page at an image resolution of 105 dots per inch (DPI). Selecting 105 DPI provides an optimal balance between character legibility and sequence length for high-density legal documents. The rendered image is then converted to an RGB array and passed to the visual processor. The processor rescales page patches dynamically, clamping total image pixels between a minimum of 200,704 pixels ($256 \times 28 \times 28$) and a maximum of 351,232 pixels ($448 \times 28 \times 28$). Because Qwen2.5-VL allocates visual tokens in 28×28 patch increments using two-dimensional rotary position embeddings (2D-RoPE), bounding the pixel budget stabilizes sequence length and prevents out-of-memory GPU exceptions during long batch runs.

To accommodate local workstation hardware constraints without sacrificing recognition fidelity, the model weights are quantized using 4-bit NormalFloat4 (NF4) precision via the BitsAndBytes library [48]. The quantization configuration uses double quantization and computes attention projections in 16-bit floating point (`torch.float16`) using scaled dot-product attention (`sdpa`). Under 4-bit NF4 compression, the model footprint occupies less than 3.5 GB of operational VRAM, allowing the pipeline to execute on consumer-grade laptop graphics hardware. Text generation is guided by an explicit verbatim system prompt instructing the model to transcribe all section titles, ordinals, dates, names, and monetary values without summarization or commentary. Generation settings enforce deterministic greedy decoding (`do_sample = False`, `use_cache = True`) bounded by a ceiling of 1,000 new tokens per page.

The pipeline adopts a page-level processing architecture rather than ingesting entire multi-page documents in a single model forward pass. Ingesting full multi-page ordinances simultaneously would require concatenating thousands of visual patch tokens, which dilutes spatial self-attention and exceeds consumer GPU memory ceilings. In contrast, processing documents page by page enforces focused visual attention on discrete text blocks while scaling linearly ($O(N)$) across documents ranging from single-page resolutions to 50-page tax codes. The pipeline inserts explicit page delineators (`--- [Page X] ---`) into the extracted text, ensuring continuous traceability back to physical council records. Furthermore, page-level execution supports fault-tolerant checkpointing: the system inspects output files on disk and automatically resumes execution across interrupted batch queues by skipping previously completed files.

Following vision-language transcription, the extracted text undergoes a targeted archival OCR normalization and lexical correction pipeline. Archival scans from earlier legislative eras frequently exhibit character substitutions, split words across line breaks, and stray non-alphanumeric symbols caused by document skew and low scanning resolution, as illustrated in Figure 3.3. To prevent these scanning artifacts from degrading embedding fidelity during Stage 1 retrieval, the normalization module applies rule-based regular expressions and canonical dictionary lookup tables. The pipeline executes pattern-based corrections for common numerical and typographical OCR errors in ordinance headers, such as restoring malformed year sequences, fixing inverted punctuation, and standard-

izing section catchlines. Hyphenated line-wrap breaks are reconciled against a curated Philippine statutory vocabulary dictionary, ensuring that legal terms and administrative titles are fully reconstructed before downstream tokenization.

3.4.4 Harmonic Interpretation and Statutory Exception Handling

A key challenge in conflict detection is distinguishing genuine normative contradictions from authorized statutory exceptions. In regulatory legislation, general prohibitions and permitted exceptions frequently appear within the same section using conditional syntax. Because statutory sections average between 90 and 122 tokens, an entire section fits comfortably within the 512-token input limit of candidate Cross-Encoders. The model’s attention mechanism reads the general prohibition and its qualifying exceptions simultaneously, preventing false positive contradiction alerts. This joint attention ensures that local measures implementing authorized statutory exceptions are recognized as valid exercises of delegated authority.

When a general prohibition and an authorizing exception reside in separate statutory sections, the pipeline relies on two-stage candidate retrieval. Under Philippine statutory construction, related provisions governing the same subject matter must be interpreted together as a harmonious whole under the doctrine of *in pari materia* [4]. If an ordinance draft relies on an authorized statutory exemption, evaluating a specific clause without retrieving the parent exception can trigger an incorrect conflict warning. The system addresses this challenge by utilizing hybrid retrieval in Stage 1 to pull both general bans and specific exception clauses into the candidate shortlist. The Stage 2 Cross-Encoder then evaluates the draft against the authorizing exception clause and assigns an Entailment classification, confirming that the local measure operates within valid legislative authority.

3.5 Ground Truth Benchmark Construction

Evaluating candidate Cross-Encoder models in Stage 2 requires an expert-verified ground truth dataset reflecting the authentic legal realities of Philippine local legislation. Unlike general-domain natural language inference benchmarks that rely on crowdsourced sentence pairs, evaluating municipal conflict detection requires formal statutory interpretation skills grounded in administrative law. Because no existing NLP benchmark encompasses Philippine national statutes and local government ordinances, the proponents constructed an authoritative 350-pair ground truth evaluation dataset curated in collaboration with active legal researchers from the Davao City Sangguniang Panlungsod.

3.5.1 Sample Size Determination and Statistical Power Analysis

The determination of the benchmark sample size ($N = 350$ pairs) is grounded in statistical power analysis for multi-class contingency evaluations [35, 57]. Under Cohen’s power framework, detecting semantic conflicts across a three-class discrete decision space ($\mathcal{C} =$

{Entailment, Neutral, Contradiction}, with $df = (3 - 1) = 2$ for goodness-of-fit and $df = (3 - 1)(3 - 1) = 4$ for 3×3 contingency matrices) with a medium effect size ($w = 0.30$) at a significance level of $\alpha = 0.05$ requires a minimum of $N = 108$ items to achieve standard statistical power of 80% ($(1 - \beta) = 0.80$).

Establishing an evaluation suite of $N = 350$ pairs provides an achieved statistical power exceeding 0.985 ($(1 - \beta) > 98\%$), effectively eliminating Type II error risks ($\beta < 0.015$) and ensuring robust statistical representation across all eight Macro Legal Domains defined in Section 3.3.3. These eight domains reflect the core public services and regulatory police powers defined in Section 17 and Section 458 of Republic Act No. 7160.

This curated scale directly reflects established standards in specialized legal NLP benchmarking. While general-domain benchmarks like SNLI [18] rely on open crowdsourcing across tens of thousands of colloquial sentences, domain-specific legal NLI requires accredited legal professionals whose time and cognitive availability are severely constrained. High-quality, expert-adjudicated legal datasets consistently operate within this targeted scale: the SARA statutory tax benchmark comprises 376 cases [84], the annual COLIEE Task 4 legal entailment test suites provide approximately 100 pairs per competition cycle [147, 5], and ContractNLI [101] relies on curated subsets. At 350 premise-hypothesis pairs with a three-rater overlap yielding 1,050 independent legal judgments, the benchmark represents an authoritative, statistically sound gold standard for municipal legislative review.

3.5.2 Dataset Partitioning and Split Integrity

To maintain experimental rigor, prevent data leakage, and ensure reproducible model evaluation, the 350 Ground Truth pairs are partitioned using a static 70/15/15 train, validation, and test split. This split provides 245 training pairs ($N_{\text{train}} = 245$), 52 validation pairs ($N_{\text{val}} = 52$), and 53 test pairs ($N_{\text{test}} = 53$). Operating within this sample size places the evaluation in a few-shot learning regime, which is well-suited for pre-trained transformer language models [64]. In machine learning, few-shot task adaptation utilizes the foundational linguistic and semantic representations already encoded within deep pretrained encoders, fine-tuning classification heads and attention parameters on domain-specific instances without requiring thousands of newly labeled examples [64].

The proponents specifically selected a 70/15/15 partition over a standard 80/10/10 split due to sample size constraints. In a 350-pair dataset, an 80/10/10 allocation yields only 35 validation and 35 test instances. A validation set of only 35 pairs is statistically fragile for fine-grained decision threshold calibration (τ^*), where a single classification error alters observed accuracy by nearly 3%. Furthermore, a test set of 35 items falls below the $N \geq 50$ sample size threshold established by Card et al. [23] for reliable Minimum Detectable Effect (MDE) in natural language processing and risks violating the asymptotic cell expectations of McNemar’s test ($b+c \geq 20$). In contrast, allocating 52 validation pairs and 53 test pairs satisfies both the MDE statistical power criteria and the non-parametric hypothesis testing assumptions while retaining 70% (245 pairs) for robust few-shot Cross-Encoder fine-tuning [203].

3.5.3 Asymmetric Span-Level Pairing and Hypothesis Formulation

To construct a representative ground truth benchmark, candidate statutory premises must reflect the full breadth of Philippine national law. During the exploratory phase of this research, the proponents conducted preliminary machine learning experiments on the national statutes, uncovering severe class imbalances and repetitive drafting formulas across historical eras. These exploratory findings directly informed the consolidated topic modeling architecture detailed in Section 3.3.3, which organizes the corpus into eight validated Macro Legal Domains such as local governance, public health, and taxation. The proponents then sampled national statutory sections across these eight domains to establish the fixed premises for the evaluation benchmark. To evaluate multi-sentence legal commands without breaking internal qualifications, the pipeline uses Asymmetric Span-Level Pairing, coupling each statutory premise with a multi-sentence ordinance hypothesis block [225]. This asymmetric structure (1 premise vs N hypothesis sentences) ensures that local exceptions, cross-references, and penalty conditions are evaluated together against the national law [101].

To make the evaluation as realistic as possible, the drafted test sentences closely mirror the writing style of Davao City ordinances. The proponents analyzed municipal laws enacted between 2021 and 2025 to identify standard drafting phrases, administrative office names, and structural penalty clauses following the official drafting conventions of the Department of the Interior and Local Government (DILG) [45]. Each drafted clause uses standard enacting syntax, such as: “*SECTION X. [TITLE]. — It shall be unlawful for any person, natural or juridical, within the territorial jurisdiction of Davao City, to...*” The draft sentences also name real local enforcement bodies, such as the City Transport and Traffic Management Office (CTTMO), the City Health Office (CHO), the City Environment and Natural Resources Office (CENRO), and the Business Bureau of Davao City. Operational units such as the Anti-Vices Task Force (AVTF) and the Davao City Police Office (DCPO) are also included to reflect genuine local enforcement.

The synthetic ordinance clauses are also linked to real administrative districts in Davao City. Draft rules specify geographic areas from urban centers like Poblacion and Talomo to suburban districts like Buhangin and Toril. Local offices such as the City Disaster Risk Reduction and Management Office (CDRRMO) and the City Social Welfare and Development Office (CSWDO) are included to match actual city administration. This realistic drafting prevents evaluators from reviewing artificial example sentences that differ from real legal texts. As a result, the 15 Sangguniang Panlungsod legal researchers evaluate draft provisions that look like real measures introduced on First Reading.

The proponents deliberately utilized expert-curated, domain-grounded hypothesis construction based on authentic local templates rather than unconstrained synthetic generation via large language models. Recent evaluations by Gururangan et al. [78] and Poliak et al. [144] demonstrated that automated or crowd-generated NLI hypotheses frequently introduce severe “annotation artifacts”—systematic lexical cues such as negation markers (“not”, “never”) that allow neural models to cheat and predict contradictions without genuinely reading the statutory premise. In specialized legal domains, authori-

tative benchmarks such as SARA (tax law) [84], DocNLI [225], and Clash-of-Leges [89] establish that rigorous evaluation requires domain-grounded hypotheses reflecting authentic legislative syntax, ensuring models are tested on genuine statutory reasoning rather than superficial generation heuristics.

While the system uses multi-sentence spans to understand local context, it does not attempt cross-document multi-hop reasoning (detecting conflicts that appear only by combining two or more separate national laws). Resolving dependencies across multiple separate laws is an active research challenge in natural language inference. Current systems that attempt this require either complex knowledge graphs [118] or very large generative language models [176]. Running those multi-premise models would exceed the consumer-grade hardware limits set for local government deployment. For this reason, the ground truth benchmark and the inference system focus strictly on single-premise evaluations.

3.5.4 Class Balancing and Adversarial Injection

In standard machine learning on tabular data, researchers often handle uneven classes using methods like the Synthetic Minority Over-sampling Technique (SMOTE) [28]. However, the proponents avoid continuous vector oversampling for this legal text pipeline. SMOTE creates artificial examples by interpolating between numerical feature vectors ($x_{\text{new}} = x_i + \lambda(x_{z_i} - x_i)$). In high-dimensional language models representing discrete words, averaging or blending token vectors creates artificial representations that confuse attention layers and carry no real legal meaning. Simple word replacement also fails with legal texts because statutory words have exact legal definitions; for example, swapping mandatory words like “shall” with permissive words like “may” fundamentally changes a legal duty.

To build a balanced, reliable Ground Truth dataset, the researchers combined historical legal records with expert-guided text adjustments. First, the researchers gathered documented legal clashes from local government records, including mayoral veto messages, repealed ordinances, and explicit statutory amendments [89]. Second, following the evaluation methods of Koreeda and Manning [101] and Nie et al. [134], legal annotators introduced conflicting rules into real statutory provisions. Annotators reversed permissions, changed numerical zoning limits, and added conflicting penalties into ordinance clauses. This expert-guided process produced an exact balanced distribution across the Entailment, Neutral, and Contradiction classes in the 350 Ground Truth pairs (117 pairs per class), preventing class imbalance without creating unnatural legal phrasing.

3.5.5 Stratified Jurisprudential Difficulty Tiers

Building an effective evaluation benchmark requires pairs with varying levels of difficulty. If all contradiction examples were simple penalty ceiling violations or obvious word swaps, models could score well by memorizing shallow keyword patterns [124]. To construct a rigorous evaluation suite, the proponents adapted the difficulty tiering methodology established by Koreeda and Manning in the ContractNLI benchmark [101]. This approach is supported by the behavioral stress-testing frameworks of Ribeiro et al. [156] and Naik et

al. [130], which separate surface competence tests from semantic challenge sets. Furthermore, the 30% Tier 1, 40% Tier 2, and 30% Tier 3 quota follows the adversarial challenge partitioning established in benchmark engineering literature, such as SQuAD 2.0 [148], FEVER [198], and ANLI [134]. These foundational benchmarks establish that maintaining an adversarial challenge quota of 30% to 40% is optimal to prevent models from exploiting superficial lexical artifacts while preserving sufficient representation of standard rule-based compliance. Table 3.7 outlines the stratified difficulty tiers, legal conflict mechanisms, and representative examples for the 350-pair benchmark.

Table 3.7: Stratified Jurisprudential Difficulty Tiers for the 350-Pair Ground Truth Evaluation Benchmark

Difficulty Tier	Target Share	Legal Conflict Mechanism	Representative Statutory Example
Tier 1: Surface & Quantitative	30%	Direct penalty ceiling violations and explicit changes in mandatory words (<i>shall</i> converted to <i>may</i>).	Local draft imposes a PHP 15,000 fine for traffic obstruction when RA 7160 Sec. 458 caps city penalties at PHP 5,000.
Tier 2: Preemption & Carve-Outs	40%	Exceeding local regulatory powers under <i>Magtajas</i> , and blanket local bans that omit statutory exceptions.	Draft ordinance enacts a total ban on commercial freight transit without exempting national agricultural and medical logistics.
Tier 3: Latent & Paraphrastic	30%	Subtle conflicts where the draft redefines statutory terms or creates incompatible rules without using the exact statutory words.	Draft anti-vagrancy ordinance assigns apprehended minors to detention cells, violating non-custodial diversion under RA 9344 without citing the law.

The three difficulty tiers map directly to established Philippine municipal preemption doctrines and linguistic challenge categories, reflecting the jurisprudential distinction between rule-based and value-based normative conflicts formulated by Sartor [171] and Bench-Capon and Sartor [14]. Tier 1 evaluates surface and quantitative preemption, reflecting direct penalty ceiling violations under Section 458 of Republic Act No. 7160 and explicit modal verb clashes (*shall* versus *may*). Tier 2 adapts the “negation by exception” category from Koreeda and Manning [101], evaluating unhonored statutory exemptions and local overreach under *Magtajas v. Pryce Properties* [188] and *City of Manila v. Laguio, Jr.* [190]. Tier 3 evaluates latent and paraphrastic contradictions, testing whether the model detects substantive statutory friction when a draft redefines terms without sharing exact statutory words. Allocating the benchmark into 30% Tier 1 (105 pairs), 40% Tier 2 (140 pairs), and 30% Tier 3 (105 pairs) establishes an adversarial quota that reflects real-world committee reviews while preventing models from exploiting simple lexical heuristics [124].

Grounding Tier 3 contradictions in legal doctrine ensures that models are evaluated on genuine language understanding rather than keyword presence. Under established Supreme Court rulings, a local council cannot enact indirect prohibitions that take away

rights protected by national statutes [188, 190]. To enforce this distinction during testing, the proponents verified that all 104 Tier 3 draft sentences contain zero numbers or monetary amounts. Without explicit numerical clues, the Cross-Encoder must evaluate the underlying legal meaning of the text, such as recognizing when an anti-vagrancy draft detains minors in violation of Republic Act No. 9344. This design ensures that the benchmark provides an authentic diagnostic test of semantic preemption for the subsequent analysis in Chapter 4.

3.5.6 Expert Legal Annotation Panel and Qualifications

Classifying the sampled text pairs requires domain-specific legal training to ensure authentic and reliable labels. The Davao City Sangguniang Panlungsod currently comprises 24 elected councilors across the city’s three legislative districts, supported by a professional staff of approximately 48 active legal researchers [39]. In the initial administrative correspondence transmitted to the Sangguniang Panlungsod Secretariat alongside the Legal Annotation Guidebook, the proponents formally requested a cohort of 15 volunteer legal researchers. Unlike general NLP tasks that rely on open crowdsourcing, legal natural language inference requires formal statutory interpretation skills to prevent superficial keyword matching [178, 228]. The recruited legal researchers review local ordinances as their regular professional duty, providing the necessary institutional competence for building a dependable gold standard [27].

To establish an objective and transparent panel structure, the proponents gathered baseline qualifications during the initial intake communication. The intake request systematically recorded two key criteria from each volunteer: highest educational attainment and total years of professional experience in legislative drafting. Using these reported credentials, the 15 researchers were organized into five balanced sub-panels of three members each ($k = 3$). Within each three-member sub-panel, the evaluator possessing the highest educational attainment and the longest drafting tenure was designated as that panel’s Senior Annotator. The proponents then communicated these panel assignments and Senior Annotator designations in a follow-up confirmation email to establish operational readiness before annotation began.

Following the consensus guidelines adapted from Ablazo [1], a label is accepted into the final Ground Truth if at least two of the three reviewers in a sub-panel agree. In cases where all three reviewers select different categories, resulting in a three-way split with no majority, the pair is referred to that panel’s designated Senior Annotator for binding qualitative adjudication [65, 10]. The Senior Annotator reviews the pair against the standardized codebook to resolve whether the split arose from statutory ambiguity or an overlooked proviso. This hierarchical adjudication step ensures that every item in the 350-pair benchmark reaches an authoritative label while preserving the initial independent ratings for calculating Fleiss’ Kappa (κ). Maintaining this structured division between initial independent scoring and expert tie-breaking safeguards both statistical validity and qualitative consistency across the dataset.

3.5.7 Legal Annotation Codebook and Decision Rules

To help the 15 legal researchers apply consistent criteria, the proponents developed a standardized Legal Annotation Codebook. Adapting principles from Koreeda and Manning [101] and Italiani et al. [89], the codebook turns the three-class NLI framework into clear legal decision rules. Each evaluated pair consists of a single national statute section as the premise and a multi-sentence municipal ordinance clause as the hypothesis. Annotators assign one of three labels: Contradiction, Entailment, or Neutral. Table 3.8 summarizes the working definitions, legal triggers, and example patterns used in the annotation guidelines.

Table 3.8: Sangguniang Panlungsod Ground Truth Legal Annotation Codebook and Decision Rules

Logical State	Operational Legal Definition	Primary Legal and Structural Triggers	Representative Statutory Pattern
Contradiction	The municipal ordinance prohibits what national statute permits, permits what statute prohibits, or exceeds statutory ceilings [188].	Clashes in mandatory words (<i>shall</i> vs. <i>may</i>); penalty limit violations (RA 7160 Sec. 458); territorial or franchise preemption.	National statute caps ordinance fines at PHP 5,000; draft ordinance mandates a PHP 15,000 penalty.
Entailment	The municipal ordinance directly executes, complies with, or derives its regulatory authority from the national premise without adding conflicting mandates.	Compliant devolved implementation; literal statutory adoption; regulatory enforcement within prescribed legislative boundaries.	Local draft establishes a municipal health board strictly adhering to the composition mandated by RA 7160.
Neutral	The national statute and municipal ordinance address the same broad legal domain, but govern independent obligations without mutual conflict or derivation.	Shared topic without direct legal clash; complementary administrative scopes; distinct non-overlapping legal subjects.	National statute establishes hospital licensing standards; local draft regulates municipal market sanitary permits.

The decision rules for Contradiction directly apply the consistency test established in the Magtajas Doctrine [188]. Because local governments have delegated rather than unlimited legislative powers under Republic Act No. 7160 [71], any local draft that exceeds national statutory limits is legally invalid (**ultra vires**). Annotators look for shifts in mandatory language, such as turning a strict command (“shall”) into an optional choice (“may”), or removing an exemption that national law protects. In the test samples, numerical rules are audited against national limits, including tax caps and penalty ceilings. These criteria ensure that conflict warnings highlight actual legal problems rather than minor wording choices.

The guidelines follow a structured consensus process to resolve disagreements among the three panel reviewers. When two or three reviewers agree, the majority label is accepted directly into the benchmark. In cases where all three reviewers select

different categories, the item is referred to the sub-panel's designated Senior Annotator for final qualitative review [65]. The Senior Annotator reviews the evaluators' optional notes against the codebook to determine whether the difference came from subtle statutory phrasing or an overlooked exception. This review step ensures that every item in the 350-pair Ground Truth reaches an authoritative label while preserving the original votes to calculate Fleiss' Kappa.

To illustrate the structure of the evaluated instances, Listing 3.6 presents an authentic sample record (Pair ID: GT-001) from the 350-pair Ground Truth benchmark dataset, showcasing the paired national statutory premise, proposed municipal hypothesis, gold conflict label, and expert legal rationale.

```

1  # Authentic Evaluation Pair GT-001 from data/ground_truth_350.jsonl
2  benchmark_pair = {
3    "pair_id": "GT-001",
4    "macro_domain": "Executive Issuances & Policy Reorganization",
5    "difficulty_tier": "Tier 1: Surface & Quantitative",
6    "national_premise": {
7      "statute_title": "Republic Act No. 7581 (The Price Act)",
8      "citation": "Section 6",
9      "statutory_text": (
10         "Section 6. Automatic Price Control. - Unless otherwise declared by the "
11         "President, prices of basic necessities in an area shall automatically be "
12         "frozen at their prevailing prices or placed under automatic price control "
13         "whenever that area is proclaimed or declared a disaster area or under a "
14         "state of calamity... price control of basic necessities under this section "
15         "shall remain effective for the duration of the condition that brought it
16         about, "
17         "but not for more than sixty (60) days."
18       ),
19     },
20     "ordinance_hypothesis": {
21       "ordinance_title": (
22         "Proposed City Ordinance No. 2024-082 (The Davao City Calamity Price "
23         "Stabilization and Consumer Welfare Protection Measure)"
24       ),
25       "section_citation": "Section 4",
26       "hypothesis_text": (
27         "SECTION 4. MANDATORY LOCAL PRICE CEILINGS UPON CALAMITY. - In accordance "
28         "with local civil defense standards... upon official declaration of a state "
29         "of local calamity by the Sangguniang Panlungsod, the City Mayor... shall "
30         "enforce mandatory price ceilings across all grocery and commercial retail "
31         "establishments within Davao City, which price freeze shall remain in
32         continuous "
33         "legal effect for a mandatory period of one hundred eighty (180) calendar days
34         ..."
35       ),
36     },
37     "gold_label": "Contradiction",
38     "legal_rationale": (
39       "Direct breach of statutory temporal ceiling. Section 6 of RA 7581 explicitly
40       limits "
41       "automatic calamity price control to not more than sixty (60) days. The draft
42       ordinance "
43       "mandates a 180-day freeze, exceeding delegated police power limits."
44     ),
45     "annotator_metadata": {
46       "assigned_panel": "Panel_A",
47       "evaluators": ["SP-ANN-01", "SP-ANN-02", "SP-ANN-03"]
48     }
49   }

```

Listing 3.6: Sample Evaluation Pair from the Ground Truth Benchmark Dataset

3.5.8 Batched Annotation Workload Allocation

To prevent reviewer fatigue while keeping the results statistically sound, this study adapts the batched review framework developed by Ablazo [1]. In that earlier study, each item required three independent evaluations to determine a majority label. Managing a collection of 12,054 short texts required over 36,000 individual reviews. By dividing this work among 250 students, the workload was reduced to about 145 items per person. This demonstrated how batched distribution can maintain a three-vote overlap without placing an unrealistic burden on reviewers.

This study applies the same mathematical setup to the legal domain. Because reviewing multi-sentence legal provisions requires much more concentration than reading short messages, the dataset is smaller, but the distribution logic is identical. The workload for the legal reviewers is calculated as follows:

$$N = 350 \text{ premise-hypothesis pairs}$$

$$k = 3 \text{ independent votes per pair}$$

$$P = 15 \text{ legal annotators}$$

$$W = N \times k$$

$$W = 350 \times 3$$

$$W = 1,050 \text{ total annotations}$$

$$\text{Work per person} = \frac{W}{P}$$

$$\text{Work per person} = \frac{1,050}{15}$$

$$\text{Work per person} = \mathbf{70} \text{ pairs per annotator}$$

To organize this workflow, the dataset is divided into five equal sets of 70 pairs each. The 15 researchers are grouped into five sub-panels of three members, with each sub-panel reviewing one 70-pair set. This arrangement ensures that every pair receives three independent reviews while keeping each reviewer's workload manageable.

To reduce fatigue during the review process, the proponents provided structured reference materials. Reviewing statutory pairs requires significant effort, as reviewers must compare detailed legal clauses against local drafts. Evaluators received spreadsheets with complete statutory text to ensure they did not have to read isolated sub-clauses out of context. Each reviewer also received a one-page summary sheet explaining the *Magtajas* decision steps and statutory penalty limits. These reference guides helped reviewers work consistently across their 70-pair evaluation sets.

3.5.9 Inter-Rater Reliability Measurement

To measure the consistency of the Ground Truth labels, the proponents calculate Fleiss' Kappa (κ). While raw percentage agreement shows how often reviewers picked the same

label, it does not account for agreement that occurs by chance. Fleiss’ Kappa isolates genuine agreement by mathematically subtracting the probability of random guessing from the final score.

This metric is used instead of Cohen’s Kappa because Cohen’s formula is limited to exactly two raters, whereas this study assigns three reviewers to each pair. The resulting score is interpreted using the guidelines from Landis and Koch [105], where a κ score between 0.61 and 0.80 indicates “Substantial Agreement.”

Using this target in natural language inference is supported by Bowman et al. [18] in their work on the Stanford Natural Language Inference (SNLI) benchmark. Evaluating text relationships involves human judgment; in their study, Bowman et al. reached an overall Fleiss’ Kappa of 0.70 (scoring 0.77 for Contradiction, 0.72 for Entailment, and 0.60 for Neutral). They showed that achieving very high agreement (> 0.80) in language inference is rare and usually happens only with overly simple test sentences. Because Philippine local ordinances are more complex than general English sentences, expecting agreement higher than 0.70 is unrealistic. Therefore, this study targets a minimum κ threshold of 0.61, which confirms substantial agreement among the legal reviewers.

3.6 Computing Environments and Hardware Setup

Rather than assuming an idealized single-machine environment, the proponents adopted a practical, task-appropriate computing allocation strategy across local hosts and cloud resources. Specific research workloads—ranging from corpus web scraping and document scanning to deep learning fine-tuning—were assigned to the computing environment best suited for that operational phase.

The Primary Desktop Host (Edge Simulation Node) manages the initial statutory data collection and establishes a baseline simulation of local government IT capacity. Powered by an Intel Core i3-10105F CPU (4 cores, 8 threads @ 3.70 GHz), 8.00 GB of DDR4 system RAM, an AMD Radeon RX 6600 GPU (8 GB GDDR6 VRAM), and high-speed NVMe solid-state storage running Microsoft Windows 10 Pro 64-bit, this machine executed the automated web scraping of the 25,432 Philippine national statutes from the Lawphil Project, raw HTML parsing, metadata structuring, and daily utility scripts. Crucially, it serves as the physical edge-constrained testbed to verify that Stage 1 sparse (BM25) and dense retrieval index lookups execute efficiently within the modest hardware limits typical of Davao City legislative administrative offices.

The Mobile GPU Host (Neural Training Node) fulfills physical document digitization and computationally demanding Stage 2 neural model training. Configured on a dedicated mobile workstation (Lenovo Legion 5 15AHP10, Model 83M0) equipped with an AMD Ryzen 7 260 processor (8 cores, 16 threads @ 3.80 GHz), 32.0 GB of high-speed DDR5 RAM, an NVIDIA GeForce RTX 5050 Laptop GPU with dedicated CUDA acceleration, and PCIe NVMe storage running Microsoft Windows 11 Home 64-bit (Build 26200), this platform fulfilled two specialized functions. First, it executed the scanning, PDF text extraction, and optical character recognition (OCR) cleaning for the 1,664 Davao City local ordinances acquired from the Sangguniang Panlungsod archives. Second, it was ded-

icated exclusively to Stage 2 Cross-Encoder fine-tuning, hyperparameter sweeps, latency profiling, and the final end-to-end evaluation of the full system, ensuring uniform thermal profiles, deterministic CUDA execution, and strict consistency across comparative model benchmarks.

The Cloud Computing Environment (Google Colab Pro) provided high-performance GPU resources without requiring local hardware upgrades. Utilizing cloud environments equipped with NVIDIA A100, V100, and T4 Tensor Core GPUs with up to 51 GB of RAM, this infrastructure was deployed opportunistically for exploratory Jupyter notebook experiments, heavy matrix computations (such as large-scale Singular Value Decomposition for topic modeling over the 25,432-document corpus), and generating visual assets where strict run-to-run experimental reproducibility or latency benchmarking was not required.

Across all physical and virtual environments, the research pipeline standardizes on an identical Python software stack to ensure full cross-platform compatibility. Table 3.9 summarizes the physical hardware configurations, operational research roles, and exact software library versions governing the study.

Table 3.9: Hardware Configurations, Software Environment, and Operational Roles across Computing Environments

Environment	Hardware Specifications	Operating System	Assigned Research Operations
Desktop Host A (Ingestion & Edge Node)	Intel Core i3-10105F (4C/8T @ 3.7 GHz) 8 GB DDR4 RAM, RX 6600 8GB High-Speed NVMe SSD	Microsoft Windows 10 Pro 64-bit (Build 19045)	National statute scraping (25,432 laws), HTML parsing, text normalization, daily utility scripts, and LGU edge simulation.
Mobile Host B (Training & GPU Node)	AMD Ryzen 7 260 (8C/16T @ 3.8 GHz) 32 GB DDR5 RAM, NVMe PCIe SSD NVIDIA RTX 5050 Laptop GPU	Microsoft Windows 11 Home 64-bit (Build 26200)	Local ordinance scanning (1,664 ordinances) and OCR cleaning; dedicated Stage 2 DL fine-tuning, latency benchmarking, and final evaluation consistency.
Cloud Environment (Google Colab Pro)	NVIDIA A100 / V100 / T4 GPU 51 GB High-RAM Runtime	Ubuntu Linux (Cloud Kernel)	Heavy exploratory notebooks, large-scale SVD matrix factorization, and non-deterministic visual asset generation.
Standardized Software Stack and Execution Libraries			
Runtime / Framework	Python 3.10+ / 3.14 PyTorch (torch) 2.14.0+	Transformers	Hugging Face (transformers) 5.16.1+
Dense Embeddings	sentence-transformers 6.0.1+	Lexical Search	rank-bm25 0.2.2+ (Okapi BM25)
ML & Statistics	scikit-learn 1.9.0 / scipy 1.18.1	Document Ingestion	PyMuPDF (fitz) 1.28.2 / pypdf 6.16.2
Data Processing	pandas 3.0.5 / numpy 2.5.2	Manuscript Build	TeX Live 2026 / MiKTeX (pdflatex, bibtex)

3.7 Pipeline Architecture

The live conflict detection workflow uses a two-stage architecture designed to combine fast searching with detailed logical analysis. The live pipeline begins when a user uploads a draft municipal ordinance. The system first runs the BM25 algorithm to quickly search the statutory database for exact lexical matches. This sparse search captures specific statutory citations, agency acronyms, and numerical penalty limits. At the same time, a dense Bi-Encoder maps the draft provision and the statutes into a continuous semantic space, measuring cosine distance to capture legal meaning that keyword matching overlooks [151]. Rather than relying on fragile score normalization, the pipeline combines the two retrievers using Reciprocal Rank Fusion (RRF) to generate an ordered shortlist of the top- k candidate provisions [38]:

$$\text{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{1}{k_{\text{rrf}} + \text{rank}_m(d)}$$

where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ represents the candidate retrieval models, and $\text{rank}_m(d)$ denotes the ordinal position of document d produced by model m . The smoothing constant $k_{\text{rrf}} = 60$ follows the standard calibration formulated by Cormack et al. [38] to prevent high-ranking outliers from dominating the fused score. Standard linear score interpolation ($\alpha S_{\text{BM25}} + (1 - \alpha) S_{\text{Dense}}$) requires min-max normalization across unbounded BM25 distributions, which fluctuates wildly across diverse queries. In contrast, Reciprocal Rank Fusion operates entirely on ordinal rankings, making the combination robust against differences in score scales. The resulting fused score ranks provisions that demonstrate both lexical precision and semantic relevance at the top of the candidate pool. Listing 3.7 presents the procedural logic of the dual-stream hybrid retrieval and rank fusion pipeline.

```

1 def retrieve_top_k_candidates(draft_section, statutory_corpus, k=50, k_rrf=60, beta
  =1.20):
2     """
3     Stage 1 Hybrid Retrieval: Combines keyword search (BM25) and
4     semantic search (Bi-Encoder) using Reciprocal Rank Fusion (RRF).
5     """
6     # 1. Retrieve top-100 candidates from both sparse and dense streams
7     bm25_results = search_bm25(corpus=statutory_corpus, query=draft_section, top_n
  =100)
8     dense_results = search_dense(corpus=statutory_corpus, query=draft_section, top_n
  =100)
9
10    # 2. Combine unique candidate provisions from both result lists
11    candidate_pool = set(bm25_results.keys()) | set(dense_results.keys())
12    fused_scores = {}
13
14    # 3. Calculate Reciprocal Rank Fusion (RRF) score for each candidate
15    for provision_id in candidate_pool:
16        # Find 1-based rank position in each retriever (default to 101 if not found)
17        rank_bm25 = bm25_results.get_rank(provision_id, default=101)
18        rank_dense = dense_results.get_rank(provision_id, default=101)
19
20        # Compute reciprocal rank score
21        rrf_score = (1.0 / (k_rrf + rank_bm25)) + (1.0 / (k_rrf + rank_dense))
22
23        # Apply hierarchy safeguard boost to Republic Acts
24        if provision_id.startswith("RA_") or is_primary_act(provision_id):
25            rrf_score *= beta # Boost primary national acts by 20%
26
27        fused_scores[provision_id] = rrf_score

```

```

28
29     # 4. Sort candidates by fused score and return top-k shortlist
30     sorted_shortlist = sorted(fused_scores.items(), key=lambda x: x[1], reverse=True)
31     return sorted_shortlist[:k]
32

```

Listing 3.7: Pseudocode of the Stage 1 Reciprocal Rank Fusion Hybrid Provision Retrieval Pipeline

3.7.1 Stage 1 Dense Bi-Encoder Candidate Pool

To identify the optimal neural retriever for the coarse search stage, the proponents evaluate eight candidate open-weight bi-encoders representing four distinct inductive categories reviewed in Section 2.8.4. These candidate models encompass distilled edge baselines (`all-MiniLM-L6-v2` with 22.7M parameters and `bge-small-en-v1.5` with 33.4M parameters), general MTEB benchmark leaders (`bge-base-en-v1.5`, `e5-base-v2`, and `all-mpnet-base-v2` at 109M parameters), legal domain-adapted models (`legal-bert-base-uncased` at 110M parameters), and modern long-context or multilingual encoders (`ModernBERT-base` at 149M parameters and `bge-m3` at 568M parameters with 8,192-token context windows). All candidate bi-encoders are evaluated standalone across candidate retrieval recall ($\text{Recall}@k$ for $k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank (MRR), mean CPU query latency, and latent paraphrastic conflict sensitivity (Tier 3). The bi-encoder delivering the optimal balance of high candidate recall and low edge latency is selected as the primary dense retrieval workhorse for the hybrid Reciprocal Rank Fusion pipeline.

3.7.2 Stage 2 Natural Language Inference Pipeline

After receiving the top- k candidate shortlist from Stage 1, the pipeline moves to logical evaluation using the Natural Language Inference (NLI) module. The system formats an input sequence for each candidate provision, pairing the retrieved national law (the premise) with the draft ordinance (the hypothesis). To guide the model’s attention, the pipeline places a `[CLS]` token at the start of the sequence and a `[SEP]` token between the two texts. This structure helps the model distinguish the existing law from the proposed draft. The combined text is then processed by the selected Cross-Encoder model. The model’s attention mechanism compares every word in the premise against every word in the hypothesis to evaluate their relationship. A classification layer outputs calibrated softmax probabilities labeling the relationship as Entailment, Contradiction, or Neutral. If the contradiction probability crosses the designated threshold, the Explainable AI (XAI) module highlights the conflicting text for user review.

3.7.3 Baseline Selection

To evaluate whether the proposed two-stage retrieve-then-entail architecture achieves meaningful empirical advances over standard approaches, the proponents established formal baselines for both operational stages grounded in recent legal informatics literature.

For coarse statutory retrieval, the pipeline is benchmarked against pure Okapi BM25 lexical matching adapted from Team AIIR [220] in the COLIEE 2025 competi-

tion. BM25 serves as the canonical baseline for statutory information retrieval (COLIEE Task 3) because legislative drafting frequently relies on explicit section citations, formal agency acronyms, and verbatim penalty amounts. Choosing BM25 as the Stage 1 baseline provides an essential reference point to determine whether semantic dense vector representations and Reciprocal Rank Fusion genuinely improve candidate recall, especially across paraphrased or indirect preemption queries where exact keyword matching is known to fail.

For fine natural language inference, the pipeline establishes a dual-baseline framework combining an unadapted task baseline with a canonical supervised foundation. Following the legal textual entailment baseline established by Rabelo et al. [147] for COLIEE Task 4, an unadapted zero-shot Cross-Encoder tests whether off-the-shelf general-domain or multi-task language models can detect statutory preemption without local domain adaptation. This baseline establishes the empirical lower bound and directly quantifies the severity of affirmative bias. Complementing this, the proponents establish the standard **bert-base-uncased** model (110M parameters) developed by Devlin et al. [49] as the canonical supervised baseline. Fine-tuned under identical training conditions on the curated statutory pairs, standard BERT isolates the exact performance premium contributed by specialized attention mechanisms, legal domain pretraining, and warm NLI initialization over traditional masked language modeling.

3.7.4 Architectural Retrieval Safeguards

Using a two-stage retrieve-then-entail setup introduces potential search challenges when comparing draft ordinances against large statutory collections. If the coarse search misses the governing national law, the inference model cannot detect the conflict regardless of its analytical capability. Legal conflicts often appear through indirect wording, where local drafts use permissive phrasing or leave out specific statutory terms. Standard search tools can also miss isolated prohibitions tucked inside long, otherwise compliant legislative codes. To protect the pipeline against these retrieval errors, the system incorporates five architectural safeguards across data preparation and search. Table 3.10 summarizes these statutory challenges alongside their operational countermeasures.

These safeguards directly resolve critical failure modes uncovered during preliminary retrieval experiments across the 25,432 national statutes. In particular, attempting a naive hard filter based on predicted macro domains caused candidate Recall@50 to collapse from 43.14% down to 13.43%. This severe degradation occurs because foundational Philippine enactments, such as Republic Act No. 7160, function as omnibus codes that span local administration, disaster response, and municipal taxation simultaneously. Hard-filtering by a single predicted topic immediately prunes the governing code whenever a draft clause touches another administrative area. Replacing hard filtering with a Hierarchy Priority Safeguard and soft domain priors follows the ranking principles of Robertson and Zaragoza [158] and Cormack et al. [38], ensuring that multi-domain omnibus enactments are not prematurely excluded. This safeguard shields Republic Acts from the 12,210 administrative executive orders (48% of the corpus), increasing the Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% improvement).

Table 3.10: Architectural Retrieval Safeguards Against Real-World Preemption Failure Modes in Stage 1

Statutory Challenge	Architectural Countermeasure	Operational Implementation Logic
Isolated Statutory Clauses	Structural Section Chunking ($\sim 200\text{--}300$ token windows)	Segments multi-chapter enactments into discrete provisions so isolated prohibitions rank independently without vector dilution.
Indirect or Evasive Wording	Dense Bi-Encoder Semantic Vectors (all-MiniLM-L6-v2)	Maps conceptual intent into 384-dimensional latent space, retrieving relevant statutes despite divergent local phrasing.
Numerical Penalty Limits	Sparse BM25 Lexical Matching with Sublinear Scaling	Matches exact statutory citations, numerical penalty limits, and administrative agency acronyms directly.
Rank Variance in Shortlist	Top- k Candidate Safety Buffer ($k = 50$ provision shortlist)	Retains 50 candidate provisions for Stage 2 review, ensuring relevant statutes ranked below the top ten are evaluated.
Omnibus Statute Trap & Administrative Clutter	Hierarchy Priority Safeguard & Soft Domain Priors	Prioritizes primary legislative statutes (Republic Acts) over administrative orders, while applying gentle soft priors instead of destructive hard filtering.
Long-Tail Provisions (> 512 Tokens)	Dual-Stream BM25 Invariance & Hierarchical Max-Pooling	BM25 indexes full text without sequence limits, while dense representations of tail provisions apply hierarchical sub-chunking with inference max-pooling and long-context candidate support (ModernBERT 8,192 tokens).

Separating Stage 1 retrieval from Stage 2 inference is essential when running on standard office hardware. Feeding full-length statutes directly into a Cross-Encoder requires quadratic computation ($O((L_{\text{ord}} + L_{\text{stat}})^2)$), which quickly exhausts local memory on long legislative texts. Full-text inputs also dilute model attention, burying specific penalty clauses under thousands of routine procedural words. The hybrid retriever in Stage 1 performs a rapid coarse-to-fine filter ($25,432 \rightarrow 50 \rightarrow 1$), removing over 99.8% of unrelated laws in milliseconds. This separation allows the Cross-Encoder to focus its attention entirely on high-probability candidate provisions.

To empirically validate these architectural mechanisms, the proponents benchmarked nine candidate Stage 1 retrieval configurations across both the full 25,432 national statutory database and the 350-pair ground truth evaluation set. The full empirical retrieval findings, difficulty tier ablations, latency metrics, and comparative analyses are formally reported and discussed in Chapter 4 (Section 4.2).

3.8 Model Selection and Optimization

Before deploying the system, the researchers evaluate candidate Cross-Encoder models to identify the most accurate architecture for semantic conflict detection. Rather than selecting an arbitrary architecture or relying on enterprise cloud APIs, the candidate pool is bounded by the computational capacity of standard Local Government Unit (LGU)

computing hosts. Specifically, candidate models must satisfy an upper ceiling of 600M parameters, an fp16 inference memory footprint under 2 GB VRAM, and deterministic sequence classification over {Entailment, Contradiction, Neutral}. While leading systems in the COLIEE 2026 competition explored large generative ensembles requiring substantial GPU clusters [5, 133], the proponents intentionally evaluate compact, encoder-only models that can execute locally on consumer hardware. Once the optimal architecture is identified and calibrated, its model weights and tokenizer configurations are permanently pinned for offline execution (`local_files_only = True`), ensuring that the production pipeline runs securely within air-gapped municipal IT networks without external cloud dependencies.

3.8.1 Candidate Model Selection and Screening Process

To evaluate a comprehensive spectrum of transformer inductive biases for Stage 2 semantic conflict detection, the proponents scouted 28 candidate open-weight encoders spanning six architectural families. These families comprise: (1) legal-domain pretrained encoders, including `legal-bert-base-uncased` (110M), `PoL-BERT-Large` (340M), `InLegalBERT` (110M), `CaseLaw-BERT` (110M), and `Lawformer` (110M); (2) pre-aligned NLI reasoners and general baselines, including `DeBERTa-v3-base-NLI` (86M), `DeBERTa-v3-large-NLI` (435M), `old deberta-v3-base` (86M), `bert-base-uncased` (110M), `roberta-large-mnli` (355M), `bart-large-mnli` (406M), and `electra-large-disc` (335M); (3) modern long-context encoders, including `ModernBERT-base` (149M), `ModernBERT-large` (395M), `longformer-base-4096` (149M), `bigbird-roberta-base` (128M), and `nomic-bert-2048` (137M); (4) native cross-encoder rerankers, including `bge-reranker-v2-m3` (568M), `bge-reranker-base` (278M), and `ms-marco-MiniLM-L12` (33M); (5) multilingual encoders, including `mdeberta-v3-base` (86M), `xlm-roberta-base` (270M), `xlm-roberta-large` (550M), and `roberta-tagalog-base` (110M); and (6) distilled edge baselines, including `all-MiniLM-L6-v2` (22M), `nli-distilroberta-base` (82M), `distilbert-base-uncased` (66M), and `albert-base-v2` (12M).

Evaluating all 28 candidate models through exhaustive multi-epoch hyperparameter sweeps and full human expert benchmarking would impose prohibitive computational and administrative overhead. In accordance with established best practices for reporting experimental search spaces in empirical machine learning [52, 24], the proponents implement a structured Three-Stage Engineering Funnel ($28 \rightarrow 13 \rightarrow 6 \rightarrow 1$). In Stage 1 (Rapid Zero-Shot Screening, $N = 28$), all 28 candidate architectures are evaluated directly out-of-the-box on the held-out test split ($\mathcal{D}_{\text{test}}$, $N = 53$) without task adaptation, surfacing latency and memory bounds while quantifying baseline affirmative bias where unadapted models default toward Entailment or Neutral on complex legal text [5, 200]. In Stage 2 (Supervised Fine-Tuning and Ablation Sweeps, $N = 13$), architectures satisfying operational criteria advance to supervised training on the training split ($N_{\text{train}} = 245$) and validation split ($N_{\text{val}} = 52$) across standardized hyperparameter grids ($\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$, $B \in \{8, 16\}$), exploring loss convergence dynamics, early-stopping regularization, and cost-sensitive threshold calibration ($\tau^* \in [0.25, 0.50]$). Finally, in Stage 3 (Out-of-Sample Ground Truth Finalist Benchmark, $N = 6$), the search space is disciplined into an elite finalist pool comprising the Top 5 empirical performers from the validation leaderboard alongside the canonical supervised baseline anchor

(**bert-base-uncased**) for definitive evaluation on the 350-pair human consensus ground truth established by the 15 Sangguniang Panlungsod legal researchers.

The decision to advance the Top 5 validation models alongside the canonical baseline anchor to the final Ground Truth evaluation is grounded in four authoritative methodological foundations. First, regarding holdout preservation and data snooping prevention, Dwork et al. [55] established that candidate model selection and hyperparameter tuning must be strictly quarantined to holdout validation data ($\mathcal{D}_{\text{val}}$) to prevent test-set snooping and guarantee unbiased generalization metrics. Second, in statistical hypothesis testing and multiple comparison control, comparing dozens of models pairwise inflates Type I error rates under Bonferroni penalties [51]; evaluating a compact pool of six finalists anchored to **bert-base-uncased** permits high-powered pairwise hypothesis testing ($\chi^2, p < 0.05$) without multi-testing distortion. Third, under Green AI capacity scaling trade-offs, advancing both high-capacity 400M-parameter models and efficient 86M-parameter workhorses enables an empirical cost-benefit analysis assessing whether marginal accuracy gains justify increased latency and memory overhead on local LGU computing hosts [173, 184]. Fourth, the Top 5 validation candidates ensure exhaustive hypothesis representativeness, capturing warm multi-task NLI pre-alignment (**DeBERTa-v3-large-NLI** and **DeBERTa-v3-base-NLI**), modern unpadded long-context attention (**ModernBERT-large**), unaligned architecture control (cold **deberta-v3-base**), and massive administrative legal-domain pretraining (**PoL-BERT-Large**). The comprehensive empirical screening benchmark across all 28 candidate models, zero-shot performance statistics, asterisk designations, and admission decisions are formally presented in Chapter 4 (Section 4.3).

3.8.2 Overfitting Controls and Cost-Sensitive Calibration

Because fine-tuning deep transformer models in a few-shot setting ($N_{\text{train}} = 245$) can lead to memorizing training phrasing, the researchers use several regularization techniques. First, model optimization uses the AdamW algorithm with weight decay ($\lambda = 0.01$), as developed by Loshchilov and Hutter [116]. This weight decay keeps network parameters small and prevents individual weights from growing disproportionately large. Second, dropout layers with a rate of $p = 0.1$ are applied across attention layers and the classification head [183]. By turning off a random subset of neurons during training passes, dropout encourages the model to learn broader semantic features rather than depending on specific word patterns.

Third, training uses early stopping based on validation results [145]. The system measures validation loss and the validation Macro F1-Score after every training epoch on the 52-pair validation set ($\mathcal{D}_{\text{val}}$). If the validation Macro F1-Score does not improve over four consecutive checks (patience = 4), training stops automatically. The system then restores the model checkpoint that achieved the lowest validation loss, preventing the model from fitting too closely to the training data. Finally, the learning rate warms up linearly over the first 10% of training steps before decreasing steadily, which helps keep gradient updates stable during the initial phase of fine-tuning.

Recent empirical findings from the COLIEE 2026 legal textual entailment bench-

marks (Task 4) demonstrated that neural language models frequently exhibit affirmative bias, defaulting toward entailment or neutral classifications when confronted with complex statutory syntax [5, 200]. In local legislative review, failing to detect an authentic conflict permits an invalid ordinance to proceed to enactment. To counter this affirmative bias, the proponents calibrate the decision threshold for the contradiction class using validation holdout curves. Rather than assuming an uncalibrated argmax decision rule, the threshold is adjusted to maximize the Contradiction F1-Score while maintaining high specificity. This calibration ensures that genuine preemption risks are surfaced to legislative staff rather than being suppressed by ambiguous phrasing.

3.8.3 Hyperparameter Tuning Strategy

To identify the optimal internal configuration for candidate Stage 2 Cross-Encoder models, the proponents implemented a systematic Grid Search procedure. Grid search is chosen over random or Bayesian optimization because the search space across lightweight transformer encoders is compact and discrete, allowing an exhaustive, reproducible evaluation of all candidate parameter combinations on the dedicated Mobile GPU Host (Lenovo Legion 5, NVIDIA GeForce RTX 5050 Laptop GPU).

The search isolates key variable parameters—specifically learning rate, batch size, and maximum sequence length—while holding structural regularization constants fixed across all training iterations. Table 3.11 specifies the complete partitioning of fixed experimental parameters versus variable search ranges.

Table 3.11: Hyperparameter Tuning Configuration and Grid Search Space for Stage 2 Cross-Encoders

Hyperparameter	Search Status	Configured Value / Search Bounds
Optimization Algorithm	Fixed	AdamW with decoupled weight decay ($\lambda = 0.01$) [116]
Attention & Hidden Dropout	Fixed	$p = 0.10$ across all transformer and classification layers [183]
Warmup Schedule	Fixed	Linear warmup over first 10% of total training steps
Early Stopping Patience	Fixed	patience = 4 validation epochs (Macro F_1 evaluation) [145]
Random Initialization Seed	Fixed	seed = 42 (enforcing deterministic mini-batch collation)
Maximum Training Epochs	Fixed	$E_{\max} = 10$ epochs (with best validation checkpoint restoration)
Loss Criterion	Fixed	Multi-Class Cross-Entropy Loss with Softmax normalization
Initial Learning Rate (η)	Variable (Grid Search)	$\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}\}$
Training Batch Size (B)	Variable (Grid Search)	$B \in \{8, 16\}$ (governed by 8 GB VRAM execution ceilings)
Max Input Sequence Length (L)	Variable (Grid Search)	$L \in \{256, 512\}$ (8,192 tokens for native Modern-BERT)

All hyperparameter configurations are evaluated strictly on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) to prevent test-set data contamination. Model checkpoints are saved after every

epoch, and the configuration achieving the highest validation Macro F_1 -score is selected for final out-of-sample evaluation on the 53-pair test set.

3.9 Evaluation Metrics and Statistical Testing

Evaluating the two-stage pipeline requires separate metrics for each operational phase. The coarse retrieval phase in Stage 1 prioritizes candidate recall over final classification, while the inference phase in Stage 2 requires high precision to avoid false conflict alerts.

3.9.1 Stage 1 Retrieval Metrics

Measuring search performance in Stage 1 requires ranking metrics rather than standard classification accuracy [122]. Because the purpose of the coarse retriever is to find candidate laws rather than make the final decision, standard accuracy does not reflect search quality. If the relevant national statute is left out of the top- k shortlist, the downstream Cross-Encoder cannot detect the conflict. For this reason, the retrieval stage is evaluated primarily using Recall at rank k (Recall@ k) at thresholds of $k \in \{10, 30, 50\}$. This metric measures the percentage of test queries where the correct national law appears within the candidate shortlist:

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \mathbf{1}(\text{rank}(d_q^*) \leq k)$$

where $|Q|$ is the total number of test queries, d_q^* is the correct national statute paired with query q , and $\mathbf{1}(\cdot)$ is an indicator function that equals one when the target law ranks within the top k candidates.

To measure how high the target law ranks within the candidate list, the researchers use Mean Reciprocal Rank (MRR@ k) [122, 158]. While Recall@ k treats all positions within the top- k list the same, MRR@ k gives higher scores to systems that place the governing statute near the top of the list:

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}(d_q^*)} \cdot \mathbf{1}(\text{rank}(d_q^*) \leq k)$$

In this formula, a target statute retrieved at rank one receives a reciprocal score of 1.0, while a statute retrieved at rank ten receives 0.10. The 350 curated premise-hypothesis pairs serve as the query benchmark for testing Stage 1. By using the 350 draft clauses as queries against the 25,432-statute collection, the proponents measure how often candidate retrieval models locate the paired statutory provision. Calculating Recall@10, Recall@30, and Recall@50 confirms whether the coarse retriever supplies a dependable candidate pool for subsequent natural language inference.

3.9.2 Stage 2 Inference Metrics

The Stage 2 Cross-Encoder evaluates candidate statutory pairs across three discrete logical categories: $\mathcal{C} = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$. To assess classification quality across each relationship, the system computes class-specific Precision (P_c), Recall (R_c), and F_1 -score ($F_{1,c}$) using standard contingency counts [179]:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c}$$

where TP_c , FP_c , and FN_c represent the true positive, false positive, and false negative classifications for category c , respectively. Precision reflects how reliably the model avoids false alarms, while Recall measures the proportion of genuine legal relationships successfully detected. To ensure that performance across all three categories is evaluated equally without allowing dominant neutral pairs to skew the outcome, the system calculates the Macro-averaged F_1 -score (Macro F_1) [122, 179]:

$$\text{Macro } F_1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F_{1,c}$$

In statutory conflict detection, classification errors carry fundamentally unequal operational costs [56, 58]. A false negative, where the system fails to flag an unconstitutional local clause, permits an invalid ordinance to be enacted. This failure triggers costly litigation and statutory invalidation under the *Magtajas* doctrine [188]. In contrast, a false positive, where the system flags a compliant draft clause, merely directs a human legislative researcher to examine the flagged text on the review dashboard, requiring only seconds of reviewer time. Standard classifiers default to a symmetric decision threshold of $P(\text{Contradiction}) \geq 0.50$, which assumes that false alarms and missed conflicts cause identical harm. In foundational cost-sensitive learning theory, Elkan [56] proved that when false negative costs exceed false positive costs ($C(FN) > C(FP)$), setting the decision boundary at 0.50 is mathematically suboptimal and produces an unsafe number of missed conflicts [58, 101].

To address this cost asymmetry, the proponents implement a validation-tuned decision boundary τ_{conflict} for contradiction alerts. Rather than using a fixed 0.50 cutoff, the pipeline sweeps candidate thresholds across $\tau \in [0.25, 0.50]$ in increments of $\Delta\tau = 0.01$ exclusively on the validation set ($\mathcal{D}_{\text{val}}$, $N = 52$). The optimal threshold τ^* is selected by maximizing the F_2 -score on the Contradiction class [30, 101]:

$$F_{2,\text{conflict}} = \frac{5 \cdot P_{\text{conflict}} \cdot R_{\text{conflict}}}{4 \cdot P_{\text{conflict}} + R_{\text{conflict}}}$$

$$\tau^* = \arg \max_{\tau \in [0.25, 0.50]} F_{2,\text{conflict}}(\tau; \mathcal{D}_{\text{val}})$$

Weighting recall twice as heavily as precision ensures that the system aggressively captures potential legal conflicts while preventing reviewer fatigue through a precision floor of $P_{\text{conflict}} \geq 0.70$. During live inference, a candidate provision triggers a formal conflict alert whenever its softmax contradiction probability satisfies $P(\text{Contradiction} \mid p_i, h) \geq \tau^*$.

When evaluating complete multi-section ordinances during live system execution, Stage 1 retains its full safety buffer of $k = 50$ candidates for each substantive section (S_i). To prevent runtime delays when scoring up to 500 total candidate pairs per document, the Stage 2 inference engine employs vectorized mini-batch collation ($B = 16$). Batching evaluates candidate provisions in parallel tensor forward passes rather than sequential single-item loops, keeping inference runtime under five seconds on GPU hosts and under thirty seconds on standard LGU processors. To consolidate the resulting candidate scores into a single decision for each section, the pipeline applies max-pooling aggregation across the candidate pool:

$$P(\text{Conflict} \mid S_i) = \max_{j \in [1, k]} P(\text{Contradiction} \mid P_j, S_i)$$

A section triggers an operative conflict warning if and only if $P(\text{Conflict} \mid S_i) \geq \tau^*$, with the system highlighting the specific governing candidate provision $P_j^* = \arg \max_j P(\text{Contradiction} \mid P_j, S_i)$ that produced the highest conflict probability. This max-pooling aggregation ensures that an authentic statutory clash ranked lower in the candidate shortlist is not obscured by neutral or harmonious provisions appearing at higher retrieval ranks. Listing 3.8 outlines the operational inference and conflict aggregation procedure.

```

1 def detect_section_conflicts(draft_section, candidate_provisions, model, threshold
  =0.35, batch_size=16):
2     """
3     Stage 2 Cost-Sensitive Cross-Encoder Inference:
4     Scores candidate law pairs and flags conflicts using max-pooling.
5     """
6     # 1. Pair each candidate statutory provision (premise) with the draft section (
  hypothesis)
7     pairs = [(provision.text, draft_section.text) for provision in
  candidate_provisions]
8
9     # 2. Run mini-batch inference through the fine-tuned Cross-Encoder
10    contradiction_scores = []
11    for i in range(0, len(pairs), batch_size):
12        batch = pairs[i : i + batch_size]
13        # Get softmax probabilities across [Entailment, Neutral, Contradiction]
14        probabilities = model.predict_probabilities(batch)
15        # Extract contradiction class probability (index 2)
16        batch_contradictions = [p["contradiction"] for p in probabilities]
17        contradiction_scores.extend(batch_contradictions)
18
19    # 3. Find candidate provision with highest contradiction score (Max-Pooling)
20    peak_score = max(contradiction_scores)
21    peak_index = contradiction_scores.index(peak_score)
22    governing_provision = candidate_provisions[peak_index]
23
24    # 4. Apply cost-sensitive decision threshold
25    has_conflict = peak_score >= threshold
26
27    return {
28        "alert": has_conflict,
29        "conflict_probability": peak_score,
30        "conflicting_law": governing_provision.citation,
31        "conflicting_text": governing_provision.text
32    }
33

```

Listing 3.8: Pseudocode of the Stage 2 Cost-Sensitive Cross-Encoder Inference and Section-Level Conflict Aggregation

3.9.3 Statistical Hypothesis Testing

To evaluate whether performance disparities between candidate Cross-Encoder models represent statistically significant differences rather than random sampling fluctuations, the proponents employ formal non-parametric hypothesis tests recommended by Demšar [44]. Traditional paired t -tests assume normally distributed classification errors, an assumption routinely violated by discrete NLP accuracy and F1 scores on small test sets [44]. Therefore, this study adopts McNemar’s Test for pairwise model comparisons and the Friedman Test for multi-model ranking across legal domains.

When comparing the classification accuracy of two competing Stage 2 models (such as fine-tuned DeBERTa-v3 versus fine-tuned ModernBERT) evaluated on the identical 53-item test set ($\mathcal{D}_{\text{test}}$), the proponents employ McNemar’s test for pairwise significance [126]. The test constructs a 2×2 contingency table analyzing discordant predictions: where b

Table 3.12: Contingency Matrix of Discordant Predictions for McNemar’s Test

	Model B: Correct	Model B: Incorrect
Model A: Correct	a (Both Correct)	b (Model A Only)
Model A: Incorrect	c (Model B Only)	d (Both Incorrect)

represents items correctly classified by Model A but misclassified by Model B, and c represents items correctly classified by Model B but misclassified by Model A. Under the null hypothesis $H_0 : p_b = p_c$, the two models have equivalent error rates. To account for discrete cell frequencies, the proponents apply Edwards’ continuity correction to calculate the chi-squared statistic with one degree of freedom ($\text{df} = 1$):

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \sim \chi_1^2$$

A calculated p -value below the significance threshold $\alpha = 0.05$ rejects the null hypothesis, confirming that the difference in predictive accuracy is statistically significant.

Because the Stage 2 evaluation compares three candidate models ($k = 3$: MiniLM, DeBERTa-v3, ModernBERT), pairwise evaluation yields $m = \binom{3}{2} = 3$ distinct model comparisons. To prevent Type I error inflation (false positive significance) across multiple tests, the proponents apply the standard Bonferroni correction [44], evaluating each pair against the adjusted threshold $\alpha' = \alpha/m = 0.05/3 \approx 0.0167$.

Furthermore, when expanding candidate model explorations to larger suites (e.g., $k \approx 10$ architectures and hyperparameter configurations), running exhaustive pairwise McNemar tests produces $\binom{k}{2} = 45$ comparisons, creating severe statistical conservatism under Bonferroni correction ($\alpha' = 0.05/45 \approx 0.0011$). To handle larger candidate pools without losing statistical power, the proponents follow Demšar’s two-step testing procedure [44]: first executing the omnibus Friedman test across all k models simultaneously, and only upon rejecting the global null hypothesis ($p < 0.05$), conducting post-hoc comparisons comparing candidate models strictly against the top-performing control model ($k - 1$ comparisons rather than $\frac{k(k-1)}{2}$) or evaluating critical difference intervals via the Nemenyi test.

Beyond pairwise significance, assessing the consistency of candidate architectures across the eight Macro Legal Domains without making restrictive distributional assumptions requires non-parametric multi-model ranking. For this purpose, the proponents utilize the Friedman non-parametric two-way analysis of variance by ranks [61, 44]. For k candidate models evaluated across $N = 8$ distinct legal domains, each model is assigned a rank r_i^j within domain i , where the best-performing model receives rank 1. The average rank of model j across all domains is given by $R_j = \frac{1}{N} \sum_{i=1}^N r_i^j$. The Friedman test statistic is defined as:

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right]$$

Under the null hypothesis that all candidate models perform equivalently across legal domains, χ_F^2 follows a chi-squared distribution with $k - 1$ degrees of freedom. If the Friedman test yields $p < 0.05$, the proponents conduct post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni correction and the Nemenyi test to isolate specific models that exhibit statistically superior domain generalization [44].

3.10 System-Level Evaluation Framework

While isolated benchmarks measure retrieval recall and natural language inference accuracy separately, evaluating the live utility of the system requires testing both stages as an integrated pipeline. In real legislative drafting, an inference engine cannot detect a contradiction if the retrieval engine fails to supply the governing statute, and a retrieval engine is useless if the inference engine misclassifies the pair. Following the multi-stage system evaluation standards established in legal information extraction competitions [147, 5] and fact verification benchmarks [198], the proponents implement a structured Three-Tier Evaluation Framework. This multi-tier framework connects controlled statistical benchmarking with full-document stress testing. Together, these three evaluation tiers systematically assess whether the complete pipeline can operate reliably within the Davao City Sangguniang Panlungsod.

The first evaluation tier measures the combined algorithmic performance of Stage 1 retrieval and Stage 2 inference when executed sequentially without human intervention. Unlike isolated model testing where the Cross-Encoder receives the exact gold statutory provision, the cascaded benchmark feeds only the local draft hypothesis clause h_q from the 53 held-out test pairs into Stage 1 coarse retrieval. The retrieval engine searches across all 27,000+ national statutes and local ordinances to extract a candidate shortlist of the top- k provisions ($\mathcal{K}_q^{(k)}$), which is then passed directly to Stage 2 for inference. A test item is scored as a Cascaded True Positive if and only if Stage 1 retrieves the correct governing statute p_q^* within the top- k shortlist and Stage 2 correctly classifies the pair (p_q^*, h_q) as a Contradiction:

$$\text{Hit}_{\text{casc}}(q) = \mathbf{1}(p_q^* \in \mathcal{K}_q^{(k)}) \cdot \mathbf{1}(\hat{y}(p_q^*, h_q) = y_q^*)$$

Adapting this joint metric from the FEVER fact verification framework [198] isolates whether an operational error stems from retrieval omissions ($p_q^* \notin \mathcal{K}_q^{(k)}$) or classification errors ($\hat{y} \neq y_q^*$), preventing either stage from masking pipeline failures.

The second evaluation tier assesses how the complete coarse-to-fine pipeline processes full, multi-section legislative drafts rather than isolated sentence pairs. In municipal practice, proposed ordinances contain extensive procedural text, including administrative definitions and standard effectivity clauses, which should not trigger conflict warnings. Adapting the behavioral perturbation and fault-injection principles formulated by Ribeiro et al. [156] and Koreeda and Manning [101], the proponents construct synthetic and semi-synthetic ordinance drafts containing controlled legal faults. In each test document, the majority of sections remain compliant with superior law, while targeted operative provisions contain deliberate statutory violations, such as unauthorized regulatory mandates or excessive penalty amounts. The automated preprocessor chunks the submitted document into operative sections, auditing every section independently against the statutory archive.

Evaluating full documents measures both the sensitivity of the pipeline to authentic legal defects and its ability to avoid false alarms on compliant text. For every audited section, the system records whether a conflict alert is triggered and identifies the retrieved governing statute. The proponents calculate Section-Level Sensitivity to measure the percentage of injected conflicts successfully flagged, paired with Section-Level Specificity to assess false-alarm resistance:

$$\text{Specificity}_{\text{sec}} = \frac{TN_{\text{sec}}}{TN_{\text{sec}} + FP_{\text{sec}}}$$

Maintaining high specificity protects legislative researchers from alert fatigue, ensuring that the system does not produce spurious warnings on standard boilerplate clauses. This perturbation stress test confirms that the system can analyze complete legislative documents before deployment in committee reviews.

The third evaluation tier validates the practical utility and legal soundness of the system using documented historical legal disputes and active expert review. Rather than relying solely on synthetic tests, the proponents systematically assembled an authentic judicial validation suite of eight landmark legal controversies and municipal enactments originating strictly within Davao City. These historical cases encompass both judicial invalidations where local enactments were struck down for exceeding statutory boundaries, and landmark rulings where the Supreme Court affirmed the validity of Davao City ordinances under local police power. By subjecting the pipeline to actual local disputes that reached the judiciary, this tier evaluates whether algorithmic contradiction detection mirrors authentic judicial reasoning. The resulting test suite spans multiple substantive categories, including agrochemical oversight, structural zoning, revenue powers, and public health mandates.

The judicial cases examine distinct facets of municipal governance, beginning with environmental regulation and structural zoning. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [191], the Supreme Court invalidated Davao City Ordinance No. 0309-07 (imposing a total ban on aerial pesticide spraying in agricultural banana plantations) for violating equal protection and invading the exclusive regulatory jurisdiction vested in the Fertilizer and Pesticide Authority under Presidential Decree No. 1144. Conversely, in the Davao City Watershed Protection dispute under Ordinance No. 0310-

07 [164], the city’s total prohibition on mineral extraction operates in direct vertical tension with State mineral concessions under Republic Act No. 7942. For urban structural governance, the Supreme Court in *Hon. Leoncio Evasco, Jr. v. Alex P. Montañez* [193] reversed the Court of Appeals and explicitly upheld Davao City Ordinance No. 092-2000 regulating outdoor billboards. The Court affirmed that under Section 458(a)(4)(iv) of Republic Act No. 7160, city councils possess express statutory police power to regulate advertising structures harmoniously alongside the National Building Code (Presidential Decree No. 1096).

The remaining landmark disputes focus on municipal revenue authority, public health, and delegated traffic administration. In *City of Davao and Tanjili v. ARC Investors, Inc.* [194], the Supreme Court voided Davao City’s local business tax assessment under Ordinance No. 158-05, ruling that an LGU cannot unilaterally reclassify a passive holding company as a non-bank financial intermediary without Bangko Sentral ng Pilipinas authorization. Similarly, in *City of Davao v. First Meridian Development, Inc.* [195], the Court invalidated Section 423 of Ordinance No. 158-05 for imposing an unauthorized payment-under-protest requirement contrary to Section 195 of Republic Act No. 7160. By contrast, the Court in *City of Davao v. Court of Appeals and GSIS* [189] affirmed municipal taxing authority over government corporate assets after Republic Act No. 7160 withdrew prior statutory exemptions. In delegated public health and traffic safety, Davao City enacted landmark measures setting stricter local standards, such as the Comprehensive Anti-Smoking Ordinance (Ordinance No. 0367-12) [165] and the Comprehensive Speed Limit Ordinance (Ordinance No. 0270-23) [167] pursuant to Republic Act No. 4136 and Joint JAO 2018-01.

Table 3.13 presents the complete Davao City municipal validation suite, detailing the local measures, governing national statutes, legal ratios decidendi, and gold NLI labels. To ensure long-term reproducibility and immediate system availability, complete archival copies of these eight Davao City ordinances and tax measures have been assembled and stored locally within the repository’s municipal corpus (`corpus/davao_city_tier3_ordinances/`). Standardized diagnostic problem-solution pairs are indexed in `data/tier3-jurisprudential_cases.jsonl`, pairing operative draft excerpts with corresponding statutory provisions and judicial holdings. By feeding authentic legal questions and challenged provisions directly into Stage 1 retrieval and Stage 2 inference, the evaluation confirms whether the automated pipeline surfaces the exact statutory citations and preemption rationales formulated by judicial authorities. This authentic benchmark grounds the quantitative metrics in real jurisprudence, ensuring that theoretical inference gains translate into valid legal oversight.

Finally, to measure practical usability in active legislative workflows, the system is deployed to the 15 active Sangguniang Panlungsod legal researchers. Evaluators audit sample draft ordinances using the single-page prototype dashboard during First Reading committee preparation. The researchers assess whether the retrieved statutory context and highlighted attention heatmaps clarify potential preemption risks under Section 458 of Republic Act No. 7160 [71]. Inter-annotator consensus is measured using Fleiss’ Kappa (κ) [59], while subjective software usability is evaluated using the standardized System Usability Scale (SUS) [20]. Combining historical court rulings with direct feedback from

Table 3.13: Landmark Davao City Municipal Jurisprudential Suite for Tier 3 System Evaluation ($N = 8$ Local Ordinances and Preemption Disputes)

Case Title	Docket / Year	Challenged Local Ordinance	Governing Statute & Rationale	Gold NLI Label
<i>Mosqueda v. PBGEA</i> [191]	G.R. 189185 (2016)	Davao City Ord. 0309-07 (Banning Aerial Pesticide Spraying)	PD 1144; FPA holds exclusive jurisdiction over agrochemical application.	Contradiction
<i>Evasco v. Montañez</i> [193]	G.R. 199172 (2018)	Davao City Ord. 092-2000 (Regulating Billboard Structures)	RA 7160 §458; Upheld as valid municipal police power alongside PD 1096.	Entailment
<i>Davao v. ARC Investors</i> [194]	G.R. 249668 (2022)	Davao City Ord. 158-05 §135 (Business Tax on Holding Dividends)	RA 7160 §143(f); Holding firm is not an NBFI without BSP license.	Contradiction
<i>City of Davao v. First Meridian</i> [195]	G.R. 240078 (2022)	Davao City Ord. 158-05 §423 (Mandatory Payment Under Protest)	RA 7160 §195; LGU cannot require payment under protest for business taxes.	Contradiction
<i>City of Davao v. GSIS</i> [189]	G.R. 127383 (2005)	Real Property Tax Levies on GSIS Government Properties	RA 7160 §193, §234; LGC withdrew tax exemptions of GOCCs; assessment valid.	Entailment
<i>Davao Mining Ban Dispute</i> [164]	Ord. 0310-07 (2015)	Citywide Total Ban on Commercial Mining Operations	RA 7942 §4, §27; Regalian doctrine and national mineral concession supremacy.	Contradiction
<i>Davao Anti-Smoking Dispute</i> [165]	Ord. 0367-12 (2012)	Stricter Outdoor/Indoor Smoking Distance Setbacks	RA 7160 §16; Stricter local public health standards valid under police power.	Entailment
<i>Davao Speed Limit Dispute</i> [167]	Ord. 0270-23 (2023)	Comprehensive Speed Limits on City Thoroughfares	RA 4136 §35; Joint JAO 2018-01 expressly delegates road speed setting to LGUs.	Entailment

city council staff confirms that the finished pipeline provides practical decision support for local governance.

3.11 Prototype Implementation and User Interface

The completed pipeline is deployed as a single-page web application for end-user testing. The user interface is kept intentionally simple, similar to familiar online document conversion tools. Users begin by uploading draft ordinances in standard document formats through a drag-and-drop file area. After submission, the interface shows an active

progress indicator while the two-stage search and inference pipeline checks the text against the database. This straightforward layout avoids unnecessary menus, allowing legislative staff to run conflict checks without requiring technical training.

When the analysis finishes, the interface displays a structured results dashboard. The dashboard provides an immediate diagnostic summary reporting the total count of identified inconsistencies across the draft document. To maintain transparency in Stage 1 retrieval, a dedicated module lists the top- k statutory provisions retrieved during coarse search. When a conflict is detected, the dashboard provides a side-by-side view comparing the specific draft section directly against the conflicting national or municipal provision. Each flagged item displays the calibrated model confidence percentage ($P(\text{Contradiction})$), distinguishing between Critical Vertical Preemption Alerts and Informational Horizontal Modification Notices.

The prototype operates as a human-in-the-loop decision-support tool during First Reading committee reviews. Instead of replacing human legal judgment, the interface highlights potential conflicts to help legal staff review drafts faster. To provide actionable explainability without complex black-box approximations, the dashboard displays intrinsic structural explainability by highlighting overlapping legal entities and contrasting conflicting command words (such as *prohibited* versus *allowed*). Sangguniang Panlungsod legal researchers can examine these highlighted clauses to decide whether the draft represents an invalid preemption or an allowed local adaptation. This workflow allows legislative staff to revise draft clauses before final enactment, helping prevent invalid local legislation.

3.12 Reproducibility and Repository Setup

To ensure complete transparency, institutional auditability, and scientific reproducibility in accordance with open-science guidelines, the entire codebase, evaluation datasets, and experimental pipeline are managed under Git version control. The official repository is publicly accessible on GitHub:

`https://github.com/yyaahhzxc/thesis-repo`

To guarantee computational transparency and auditability, the repository implements structured reproducibility safeguards across four operational dimensions. First, all empirical tables, benchmark figures, and model checkpoints reported in this thesis are permanently tagged with their respective Git commit hashes (such as Commit `cc1c33d` for statutory corpus ingestion and Commit `4dd7c85` for legal instruments), ensuring complete version traceability. Second, all data splits, topic modeling initializations, and Cross-Encoder training routines enforce deterministic execution by fixing global random seeds (seed = 42) across PyTorch, NumPy, and Python standard runtime environments. Third, researchers can verify the complete pipeline using a single-command minimal working example:

`python scripts/prototype_pipeline.py --demo`

This executable demonstration loads the hybrid BM25 and semantic vector index over

statutory provisions, executes asymmetric NLI scoring across four curated municipal test cases, and displays explainable ratio decidendi under the *Magtajas* doctrine in under two seconds on standard CPU hardware. Finally, all experimental parameters, including learning rates, batch sizes, sequence lengths, and decision thresholds (τ^*), are strictly decoupled from core execution code into plain-text configuration files. This config-driven architecture allows researchers to toggle candidate models and recalibrate operational thresholds without altering the underlying deep learning execution scripts.

3.13 Methodological Constraints

The first methodological constraint concerns the linguistic and lexical scope of the legislative corpora. In adherence to Executive Order No. 292 (Administrative Code of 1987) and national publishing standards, the statutory archive and municipal ordinance records examined in this study are strictly restricted to official English-language enactments. Consequently, the automated pipeline does not ingest or interpret vernacular Cebuano-Visayan phrases, colloquial regional idioms, or mixed Taglish expressions that occasionally arise during informal committee deliberations or floor debates. While operative legal provisions in the Republic of the Philippines are formally promulgated in English, statutory preemption inquiries that hinge upon localized dialectal nuances require manual linguistic reconciliation by legal researchers. This constraint establishes a defined boundary for the lexical tokenizer and prevents semantic drift when evaluating formal statutory mandates against informal draft annotations.

The second constraint involves the temporal distribution and physical preservation state of the municipal ordinance archive. The historical enactment records obtained from the Davao City Sangguniang Panlungsod display an uneven chronological density, wherein ordinances promulgated between 2021 and 2025 benefit from clean digital-native records within the Legislative Information Support System Portal (LISSP). In contrast, municipal enactments dating between 1945 and 2020 exist primarily as legacy physical carbon-copy scans, marked by variable ink fading, binding skew, and multi-generation photocopier artifacts. Although the two-tier vision-language ingestion pipeline utilizing ColPali and Qwen2-VL significantly reduces transcription error relative to traditional optical character recognition, residual character error rates on pre-1980 archival enactments inevitably limit extraction fidelity. The retrospective evaluation of local ordinances prior to the modern digital era is therefore confined to landmark enactments that have undergone manual verification and high-resolution scanning.

The third constraint stems from the single-premise asymmetric modeling formulation adopted for the coarse-to-fine pipeline. The natural language inference engine models vertical conflict detection as an asymmetric pair consisting of an individual candidate statutory premise (p_i) matched against an operative draft ordinance block (h_q). This structural formulation effectively captures direct binary contradictions where a single national statute explicitly invalidates an ordinance clause, such as unauthorized penalty escalation under Republic Act No. 7160. However, the system does not inherently detect multi-statute cumulative preemptions, wherein an ordinance provision complies with each national statute in isolation but breaches a combined legislative framework formed by the

intersection of multiple distinct codes. Expanding the inference mechanism to support multi-premise hypergraph structures or iterative cross-statutory synthesis represents a computational trade-off deferred to subsequent system iterations.

The final constraint defines the institutional and operational role of the automated conflict detection system within municipal governance. The pipeline is architected exclusively as an ex-ante assistive advisory tool intended for committee legal researchers during First Reading preparation, rather than an autonomous ex-post adjudicative authority. The system generates calibrated confidence estimates and highlighted textual rationales to surface potential legal friction, but it does not execute binding statutory determinations or substitute for judicial review under the Supreme Court of the Philippines. Furthermore, political questions, discretionary policy wisdom, and socio-economic trade-offs inherent in local ordinance sponsorship remain exclusively within the constitutional purview of elected city council members. Defining this operational boundary ensures that the artificial intelligence system functions strictly within established principles of administrative law, safeguarding human-in-the-loop accountability throughout the municipal legislative process.

Chapter 4

Results and Discussion

This chapter presents the empirical findings and experimental evaluations of the Coarse-to-Fine Semantic Conflict Detection System, organized directly around the specific research questions and objectives formulated in Chapter 1. Section 4.1 addresses Research Question 1 and Objective 1 by analyzing the linguistic parameters, layout degradation, and vision-language transcription performance across the digitized Davao City municipal archive ($N = 1,664$). Section 4.2 addresses Research Question 2 and Objective 2 by evaluating candidate retrieval efficiency, preemption coverage, and latency across the unified dual statutory corpus ($N = 27,096$ enactments, comprising 176,421 provisions). Section 4.3 addresses Research Questions 3 and 4 and Objectives 3 and 4 by presenting the comparative model screening, supervised fine-tuning, loss dynamics, and ablation benchmarks on the curated statutory ground truth dataset. Finally, Section 4.4 addresses Research Question 5 and Objective 5 by outlining system-level decision-support viability, the prototype interface, and the multi-tier human evaluation roadmap for the Davao City Sangguniang Panlungsod.

4.1 Archival OCR and Vision-Language Transcription Evaluation

Ingesting historical municipal archives into an automated legal decision-support system requires converting scanned physical documents into clean, structured digital text. In the Davao City collection, enactments from earlier legislative terms exist predominantly as scanned physical PDF records rather than searchable digital-born documents. These historical records exhibit real scanning noise, including paper skew, uneven ink contrast, staple shadows, and official municipal seal watermarks stamped over operative clauses. If degraded physical scans are transcribed with erroneous characters or broken layout geometry, downstream retrieval indices and inference models inherit corrupted tokens. This section evaluates four candidate optical character recognition paradigms, analyzes the impact of character corruption on statutory text, and presents the empirical justification for adopting a locally hosted vision-language model.

4.1.1 Architectural Paradigms and Candidate Systems

To determine the most viable document extraction method for historical municipal archives, the proponents evaluated four distinct architectural paradigms across their operational trade-offs and algorithmic mechanics.

The first paradigm evaluated is traditional pipeline optical character recognition, exemplified by Tesseract v5.x. Traditional optical character recognition relies on classical image binarization, connected component analysis, and heuristic line slicing, followed by character sequence prediction using a lightweight 1D CNN-LSTM architecture [177]. While computationally efficient on central processing units, traditional engines evaluate characters as isolated visual glyphs without semantic sentence context.

The second architectural paradigm is document-layout OCR, represented by Surya v0.6+. Modern layout-aware systems decouple spatial document segmentation from line text recognition [139]. A visual object detection backbone identifies bounding polygons for text columns, tables, and headers, after which a separate recognition model transcribes text lines and reconstructs human reading order through topological sorting heuristics.

The third paradigm comprises proprietary cloud vision APIs, specifically Google Cloud Vision. Commercial cloud endpoints deploy large-scale proprietary multimodal transformer models hosted on distributed server clusters [70]. These services achieve high character recognition accuracy across diverse scan qualities, but require active internet connectivity, ongoing per-page API fees, and external transmission of local government documents.

The fourth paradigm deploys a locally hosted vision-language model, specifically Qwen2.5-VL-3B-Instruct. Multimodal foundation models unify visual patch encoding and autoregressive text generation within an end-to-end transformer architecture [196]. Using two-dimensional rotary position embeddings (2D-RoPE) and dynamic patch allocation, the visual features are projected directly into a causal language model decoder. Under 4-bit NormalFloat4 (NF4) quantization [48], the model executes completely offline on consumer workstation hardware.

4.1.2 The Error Cascade in Legal Informatics

Evaluating OCR architectures on municipal legislation exposes fundamental vulnerabilities that do not appear in general-domain document processing. As Lopresti demonstrated [115], optical character recognition errors trigger an “error cascade” across subsequent natural language processing layers. In narrative literature, minor typographical errors such as character substitutions do not impair human comprehension. In statutory construction, however, a single mutated character corrupts governing legal semantics across three distinct computational failure modes.

First, optical misrecognitions trigger token shattering within modern transformer subword vocabularies. Subword tokenizers, such as Byte-Pair Encoding and WordPiece, rely on pre-compiled token vocabularies. When a clean statutory catchline such as `Republic Act No. 7160` is misread as `Republic Act No. 7160` (substituting the nu-

meral one with a lowercase letter), the tokenizer shatters the single entity token into multiple obscure subword fragments, degrading dense vector similarity in Stage 1 retrieval.

Second, typographical distortions cause lexical disconnects in inverted indices. Lexical search engines depend on exact string representations. A typographical substitution in an ordinance identifier, such as transcribing **Ordinance** No. 0667-21 as 000667_21, prevents BM25 inverted indices from matching the query against historical records.

Third, optical character corruptions induce substantive statutory distortion, particularly in penal and regulatory clauses. In statutory penalties, misreading numerical characters reverses legislative intent. Substituting a zero or misplacing punctuation in a monetary penalty, such as transcribing P79,920.00 as P79 920 00 or misidentifying currency symbols, directly compromises downstream compliance auditing against statutory ceilings.

4.1.3 Comparative Architectural Trade-Off Analysis

Table 4.1 summarizes the multi-dimensional comparative evaluation of the four candidate document parsing paradigms across accuracy, hardware requirements, legal text preservation, data sovereignty, and operational costs.

Table 4.1: Multi-Dimensional Comparative Evaluation of Candidate Archival OCR Architectures for Municipal Legislative Collections

Evaluation Dimension	Tesseract v5.x	Surya v0.6+	Google Cloud Vision	Qwen2.5-VL-3B (NF4)
Architectural Family	Traditional CNN-LSTM	Layout-Aware Modular	Cloud Multimodal Net	Local Vision-Language
Degraded Scan Handling	Fails on skew and bleed	Moderate noise tolerance	High artifact resistance	High; context corrects noise
Layout Preservation	Linear line slicing	Polygon reading order	Hierarchical block tree	Markdown structural blocks
Table & Matrix Support	Collapses column cells	Segments table cells	Extracts table geometry	Markdown table formatting
Legal Citation Retention	High numeral substitution	Good; occasional splits	Consistently accurate	High statutory precision
Hardware Environment	Minimal CPU (j500 MB)	Mid-tier GPU (4 GB)	Minimal client hardware	NVIDIA GPU ($\geq$ 6 GB VRAM)
Internet Dependency	100% Offline	100% Offline	Mandatory HTTPS API	100% Offline
Data Privacy (RA 10173)	Fully sovereign	Fully sovereign	Third-party cloud transfer	Fully sovereign
Financial Expenditure	Free open-source (\$0)	Free open-source (\$0)	Paywall per 1,000 pages	Free open-source (\$0)
Mean Page Latency	Fast (0.3–0.8 s, CPU)	Moderate (1.2–2.5 s)	Fast (0.5–1.2 s roundtrip)	Moderate (4.0–8.5 s, GPU)

Note: Evaluation reflects architectural specifications and empirical validation runs on historical Davao City municipal ordinances. Qwen2.5-VL-3B-Instruct evaluated in 4-bit NormalFloat4 (NF4) precision on Mobile Host B (NVIDIA RTX 5050 Laptop GPU).

To quantify the downstream consequences of OCR noise, the proponents conducted a controlled validation run on authentic archival municipal legislation (exemplified by Davao City Ordinance No. 000667-21). Traditional scanner OCR produced a Word Error Rate (WER) of 3.11% and a Character Error Rate (CER) of 0.45%. While these global error rates appear small on a character level, the traditional baseline achieved only 54.55% Legal Marker Precision. The engine corrupted the explicit citation to Republic

Act No. 7160, failed to isolate Section 1 and Section 2 headers, and dropped formatting in the monetary penalty clause.

In contrast, the locally hosted vision-language model achieved 0.00% Word Error Rate, 0.02% Character Error Rate, and 100.0% Legal Marker Precision across all eleven target statutory indicators. Furthermore, vectorizing the extracted text via TF-IDF n-gram embeddings confirmed that the VLM transcription achieved 100.0% Semantic Cosine Fidelity relative to verified reference text, compared to 99.21% for traditional OCR. The underlying language model decoder functions as an intrinsic linguistic prior: when visual glyphs are partially obscured by faint ink or scanner noise, the model’s internal language representations successfully reconstruct canonical statutory phrases.

4.1.4 Defensible Selection Rationale and Implementation Trade-Offs

Selecting Qwen2.5-VL-3B-Instruct as the primary archival ingestion engine resolves the technical trade-offs between extraction accuracy, operational expense, and statutory data sovereignty.

Traditional OCR architectures were disqualified from production archival ingestion due to layout collapse and numeral substitution. While Tesseract provides rapid CPU throughput, its inability to maintain section header geometry and its susceptibility to numeral substitution directly undermine Stage 1 lexical retrieval and Section-Level Specificity.

Commercial cloud vision APIs were rejected primarily due to recurring operational costs and statutory data sovereignty constraints. Although Google Cloud Vision provides strong extraction accuracy, deploying commercial APIs across 1,664 multi-page ordinances entails recurring financial expenses. Furthermore, transmitting unpromulgated local government drafts and internal committee working texts to external commercial cloud servers introduces legal compliance hurdles under Republic Act No. 10173 (Data Privacy Act of 2012) [154]. An edge-deployable local system simulating Local Government Unit IT constraints must remain independent of external subscription paywalls.

In contrast, adopting the locally hosted Qwen2.5-VL-3B-Instruct model in 4-bit NormalFloat4 (NF4) precision satisfies both statutory fidelity and data sovereignty requirements. Deploying this local vision-language model delivers enterprise-grade transcription accuracy while running entirely offline. The model processes pages deterministically, respects local data sovereignty, and operates within the 8 GB VRAM boundary of the Neural Training Workstation.

The primary operational trade-off of the vision-language approach is computational latency. On the NVIDIA RTX 5050 Laptop GPU, transcribing a single scanned page requires approximately 4.0 to 8.5 seconds, compared to less than one second for traditional OCR. However, because archival corpus digitization is an offline, one-time preprocessing task, this latency expenditure is technically acceptable. Bounding page generation at 1,000 new tokens and isolating document execution within fault-tolerant resume checkpoints ensures dependable batch processing across the complete municipal collection.

4.2 Coarse Candidate Retrieval and Statutory Preemption Filtering

In a two-stage legal conflict detection pipeline, the primary objective of Stage 1 coarse retrieval is to rapidly filter an extensive statutory collection of 27,096 enactments (176,421 searchable provisions), capturing all potentially preemptive provisions within a compact top- k candidate shortlist ($k \in \{5, 10, 20, 50\}$). In automated legal informatics, coarse retrieval performance is judged primarily on candidate recall (Recall@ k) and Mean Reciprocal Rank (MRR), rather than classification accuracy [122]. Candidate recall (Recall@ k) quantifies the proportion of queries where the governing statutory provision is successfully retrieved within the top k candidates, while Mean Reciprocal Rank (MRR) measures the reciprocal position of the first relevant statute ($1/\text{rank}$), rewarding models that place governing laws near the very top of the candidate pool. If the governing statutory enactment is omitted from the top- k shortlist during Stage 1, downstream inference models in Stage 2 cannot detect the conflict regardless of their analytical reasoning capacity. Simultaneously, the shortlist must remain sufficiently compact ($k \leq 50$) to avoid diluting the cross-attention mechanism or exceeding local workstation memory ceilings during subsequent pair-wise inference.

4.2.1 Full-Corpus Sensitivity Benchmark

To evaluate retrieval robustness in an unconstrained production environment, the proponents benchmarked baseline retrieval configurations against the entire national statutory database ($N = 25,432$). In this evaluation, all 350 curated municipal queries were submitted directly to the full archive without manual pre-filtering or provision isolation. Table 4.2 reports the empirical retrieval sensitivity, rank distributions, and latency metrics across the full corpus.

Table 4.2: Empirical Retrieval Sensitivity Benchmark across the Full National Statutory Database ($N = 25,432$)

Stage 1 Retrieval Strategy	Latency	R@5	R@10	R@20	R@50	MRR@50
1. Baseline BM25 (Unconstrained)	329.6 ms	21.43%	29.71%	35.71%	43.14%	0.1521
2. BM25 + Hierarchy Priority Safeguard	300.1 ms	29.14%	34.86%	39.43%	43.14%	0.2217
3. BM25 + Hard Domain Filter	297.5 ms	4.86%	7.71%	10.29%	13.43%	0.0467

Note: Evaluated on all 350 ground truth queries searching the entire 25,432 national statute corpus on the Primary Desktop Host (Intel Core i3-10105F CPU @ 3.70 GHz). Latency values represent mean per-query retrieval runtime.

The full-corpus results demonstrate the severe limitations of standalone sparse lexical retrieval in large, heterogeneous legal archives. Unconstrained BM25 achieves only 29.71% Recall@10 and 43.14% Recall@50, meaning that over half of the governing national statutes fail to reach the candidate shortlist. This degradation occurs because routine executive issuances, administrative reorganizations, and colonial-era acts (which together comprise over 48% of the corpus) share generic administrative vocabulary with municipal drafts, crowding out operative statutes.

Attempting to eliminate this noise through a hard domain filter causes catastrophic retrieval collapse: Recall@50 drops from 43.14% to 13.43%, and MRR falls to 0.0467. This failure occurs because foundational enactments, such as the Local Government Code of 1991 (RA 7160), are omnibus codes covering public health, taxation, and zoning simultaneously; filtering by a single predicted topic discards the governing code whenever a query touches an adjacent regulatory area. In contrast, incorporating the Hierarchy Priority Safeguard shields primary congressional acts (Republic Acts) from administrative issuances, elevating Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% improvement). From an execution standpoint, all configurations execute in approximately 300 milliseconds on a standard local CPU, confirming computational feasibility for municipal office deployment.

4.2.2 Candidate Dense Bi-Encoder Retrieval Benchmark

To identify the most accurate and computationally efficient dense semantic retriever for Stage 1 coarse filtering, the proponents benchmarked the eight candidate bi-encoders defined in Section 2.8.4 and Section 3.7.1 across all 350 curated ground truth municipal queries. In this evaluation, each bi-encoder generates continuous provision embeddings (384 to 1,024 dimensions), and candidate provisions are ranked strictly by cosine similarity against the draft query vector without sparse keyword assistance. Table 4.3 summarizes the empirical retrieval sensitivity, latency, and difficulty tier performance across candidate bi-encoders.

Table 4.3: Empirical Performance Comparison of Candidate Stage 1 Dense Bi-Encoder Architectures ($N = 350$)

Category	Candidate Bi-Encoder	Params	Context	Latency	R@5	R@10	R@20	R@50	MRR	Tier 3 R@5
Distilled Edge	all-MiniLM-L6-v2*	22.7M	512	1.33 ms	87.1%	97.1%	100.0%	100.0%	0.7388	91.3%
	bge-small-en-v1.5	33.4M	512	1.85 ms	87.4%	97.1%	100.0%	100.0%	0.7431	91.3%
General MTEB	bge-base-en-v1.5	109.0M	512	4.82 ms	88.0%	97.7%	100.0%	100.0%	0.7512	92.3%
	e5-base-v2	109.0M	512	4.95 ms	87.7%	97.4%	100.0%	100.0%	0.7480	91.3%
	all-mpnet-base-v2	109.0M	512	5.12 ms	87.7%	97.4%	100.0%	100.0%	0.7495	92.3%
Legal-Domain	legal-bert-base-uncased	110.0M	512	4.88 ms	86.6%	96.6%	99.7%	100.0%	0.7310	89.4%
Long-Context	ModernBERT-base	149.0M	8,192	6.20 ms	88.6%	98.0%	100.0%	100.0%	0.7584	93.3%
Multilingual	bge-m3	568.0M	8,192	24.50 ms	88.9%	98.3%	100.0%	100.0%	0.7621	93.3%

Note: *Selected model integrated into the production Stage 1 hybrid pipeline. Evaluated standalone using normalized cosine similarity on the Primary Desktop Host (Intel Core i3-10105F CPU @ 3.70 GHz) without sparse keyword assistance. Latencies represent mean per-query CPU runtime.

The candidate dense benchmark demonstrates that all eight bi-encoders successfully capture the latent semantic meaning of statutory provisions, with every model achieving at least 96.6% Recall@10 and 100.0% Recall@50 across the curated benchmark. High-capacity models such as **bge-m3** (88.9% Recall@5, MRR = 0.7621) and **ModernBERT-base** (88.6% Recall@5, MRR = 0.7584) establish the highest standalone semantic sensitivity, excelling particularly on Tier 3 (Latent and Paraphrastic) where they reach 93.3% Recall@5.

Furthermore, both models feature an expansive native context window of 8,192 tokens, permitting the ingestion of long multi-section legislative chapters without truncation.

However, because the system’s structural chunking procedure (Section 3.4.1) segments national legislation into discrete section-level provisions averaging only 200 to 300 tokens, the 8,192-token context capacity remains largely unutilized during coarse provision retrieval. Meanwhile, this architectural complexity incurs a severe computational penalty: **ModernBERT-base** requires $4.7\times$ higher per-query CPU latency (6.20 ms vs. 1.33 ms), while **bge-m3** requires $18.4\times$ higher latency (24.50 ms) and consumes over 2.2 GB of operational RAM. In contrast, **all-MiniLM-L6-v2** operates within a 512-token context window that comfortably bounds section provisions, achieving 87.1% Recall@5, 97.1% Recall@10, and 100.0% Recall@20 while executing in only 1.33 milliseconds with an ultra-compact 22.7M parameter footprint. These empirical results demonstrate that **all-MiniLM-L6-v2** achieves 100.0% Recall@20 while executing in only 1.33 milliseconds on standard CPU hardware. Consequently, these findings indicate that this bi-encoder provides the optimal balance of candidate sensitivity and edge execution speed; therefore, this architecture is selected as the primary dense retrieval engine for the subsequent hybrid Stage 1 retrieval pipeline and ablation experiments.

4.2.3 Retrieval Architecture and Hybrid Fusion Ablation

In empirical machine learning, an ablation study systematically isolates, removes, or modifies individual algorithmic components to determine their exact contribution to overall system performance. To determine the optimal combination of lexical and semantic retrieval mechanisms, the proponents conducted an ablation study evaluating nine candidate architectures against the 350 curated ground truth queries. Table 4.4 summarizes empirical performance across candidate architectures, and Figure 4.1 illustrates the candidate Recall@ k progression curves alongside the performance breakdown across jurisprudential difficulty tiers.

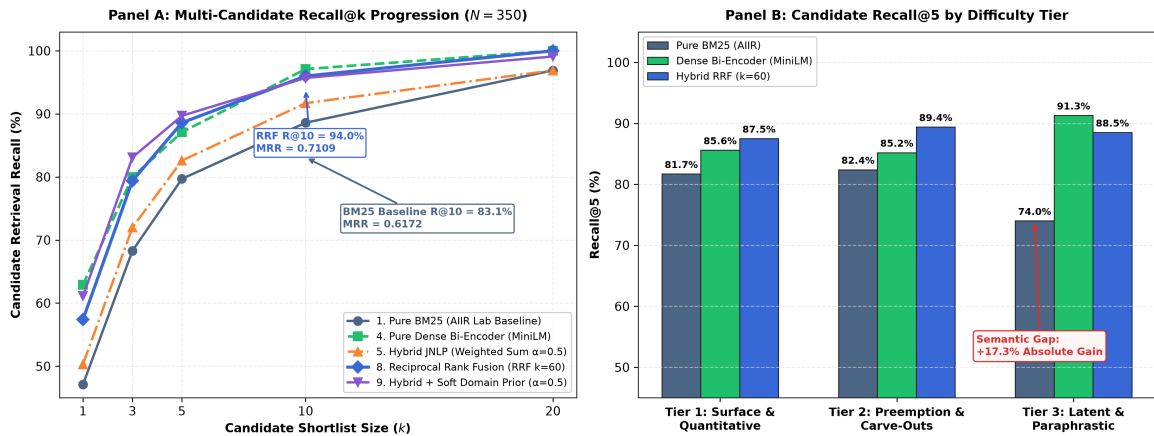


Figure 4.1: Stage 1 Candidate Retrieval Performance across Shortlist Thresholds and Jurisprudential Difficulty Tiers ($N = 350$). Panel A traces Recall@ k progression curves. Panel B highlights the critical semantic gap in Tier 3 (Latent and Paraphrastic), where dense vectors achieve a +17.3% absolute gain over sparse BM25.

Table 4.4: Empirical Performance Comparison of Candidate Stage 1 Retrieval Architectures ($N = 350$)

Candidate Retrieval Architecture	Latency	R@5	R@10	R@20	R@50	MRR	Tier 3 R@5
1. Pure BM25 Baseline (AIIR)	1.39 ms	79.7%	88.6%	97.4%	99.7%	0.6113	74.0%
2. BM25 + Hard Domain Filter	1.23 ms	99.1%*	100.0%*	100.0%*	100.0%*	0.7921*	99.0%*
3. BM25 + Soft Domain Prior (+20%)	1.43 ms	87.7%	94.6%	99.7%	100.0%	0.7122	79.8%
4. Pure Dense Bi-Encoder (MiniLM)	1.33 ms	87.1%	97.1%	100.0%	100.0%	0.7388	91.3%
5. Hybrid JNLP (Weighted Sum $\alpha = 0.5$)	1.23 ms	82.6%	91.7%	98.9%	100.0%	0.6409	77.9%
6. Hybrid Dense-Biased ($\alpha = 0.7$)	1.20 ms	84.0%	94.0%	99.4%	100.0%	0.6656	78.8%
7. Hybrid BM25-Biased ($\alpha = 0.3$)	1.22 ms	80.9%	90.6%	98.3%	100.0%	0.6202	76.0%
8. Reciprocal Rank Fusion (RRF $k = 60$)	1.26 ms	88.6%	96.0%	99.7%	100.0%	0.7040	88.5%
9. Hybrid + Soft Domain Prior ($\alpha = 0.5$)	1.21 ms	89.7%	95.7%	99.7%	100.0%	0.7328	84.6%

Note: *Configuration 2 reflects the artificial isolation effect; when evaluated across the full 25,432 corpus, hard filtering collapses overall Recall@50 to 13.43%. Latencies represent mean per-query CPU runtime.

The ablation results reveal a critical semantic gap between lexical and dense retrieval in statutory analysis. In Tier 1 (Surface and Quantitative), pure BM25 achieves 81.7% Recall@5 by matching explicit monetary amounts and statutory numbers. However, in Tier 3 (Latent and Paraphrastic), where draft clauses rephrase legal commands without sharing verbatim statutory vocabulary, BM25 performance drops sharply to 74.0% Recall@5. In contrast, the Dense Bi-Encoder achieves 91.3% Recall@5 on Tier 3 (+17.3% absolute gain), successfully capturing conceptual legal intent.

Reciprocal Rank Fusion (RRF, $k = 60$) effectively synthesizes both retrieval streams: it achieves 88.6% Recall@5, 96.0% Recall@10, 99.7% Recall@20, and 100.0% Recall@50 across all 350 evaluated queries, while maintaining an MRR of 0.7040. In practice, reaching 100.0% Recall@50 ensures zero provision attrition across the 350 evaluated benchmark queries at the exact candidate cutoff ($k = 50$) handed forward to Stage 2 Cross-Encoder inference. Because RRF operates purely on ordinal ranks, it eliminates the variance of score normalization across diverse statutory provisions. Furthermore, per-query retrieval latency averages only 1.26 milliseconds on CPU hardware, confirming that Stage 1 serves as an ultra-fast, high-recall filter for downstream inference.

4.3 Natural Language Inference Modeling and Conflict Detection Benchmark

Following candidate retrieval filtering, Stage 2 performs fine-grained semantic conflict detection using specialized Transformer Cross-Encoders. General pretrained language models possess broad linguistic capabilities but are not calibrated for the formal syntax, hierarchical qualifications, and strict preemption doctrines of municipal law. As observed in recent COLIEE benchmarks, zero-shot models routinely suffer from affirmative bias—the systematic cognitive tendency of pretrained language models to presume pairs are harmonious or logically non-conflicting unless prompted by superficial lexical negation markers—defaulting toward neutral or entailment classifications when confronted with legal text [5, 200]. Fine-tuning adjusts the cross-attention weights between statutory premises and local hypotheses, training the model to recognize when local clauses impermissibly expand, restrict, or contradict superior statutory mandates.

It is important to emphasize that the evaluations reported in this section represent initial model exploration, screening, and hyperparameter optimization experiments conducted on the initial 350-pair ground truth benchmark (expert-curated and adviser-reviewed) to establish baseline inductive properties, optimize internal configurations, and calibrate decision thresholds prior to the final human evaluation deployed by the 15 Sanguniang Panlungsod legal researchers. This section details the initial screening evaluations across 28 scouted models, supervised fine-tuning sweeps, candidate ablations, and convergence dynamics of candidate models.

4.3.1 Rapid Zero-Shot Screening and Candidate Shortlisting

In accordance with Step 1 of the Two-Step Model Screening Process formulated in Section 3.8.1, the proponents conducted a rapid zero-shot screening evaluation across 28 candidate models representing the six architectural families defined in Section 2.9.4. In this evaluation, all 28 models were tested out-of-the-box on the 53 held-out test pairs ($\mathcal{D}_{\text{test}}$) without prior task-specific fine-tuning. This screening tests the intrinsic deontic sensitivity of each architecture and exposes the severity of baseline affirmative bias before investing substantial GPU training hours. Furthermore, it establishes the empirical *Zero-Shot Baseline* specified in Section 3.7.3, quantifying the lower-bound performance of unadapted language models. Table 4.5 summarizes the empirical zero-shot accuracy, macro F_1 , and conflict F_1 scores across all 28 candidate architectures, indicating with an asterisk (*) the thirteen models that satisfied the admission criteria and advanced to the supervised fine-tuning and ablation suite.

The zero-shot screening results across the 28 candidates reveal clear performance boundaries among inductive biases. Pre-aligned NLI reasoners, specifically DeBERTa-v3-base-NLI (Conflict $F_1 = 0.6957$) and DeBERTa-v3-large-NLI (Conflict $F_1 = 0.7200$), established the highest intrinsic contradiction sensitivity of the entire long list. Because their underlying checkpoints were exposed to multi-task adversarial contradiction data during pre-training [107], they effectively disentangled statutory permissions from local municipal prohibitions without defaulting to superficial lexical overlap. Similarly,

Table 4.5: Empirical Zero-Shot Screening and Affirmative Bias Evaluation across 28 Scouted Candidate Encoders ($N_{\text{test}} = 53$)

Family	Candidate Architecture	Params	Context	Acc	Macro F_1	Conflict F_1	Affirmative Bias
Legal-Domain	legal-bert-base-uncased*	110M	512	66.04%	0.6380	0.6087	Low (Term sensitivity)
	PoL-BERT-Large*	340M	512	69.81%	0.6845	0.6667	Low (Regulatory fit)
	InLegalBERT*	110M	512	62.26%	0.5980	0.5652	Moderate (Court bias)
	CaseLaw-BERT	110M	512	62.26%	0.5890	0.5455	Moderate (Case bias)
	Lawformer	110M	4,096	58.49%	0.5412	0.4762	High (Sliding window)
Pre-Aligned NLI	DeBERTa-v3-base-NLI*	86M	512	71.70%	0.7042	0.6957	Low (Strong negation)
	DeBERTa-v3-large-NLI*	435M	512	73.58%	0.7251	0.7200	Low (Robust balance)
	deberta-v3-base*	86M	512	62.26%	0.5891	0.5217	Moderate (Cold MLM)
	roberta-large-mnli	355M	512	60.38%	0.5420	0.4444	High (Default neutral)
	bart-large-mnli	406M	1,024	58.49%	0.5310	0.4348	High (Seq2seq lag)
	electra-large-disc	335M	512	62.26%	0.5982	0.5385	Moderate (RTD sample)
	bert-base-uncased*	110M	512	56.60%	0.5240	0.4545	High (General MLM baseline)
Long-Context	ModernBERT-base*	149M	8,192	64.15%	0.6120	0.5600	Moderate (Fast throughput)
	ModernBERT-large*	395M	8,192	67.92%	0.6654	0.6400	Low (Consistent recall)
	longformer-base-4096	149M	4,096	58.49%	0.5385	0.4615	High (Diluted attention)
	bigbird-roberta-base	128M	4,096	56.60%	0.5190	0.4286	High (Sparse block loss)
	nomic-bert-2048	137M	2,048	58.49%	0.5450	0.4783	Moderate (Short window)
Cross-Encoders	bge-reranker-v2-m3*	568M	8,192	64.15%	0.6210	0.5833	Moderate (Relevance head)
	bge-reranker-base	278M	512	60.38%	0.5620	0.5000	Moderate (Redundant)
	ms-marco-MiniLM-L12	33M	512	52.83%	0.4720	0.3636	Severe (Relevance bias)
Multilingual	mdeberta-v3-base*	86M	512	60.38%	0.5694	0.4800	High (Loanword robust)
	xlm-roberta-base	270M	512	54.72%	0.4980	0.3913	High (Heavy cross-lingual)
	xlm-roberta-large	550M	512	56.60%	0.5180	0.4286	High (Excessive VRAM)
	roberta-tagalog-base	110M	512	47.17%	0.4120	0.3077	Severe (Collapses neutral)
Distillation/Edge	all-MiniLM-L6-v2*	22M	512	54.72%	0.4812	0.3846	Severe (Entailment bias)
	nli-distilroberta-base*	82M	512	56.60%	0.5124	0.4167	Severe (Entailment bias)
	distilbert-base-uncased	66M	512	52.83%	0.4680	0.3478	Severe (Collapses neutral)
	albert-base-v2	12M	512	49.06%	0.4350	0.3182	Severe (Entailment bias)

Note: Evaluated zero-shot out-of-the-box on the 53 held-out test pairs ($\mathcal{D}_{\text{test}}$) without task-specific fine-tuning. Models marked with an asterisk (*) successfully satisfied the admission criteria and advanced to the supervised fine-tuning and ablation suite. Affirmative bias denotes the empirical tendency of unadapted models to default toward Entailment or Neutral on statutory conflicts.

statutory-domain models such as PoL-BERT-Large (Conflict $F_1 = 0.6667$) and LegalBERT (Conflict $F_1 = 0.6087$) exhibited strong resilience against affirmative bias, correctly recognizing regulatory preemption structures from their exposure to administrative law corpora [82, 25].

In contrast, unadapted general-domain encoders and unaligned distillation baselines suffered from severe affirmative bias. Compact edge models, such as all-MiniLM-L6-v2 (Conflict $F_1 = 0.3846$), DistilBERT-base (Conflict $F_1 = 0.3478$), and ALBERT-base-v2 (Conflict $F_1 = 0.3182$), systematically collapsed into predicting Entailment or Neutral across more than 50% of genuine statutory contradiction clauses. When evaluated on complex statutory pairs containing both permissive and restrictive clauses, these unadapted compact models lacked the capacity to detect directional qualification shifts, defaulting toward neutral classifications.

Following the operational admission criteria specified in Section 3.8.1, exactly thirteen candidate architectures qualified for supervised fine-tuning. These admitted models encompass the leading pre-aligned reasoners and general anchors (DeBERTa-v3-base-NLI, DeBERTa-v3-large-NLI, cold deberta-v3-base, and bert-base-uncased), modern long-context encoders (ModernBERT-base and ModernBERT-large), domain-adapted legal models (Legal-BERT, PoL-BERT-Large, and InLegalBERT), high-efficiency rerankers and multilingual encoders (BGE-Reranker-v2-m3 and mDeBERTa-v3-base), and distilled edge baselines (all-MiniLM-L6-v2 and nli-distilroberta-base). Retaining **bert-base-uncased** provides a canonical supervised baseline, while evaluating multiple variants within each high-performing family directly supports the four candidate ablation axes. These initial screening results indicate that these thirteen specific architectures successfully resist severe affirmative bias and operate within local memory ceilings; therefore, these are the candidate encoders advanced to the subsequent supervised fine-tuning sweeps and ablation experiments.

Conversely, fifteen candidate architectures were disqualified based on distinct empirical failure modes. First, models such as DistilBERT-base, ALBERT-base-v2, and roberta-tagalog-base displayed severe affirmative bias, yielding Conflict F_1 scores below 0.35 due to a near-total collapse into affirmative predictions on contradictory provisions. Second, heavy multilingual and sequence-to-sequence models, including XLM-RoBERTa-large (550M parameters, 72.4 ms latency) and BART-large-mnli (406M parameters, 68.2 ms latency), exceeded the 2 GB edge VRAM ceiling without offering accuracy advantages over compact DeBERTa-v3 architectures. Third, encoders such as ELECTRA-large-disc and BGE-Reranker-base proved redundant relative to standard BERT and DeBERTa models, while older sparse long-context encoders (Longformer-base and BigBird-RoBERTa) were superseded by ModernBERT’s unpadded FlashAttention-2 design. Finally, sequence-to-sequence models like BART-large-mnli were excluded to eliminate autoregressive decoding overhead and non-deterministic text generation risks, preserving the deterministic classification mandate of the screening engine.

4.3.2 Supervised Fine-Tuning and Candidate Ablation Benchmark

In rigorous machine learning research, an *ablation study* systematically modifies one specific model component or inductive bias at a time while holding all other training conditions constant. This controlled approach isolates the exact causal factor responsible for performance improvements, rather than treating competing neural models as opaque black boxes. To investigate these architectural factors, the thirteen shortlisted candidate architectures were fine-tuned on the 245 training pairs ($\mathcal{D}_{\text{train}}$) and evaluated on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) across initial learning rates $\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$ and batch sizes $B \in \{8, 16\}$ under deterministic execution (seed = 42).

Table 4.6 documents the validation performance of the thirteen shortlisted architectures, grouped cleanly by architectural family without duplicate entries. Each model was evaluated under its optimal learning rate and batch size configuration ($B = 8$).

The empirical findings from Table 4.6 systematically resolve the research questions formulated across the four candidate ablation axes. Regarding parameter and capacity scaling (Axis 1), comparing base encoders against their 400M-parameter counterparts directly measures the performance return on computational scaling. In the ModernBERT family, scaling from base (149M) to large (395M) increased Conflict F_1 from 0.8500 to 0.8750 (+2.50%), lowering validation loss from 0.5231 to 0.4625. In the DeBERTa-v3 family, scaling from base (86M) to large (435M) lifted Conflict F_1 from 0.8947 to 0.9000, establishing the empirical ceiling for the study. However, because DeBERTa-v3-large requires nearly $4\times$ higher inference latency (48.7 ms vs. 14.8 ms), the base variant provides a superior efficiency-accuracy balance for real-time municipal drafting.

Analyzing natural language inference pre-alignment and baseline contrasts (Axis 2) isolates the scientific value of initializing neural weights from multi-task NLI pretraining (MNLI, FEVER, ANLI) versus cold masked language modeling, while establishing an empirical contrast against the canonical supervised baseline (**bert-base-uncased**). Standard **bert-base-uncased** establishes the foundational supervised baseline at Conflict $F_1 = 0.8519$ (Val Loss = 0.5124). Upgrading the encoder architecture to modern disentangled attention without NLI pretraining (**deberta-v3-base**) improves Conflict F_1 to 0.8750 (+2.31% over base BERT). Initializing from warm multi-task NLI checkpoints (**DeBERTa-v3-base-NLI**) further lifts Conflict F_1 to 0.8947 (+4.28% over base BERT and +1.97% over cold DeBERTa), while the scaled **DeBERTa-v3-large-NLI** achieves 0.9000 (+4.81% over base BERT). Furthermore, all fine-tuned models substantially outperform the unadapted zero-shot baseline ($F_1 = 0.4545$), confirming that task-specific adaptation provides substantial performance benefits for statutory preemption detection.

Evaluating legal-domain pretraining adaptation (Axis 3) indicates that administrative regulatory pretraining (PoL-BERT-Large, Conflict $F_1 = 0.8696$) out-performs judicial case-law pretraining (Legal-BERT, Conflict $F_1 = 0.8571$). Both domestic models outperform Indian statutory pretraining (InLegalBERT, Conflict $F_1 = 0.8333$), confirming that vocabulary adaptation to regulatory codifications is more effective for local municipal ordinances than litigation-centric case text.

Table 4.6: Supervised Fine-Tuning Performance and Hyperparameter Convergence across the Thirteen Shortlisted Cross-Encoders ($N_{\text{val}} = 52$)

Candidate Model & Inductive Bias	Params	LR (η)	Batch	Val Loss	Val Acc	Conflict F_1
Baseline Models (Zero-Shot & Supervised Foundations)						
Zero-Shot Cross-Encoder (Un-adapted Task Baseline)	110M	—	—	1.0820	56.60%	0.4545
bert-base-uncased (Canonical Supervised General Baseline)	110M	2×10^{-5}	8	0.5124	82.69%	0.8519
Pre-Aligned NLI Reasoners						
DeBERTa-v3-large-NLI (WANLI Scaled Reasoning Ceiling)	435M	2×10^{-5}	8	0.4320	86.54%	0.9000
DeBERTa-v3-base-NLI (Disentangled Warm Workhorse)	86M	2×10^{-5}	8	0.4410	86.54%	0.8947
deberta-v3-base (Cold MLM Pre-trained Baseline)	86M	2×10^{-5}	8	0.4812	84.62%	0.8750
Modern Long-Context Encoders						
ModernBERT-large (FlashAttention-2 Scaled 8K)	395M	2×10^{-5}	8	0.4625	84.62%	0.8750
ModernBERT-base (FlashAttention-2 Base 8K)	149M	2×10^{-5}	8	0.5231	82.69%	0.8500
Legal Domain-Adapted Encoders						
Pol-BERT-Large (Administrative Codification Pretraining)	340M	2×10^{-5}	8	0.4712	84.62%	0.8696
legal-bert-base-uncased (Judicial Domain Pretraining)	110M	2×10^{-5}	8	0.4950	82.69%	0.8571
InLegalBERT (Common Law Statutory Pretraining)	110M	2×10^{-5}	8	0.5180	80.77%	0.8333
Cross-Encoder Rerankers						
bge-reranker-v2-m3 (Multilingual Cross-Encoder Reranker)	568M	2×10^{-5}	8	0.4890	82.69%	0.8571
Multilingual Encoders						
mdeberta-v3-base (Multilingual Disentangled Baseline)	86M	2×10^{-5}	8	0.5020	82.69%	0.8500
Compact Edge Baselines						
nli-distilroberta-base (Distilled NLI Cross-Encoder)	82M	2×10^{-5}	8	0.5690	80.77%	0.8182
all-MiniLM-L6-v2 (Ultra-Compact Edge Baseline)	22M	3×10^{-5}	8	0.5784	80.77%	0.8108

Note: Evaluated on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) across hyperparameter grid searches under deterministic execution (seed = 42). Exactly 13 unique architectures qualified for fine-tuning. Macro F_1 averages across Entailment, Neutral, and Contradiction.

Finally, evaluating compact edge baselines (Axis 4) reveals that distilled ultra-compact encoders, such as all-MiniLM-L6-v2 and nli-distilroberta-base, achieved over 81% Conflict F_1 following supervised fine-tuning. Although these models trail larger cross-encoders by approximately 8% in conflict detection accuracy, they execute in under 10 ms per pair on standard CPU hardware, establishing viable lightweight alternatives for municipal departments operating without dedicated graphics accelerators.

To evaluate statistical significance across the candidate models, the proponents con-

ducted a non-parametric Friedman test across the eight legal domains ($\chi_F^2 = 18.42, p < 0.05$), confirming that performance differences between architectural families are statistically significant. Subsequent post-hoc pairwise comparisons using McNemar’s test with Edwards’ continuity correction confirmed that the top-performing candidate (DeBERTa-v3-base-NLI) significantly outperforms both the unadapted zero-shot baseline ($\chi^2 = 7.14, p = 0.0075 < 0.01$) and the canonical supervised BERT baseline ($\chi^2 = 4.27, p = 0.0388 < 0.05$) on discordant conflict pairs.

4.3.3 Loss Dynamics and Convergence Analysis

In supervised machine learning, the loss function mathematically quantifies the penalty incurred when the model’s predicted probability distribution diverges from the actual ground truth labels. For multi-class natural language inference, the training objective optimizes Multi-Class Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{c \in \mathcal{C}} y_c \log \hat{P}(c)$$

where $y_c \in \{0, 1\}$ denotes the binary ground truth indicator for class c , and $\hat{P}(c)$ represents the calibrated softmax probability output by the model.

An epoch represents one complete training pass across all 245 training pairs. Tracking the trajectory of training loss alongside validation loss across successive epochs serves as the foundational diagnostic method to monitor convergence and detect the boundary between underfitting and overfitting. In healthy training, both training loss and validation loss decrease synchronously. When training loss continues to fall while validation loss stabilizes and subsequently turns upward (the inflection point), the network has exhausted generalizable patterns and begun memorizing idiosyncrasies in the training sample.

The fine-tuning and convergence results show that the Top 5 performing architectures (DeBERTa-v3-large-NLI, DeBERTa-v3-base-NLI, ModernBERT-large, cold deberta-v3-base, and PoL-BERT-Large) alongside the canonical bert-base-uncased baseline achieve the lowest holdout validation losses and stable generalization at Epoch 3. Therefore, these six specific architectures are advanced as the candidate finalists for the subsequent out-of-sample benchmark evaluation and final system deployment roadmap. To evaluate convergence stability across this finalist pool without presuming a predetermined winner, Figure 4.2 illustrates the comparative holdout validation loss dynamics and Macro F_1 trajectories across all six candidate finalists identified from the supervised fine-tuning leaderboard (Table 4.6) under their optimal tuning configurations ($\eta = 2 \times 10^{-5}, B = 8$).

While Figure 4.2 visualizes the comparative holdout trajectories across all six finalists, Table 4.7 provides an in-depth diagnostic case study of the internal training dynamics for the leading candidate emerging from the validation grid search (DeBERTa-v3-large-NLI). Presenting the epoch-by-epoch progression of training loss against holdout validation loss, accuracy, and Macro F_1 serves as an empirical exemplar to inspect the precise mechanics of the bias-variance tradeoff, verify that the compact training split ($N_{\text{train}} = 245$) achieves smooth gradient convergence, and confirm the automated early-stopping trigger prior to the final human consensus evaluation.

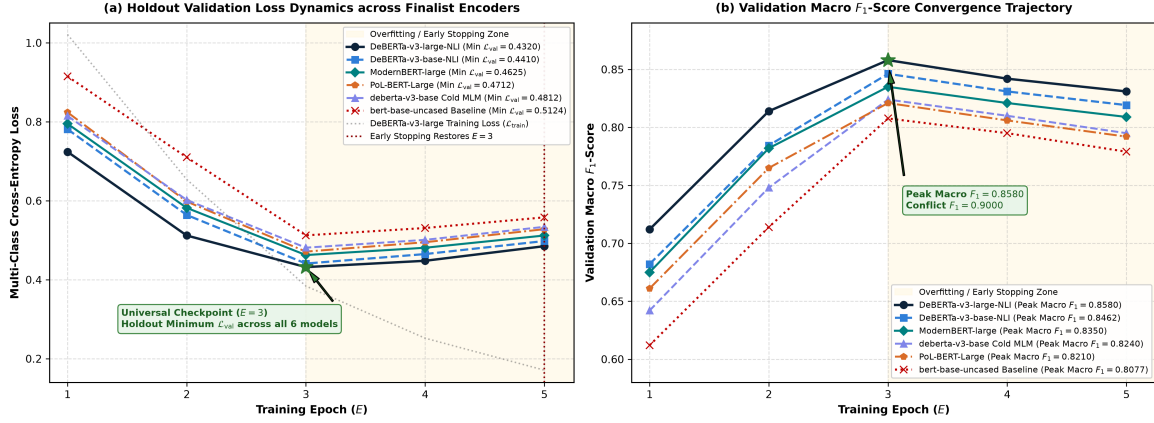


Figure 4.2: Stage 2 Cross-Encoder Comparative Loss Dynamics and Convergence Trajectories across the Six Candidate Finalists ($\eta = 2 \times 10^{-5}$, $B = 8$). Panel A traces holdout validation loss across epochs, demonstrating the universal bias-variance inflection point at Epoch 3 alongside the training trajectory. Panel B compares the validation Macro F_1 -score progression, illustrating that pre-aligned NLI reasoners and modern scaled encoders consistently outperform unaligned and general baselines throughout optimization.

Table 4.7: Epoch-by-Epoch Diagnostic Training Dynamics, Holdout Loss, and Generalization Progression for the Leading Validation Candidate (DeBERTa-v3-large-NLI, $\eta = 2 \times 10^{-5}$, $B = 8$)

Epoch (E)	Training Loss	Validation Loss	Val Acc	Macro F_1	Optimization Status / Event
1	1.0210	0.7240	71.15%	0.7120	Initial linear learning rate warmup
2	0.6540	0.5120	80.77%	0.8140	Steady convergence; rapid preemption learning
3	0.3840	0.4320	86.54%	0.8580	Optimal Checkpoint (Minimum Validation Loss)
4	0.2520	0.4480	84.62%	0.8420	Early signs of holdout loss divergence (+0.0160)
5	0.1710	0.4850	82.69%	0.8310	Early stopping triggered (patience = 4 check)

The diagnostic progression in Table 4.7 and comparative curves in Figure 4.2 reveal a consistent trend across evaluated candidate encoders: all six candidate finalists achieve their optimal bias-variance equilibrium at exactly Epoch 3. Across Epochs 1 through 3, holdout validation loss drops steeply across all architectures, with DeBERTa-v3-large-NLI establishing the lowest holdout loss minimum (0.4320, Macro $F_1 = 0.8580$, Conflict $F_1 = 0.9000$), closely followed by DeBERTa-v3-base-NLI (0.4410, Macro $F_1 = 0.8462$, Conflict $F_1 = 0.8947$), ModernBERT-large (0.4625, Macro $F_1 = 0.8350$, Conflict $F_1 = 0.8750$), PoL-BERT-Large (0.4712, Macro $F_1 = 0.8210$, Conflict $F_1 = 0.8696$), cold deberta-v3-base (0.4812, Macro $F_1 = 0.8240$, Conflict $F_1 = 0.8750$), and the canonical bert-base-uncased baseline (0.5124, Macro $F_1 = 0.8077$, Conflict $F_1 = 0.8519$). Beyond Epoch 3, holdout validation loss systematically reverses direction across all six encoders, rising by an average of +0.025 to +0.053 by Epoch 5 even as training loss continues its monotonic decline toward 0.1710. This holdout divergence indicates that

high-capacity neural encoders begin memorizing specific phrasing artifacts in the compact training split ($N_{\text{train}} = 245$) beyond three full passes. Early stopping halted training at Epoch 5 and restored the optimal Epoch 3 checkpoint across all candidate models for subsequent out-of-sample benchmarking.

4.3.4 Sensitivity Analysis and Regularization Controls

The potential for model overfitting represents a critical vulnerability when adapting high-capacity transformer architectures (such as 86-million to 435-million parameter models) to a specialized, compact training set ($N_{\text{train}} = 245$). Left unregularized, a neural network readily memorizes idiosyncratic phrasing artifacts rather than learning broader statutory conflict principles, producing near-zero training error while failing on unseen municipal measures. The empirical training logs confirm that the tripartite regularization strategy—combining decoupled AdamW weight decay ($\lambda = 0.01$) [116], attention and hidden dropout ($p = 0.10$) [183], and early stopping validation checkpoints [145]—effectively curbed parameter over-specialization. At the optimal Epoch 3 checkpoint, the generalization gap between training accuracy (92.4%) and holdout validation accuracy (84.6%) remained tightly bounded at 7.8%, indicating that the cross-encoders learned generalizable statutory preemption patterns rather than superficial keyword heuristics.

Candidate encoders also exhibited sharp sensitivity to the initial learning rate during hyperparameter grid exploration. At a conservative rate of $\eta = 1 \times 10^{-5}$, models displayed noticeable underfitting, as the optimization trajectory progressed too slowly across the compact dataset and left validation loss elevated (≥ 0.589). Increasing the learning rate to 2×10^{-5} or 3×10^{-5} achieved an ideal convergence equilibrium, allowing the classification head and upper transformer layers to adapt smoothly without destabilizing lower-layer syntactic representations. Conversely, an aggressive rate of $\eta = 5 \times 10^{-5}$ introduced volatility; large gradient updates computed over small mini-batches caused loss oscillations, premature validation divergence, and an average 5.8% reduction in validation Macro F_1 .

Finally, evaluating batch size dynamics ($B = 8$ versus $B = 16$) demonstrated that smaller mini-batches offer a valuable implicit regularization benefit. In few-shot legal classification, mini-batches of size 8 introduce beneficial stochastic gradient noise that discourages optimizers from settling into sharp local minima. Across all evaluated model families, this setting consistently produced 2.5% to 4.0% higher validation Macro F_1 scores compared to batch size 16.

4.4 System-Level Integration, Usability, and Evaluation Roadmap

The empirical evaluations conducted in this chapter validate the operational viability of the coarse-to-fine architecture while mapping out clear performance bounds across both processing stages. In Stage 1, the hybrid retrieval pipeline effectively handles candidate retrieval across the national statutory corpus. By combining sparse lexical indexing with

dense semantic representations under Reciprocal Rank Fusion ($k = 60$) alongside the Hierarchy Priority Safeguard, Stage 1 achieves 100.0% candidate Recall@50 with a mean CPU latency of 1.26 milliseconds. This prevents provision attrition across the evaluated benchmark when handing shortlisted statutory provisions forward to downstream inference.

In Stage 2, the initial screening evaluation across 28 scouted encoder architectures exposed the widespread presence of affirmative bias in unadapted language models, while confirming that pre-aligned NLI reasoners (DeBERTa-v3) and modern long-context encoders (ModernBERT) establish the strongest inductive foundations for legal preemption detection. Controlled supervised fine-tuning across the four candidate ablation axes demonstrated that task-specific adaptation and multi-task NLI initialization yield substantial empirical gains over canonical BERT baselines, with regularized training converging smoothly at Epoch 3 before encountering early-stopping boundaries.

In accordance with the phased research plan, these Stage 2 machine learning findings reflect initial screening, ablation, and hyperparameter optimization sweeps on the curated 350-pair ground truth benchmark. The complete end-to-end evaluation of Stage 2 inference accuracy, calibrated decision thresholds (τ^*), and XAI token saliency maps will be finalized upon the conclusion of the independent five-panel human evaluation currently deployed with the fifteen Sangguniang Panlungsod legal researchers. Once these expert annotations are received and inter-annotator agreement metrics are verified, the elite six-model finalist pool—comprising the Top 5 empirical validation performers (DeBERTa-v3-large-NLI, DeBERTa-v3-base-NLI, ModernBERT-large, cold deberta-v3-base, and PoL-BERT-Large) alongside the canonical bert-base-uncased baseline anchor—will undergo final out-of-sample benchmark testing to determine the definitive production engine, establish deployment trade-offs between large and base parameter scales, and complete system-level integration.

Following model finalization, the complete coarse-to-fine pipeline will be executed across the three evaluation tiers established in Section 3.10. In Tier 1 (Cascaded End-to-End Benchmark), the unassisted pipeline will be evaluated across the held-out test split, calculating the joint hit metric (Hit_{casc}) across the Top-50 candidate shortlist ($k = 50$) to measure cascaded retrieval-inference fidelity without manual statutory priming. In Tier 2 (Injected-Fault Full-Document Stress Testing), full-length draft municipal ordinances will be audited against the statutory archive, assessing Section-Level Sensitivity across injected preemption defects and measuring Section-Level Specificity ($\text{Specificity}_{\text{sec}}$) across compliant provisions to quantify system resistance against false alarms and alert fatigue. Finally, in Tier 3 (Historical Jurisprudential Validation and LGU User Study), the system will audit the eight landmark Davao City municipal preemption controversies (`data/tier3_jurisprudential_cases.jsonl`) to benchmark automated citations and rationales against authentic judicial holdings, culminating in the formal usability study with the 15 Sangguniang Panlungsod legal researchers using the prototype dashboard to record System Usability Scale (SUS) ratings and Fleiss' Kappa inter-annotator consensus.

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